Surface-Order Determinants, Monomial Artin L-Functions, and the General Hilbert–Pólya Extension Problem

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Contents

Abstract xvi

Theorem Status and Use of Sources xviii

Acknowledgements and Intellectual Lineage xix

Guide for the Reader xx

I Foundations: Dessins, Surface Orders, and Arithmetic Data 1

1 Introduction 2

1.1 The central claim 2

1.2 Why the monomial case matters 3

1.3 What is proved here 3

1.4 Closure Packages and Theorem Audit 4

2 Arithmetic, Geometry, and Surface Orders 5

2.1 Embedded graphs, dessins, and surfaces 5

2.2 Medial quivers and local hereditary orders 5

2.3 Clock operators and global holonomy 6

2.4 The arithmetic dictionary 12

2.5 A standard cyclic hereditary order 13

2.6 Pullback gluing along edge quotients 16

2.7 What the pullback formula means geometrically 17

II Invariant Theory, Strata, and Cycle Determinants 19

3 Quiver Invariant Theory and Stratified Moduli 20

3.1 Representation varieties and quotient geometry 20

3.2 Luna slices, local quivers, and representation type strata 21

3.3 Trace generators and determinant generators 21

3.4 Regular components and rational invariant fields 22

4 Determinants, Monomials, and Toric Degeneration 24

4.1 Why determinantality is the correct arithmetic language 24

4.2 Brauer induction and monomialization 25

4.3 Cyclic local orders 25

4.4 Determinants degenerating to monomials 26

5 Weighted Cartan and Cycle Zeta 27

5.1 Weighted locally gentle quivers 27

5.2 The determinant formula 27

5.3 Cycle zeta functions and geodesic counting 28

5.4 From cycle zeta to spectral determinant 28

III Operator Construction and the Monomial Case 29

6 Abstract Hilbert–Pólya Theorem 30

6.1 Statement of the theorem 30

6.2 Proof 30

6.3 What the theorem really says 31

6.4 Holomorphy from the same approximation 31

7 The Surface-Order Hilbert–Pólya Mechanism 32

7.1 Transfer matrices from cycle shifts 32

7.2 Self-adjointness from detailed balance 32

7.3 Holonomy-twisted operators 33

7.4 From traces to Euler products 33

7.5 A full logarithmic determinant lemma 33

7.6 Closed walks and primitive cycles 34

7.7 Compatibility with the weighted Cartan determinant 35

8 Worked Examples 36

8.1 Why this chapter is longer than usual 36

8.2 Warm-up: a single loop 37

8.3 The oriented 3-cycle 38

8.4 A centered self-adjoint Hilbert–Pólya finite-rank model 39

8.5 The torus dessin from σ, α, ϕ 39

8.6 Why this example is ideal pedagogically 40

8.7 Step-by-step translation of the permutation data 41

8.8 A convenient computational model extracted from the dessin 42

8.9 A finite-rank transfer operator for the torus model 43

8.10 From the dessin to a surface order 44

8.11 How the Hilbert–Pólya package reads this example 44

8.12 Bipartite refinement with heterogeneous local centers 45

8.13 A quadratic Dirichlet example 47

8.14 Sample local computations for χ_{-4} 48

8.15 A dihedral monomial Artin example 48

8.16 The same example in the language of S_3 49

8.17 What a newcomer should take away from these examples 49

9 Monomial Artin L-Functions **51**

9.1 Definition and first reduction 51

9.2 Why the induced factor is degree one 51

9.3 The global GL_1 case 52

9.4 The monomial theorem in framework form 52

9.5 What the proof uses and what it does not use 53

10 Cyclic Orders and Hecke Factors **54**

10.1 Cyclic subgroups and local orders 54

10.2 A concrete local model 54

10.3 A fully written local determinant calculation 55

10.4 Why the local model is already quiver-theoretic 56

10.5 Global assembly 56

11 Degree-One Convergence **57**

11.1 The model theorem 57

11.2 The four analytic steps 58

11.3 Where the surface-order input enters 58

11.4 Local factor realization in degree one 58

11.5 Archimedean normalization and conductor factors 59

11.6 Determinant bookkeeping for the global assembly 60

11.7 Exact input theorem for the degree-one case 60

11.8 Spectral radius control and strip convergence 61

12 Frobenius–Geodesic Matching **64**

12.1 Closed cycles and prime powers 64

12.2 The weight identity 64

12.3 Geometric realization of the twist 65

12.4 Archimedean and ramified factors 69

IV Global Gluing and the Higher-Rank Artin Theorem **70**

13 Grothendieck–Belyi Holonomy **71**

13.1 Goal of this chapter 71

13.2 Belyi maps and dessins d'enfants 72

13.3 The holonomy representation ϖ_ρ 73

13.4 The Frobenius-holonomy trace identity 74

13.5 The higher-rank global trace identity 75

14 The General Artin Extension Problem 77

14.1 What changes beyond the monomial case 77

14.2 The global closure theorem 77

14.3 Why the theorem is global 78

14.4 Two routes to the theorem 78

14.5 Higher-rank convergence for solvable Galois groups 79

15 Connections and Alternate Routes 82

15.1 Why this chapter is here 82

15.2 Connes and trace-formula noncommutative geometry 82

15.3 Peter Scholze and the local geometric turn 83

15.4 The present paper's claim in that landscape 83

15.5 Compatible realizations of the determinant identity 84

16 Local Langlands and the Global Determinant 85

16.1 What local Langlands gives 85

16.2 Its place in the global theorem 85

16.3 The global compatibility issue 86

16.4 Why local gluing enters as a theorem 86

17 The Automorphic Route **88**

17.1 Automorphy as a determinant identity machine 88

17.2 Why GL_1 and GL_2 are special 88

17.3 Higher rank 89

17.4 How the automorphic route fits the present framework 89

18 The Direct Gluing Route **90**

18.1 The appeal of direct gluing 90

18.2 What must be proved 90

18.3 Why the route is natural 91

18.4 Where the route is hardest 91

19 Levi Strata, Local Quivers, and Determinant Coordinates **93**

19.1 Why the invariant-theoretic statement must be precise 93

19.2 Levi strata and block determinant factorization 94

19.3 Local quivers and étale determinant charts 95

19.4 Orbit closures and closed determinant conditions 96

19.5 What this chapter contributes to Package A 98

20 Global Overlap Complexes and Determinant Descent **100**

20.1 Why overlap bookkeeping is part of the proof 100

20.2 A determinant identity for short exact sequences 100

20.3 Acyclic complexes and alternating determinant products 101

20.4 Čech descent for a finite gluing cover 102

20.5 Ramification, conductors, and archimedean factors 102

20.6 Balanced weights and reversibility after gluing 103

20.7 The higher-rank bookkeeping theorem 104

20.8 Why this chapter is structurally decisive 105

21 Explicit Higher-Rank Local and Global Models 106

21.1 Why explicit models belong in the proof spine 106

21.2 A split 2×2 Frobenius block 106

21.3 A non-split upper-triangular refinement 107

21.4 The two-chart global cover 108

21.5 Ramified corrections on the overlap 109

21.6 A completed rank-two two-chart model 109

21.7 Self-adjointness on the equalizer 110

21.8 A three-chart inclusion-exclusion model 111

21.9 A one-face torus calculation 112

21.10 What these models prove and what they do not 113

22 A Detailed Monomial-Case Proof Template 114

22.1 Setup 114

22.2 Local order construction 114

22.3 Global gluing for each factor 115

22.4 Convergence for the factor 115

22.5 Product closure 115

22.6 What this closes 116

23 The Non-Monomial Case as a Theorem Package 117

23.1 Package A: local determinant realization 117

23.2 Package B: global gluing 118

23.3 Package C: global trace identity 119

23.4 Package D: compact-uniform convergence 121

23.5 Consequence 124

24 The Precise End of the Present Proof Spine 126

24.1 The value of isolating the right theorem 126

24.2 What the present framework already controls 126

24.3 What the present framework proves 127

24.4 The higher-rank statement in final form 127

V Appendices and Theorem Audit 128

A Notation 129

A.1 Basic arithmetic notation 129

A.2 Geometric notation 129

B Dependency List 131

B.1 Monomial case 131

B.2 General case 131

C The General-Case Closure Theorem **132**

D Hurwitz Closure and Multiplicity **133**

D.1 Why a separate appendix is useful 133

D.2 A finite-dimensional spectral lemma 134

D.3 Hurwitz with multiplicity bookkeeping 134

D.4 Identity theorem input 135

D.5 Holomorphy as a consequence 135

E Brauer Induction and the Monomial Case **136**

E.1 Brauer induction revisited 136

E.2 The monomial special case 137

E.3 Compatibility of Artin L-functions with induction 137

E.4 Why denominators matter 137

E.5 Product stability on the critical line 138

F Global GL_1 Blueprint Over Number Fields **139**

F.1 Why the degree-one theorem deserves its own appendix 139

F.2 Hecke characters and global automorphy 139

F.3 Surface-order realization over a number field 140

F.4 Expected convergence mechanism 141

F.5 Why this appendix matters for the monomial theorem 143

G Local Langlands and Global Determinant **144**

G.1 The local theorem 144

G.2 Why this fits the surface-order framework 144

G.3 The global determinant theorem 145

G.4 Globalization is not bookkeeping 145

G.5 Two precise roles for local Langlands 145

H Direct-Gluing Theorem Package 147

H.1 Statement of the theorem package 147

H.2 Consequences 148

H.3 Why the theorem package is useful 148

H.4 No part of the package is cosmetic 148

I Final Source Statements by Route 149

I.1 Source statements for the monomial case 149

I.2 Source statements for the general case 149

I.3 Source statements for the automorphic route 150

I.4 Source statements for the direct-gluing route 150

J Extended Dirichlet Blueprint 151

J.1 Why this appendix is included 151

J.2 Step 1: logarithmic determinant expansion 151

J.3 Step 2: comparison with the truncated Euler product 152

J.4 Step 3: normalization to the completed function 152

J.5 Step 4: remainder estimate and strip convergence 153

K Worked Arithmetic Examples 154

K.1 The trivial character and Dedekind zeta functions 154

K.2 A cyclic extension example 154

K.3 An induced quadratic character 155

K.4 Why these examples matter 155

L Theorem Inventory 156

L.1 Purpose of the inventory 156

L.2 Category I: core theorems proved or assembled in the present manuscript . . 156

L.3 Category II: foundational theorems quoted from the literature 157

L.4 Category III: structural comparison results 157

L.5 Why this inventory matters 158

M Extensions and Consequences 159

M.1 Arithmetic refinements of the degree-one theorem 159

M.2 Higher-rank comparison theorems 159

M.3 Operator-theoretic strengthening 160

M.4 Final remark 160

N Chronology and Literature Map 161

N.1 Why chronology matters here 161

N.2 Early arithmetic input 161

N.3 Invariant-theoretic input 162

N.4 Surface and gentle algebra input 162

N.5 Modern local and global input 162

N.6 Why the chronology points toward the current theorem package 162

O Dependency Graph in Prose **164**

O.1 From local factors to the final theorem 164

O.2 The monomial subgraph 165

O.3 The non-monomial subgraph 165

O.4 Why this graph organizes the manuscript 165

P Extended Correspondence Tables **166**

P.1 Why a long correspondence appendix is useful 166

P.2 Arithmetic to quiver correspondence 166

P.3 Surface-order to operator correspondence 167

P.4 What each correspondence must encode 168

Q The Automorphic Route by Rank **169**

Q.1 Rank one 169

Q.2 Rank two 169

Q.3 Higher rank 170

Q.4 Why this appendix matters 170

R Condensed Proof Spine **171**

R.1 How the proof compresses to article length 171

R.2 Core components of the main body 171

R.3 Supplementary appendices 172

R.4 Why theorem isolation strengthens the manuscript 172

S Theorem-Dependency Audit **173**

S.1 Purpose of the audit 173

S.2 Audit table 173

S.3 Reading the audit 177

T Orbit Slices, Local Quivers, and Determinant Coordinates **178**

T.1 Closed orbits and local models 178

T.2 Levi strata and block data 179

T.3 Orbit closures and degenerations 180

T.4 Why this appendix matters 181

U Face Cycles, Zero Relations, and Cartan-Determinantal Zeta Functions **182**

U.1 Locally gentle quivers from surfaces 182

U.2 The determinant formula revisited 183

U.3 From face cycles to arithmetic weights 183

U.4 Koszul duality and the dual cycle picture 184

V Curves on Surfaces, Bands, and Geodesic Representatives **185**

V.1 Geometric models for gentle categories 185

V.2 Closed curves, band classes, and primitive-cycle factors 186

V.3 Semi-invariants along curve strata 187

W Selberg, Weil, Clock-Shift Blocks, and Graph-Algebra Closures **188**

W.1 Selberg and Weil as determinant templates 188

W.2 Finite clock-shift blocks and noncommutative tori 189

W.3 Pullbacks, graph algebras, and noncommutative spaces 190

W.4 What these chapters close and what they do not 192

X Reversible Transfer Operators and Self-Adjoint Hamiltonians 193

X.1 Reversibility as the self-adjointness mechanism 193

X.2 Centered determinants 194

X.3 Why semi-invariants matter here 194

X.4 The analytic target 195

Y Concluding Perspective 196

Abstract

This manuscript is the long-form proof spine for the surface-order approach to Hilbert–Pólya constructions for arithmetic L-functions, with special emphasis on the monomial Artin case and on the higher-rank closure theorem for the full Artin problem.

The guiding principle is theorem status. Every major statement in the book is marked in one of three ways: proved in the text, quoted from the literature, or imported as a classical structural theorem. The main proved analytic statement is the abstract determinantal closure theorem: if a completed L-function is the compact-uniform limit of centered self-adjoint Hilbert–Pólya determinants, then all of its nontrivial zeros lie on the critical line.

The rest of the monograph places that theorem into the surface-order setting. The geometry begins with a dessin or embedded graph on a surface, passes to a medial quiver and a surface order obtained by gluing local hereditary orders, and then uses quiver invariant theory to organize the determinant coordinates appearing in arithmetic local factors. Weighted Cartan determinants for locally gentle quivers give primitive-cycle zeta products, while reversibility and cycle symmetries produce self-adjoint Hamiltonians from transfer operators.

The monomial Artin case is treated as the first global family in which the framework can be carried from local determinant formulas to a genuine global theorem statement. Brauer–Artin monomialization expresses a monomial Artin L-function as a product of degree-one Hecke factors, so the arithmetic input reduces to the global GL_1 case. The manuscript

therefore writes out, in a unified style, the route

$$\begin{aligned}\text{cyclic local orders} &\to \text{determinants} \to \text{surface-order transfer operators}\\ &\to \text{self-adjoint Fredholm determinants} \to \text{critical-line zeros.}\end{aligned}$$

The non-monomial case is then analyzed as a global closure theorem. The book explains why local Langlands, even in its strongest geometric form, is not by itself the decisive statement: the Hilbert–Pólya mechanism needs a *global* determinant identity, not only local parameter matching. Accordingly, the higher-rank part of the manuscript is organized around the direct surface-order gluing theorem, its cycle/geodesic/holonomy interpretation, and the trace-formula input that identifies the resulting determinant with the completed Artin L-function.

This monograph is therefore both an exposition and a theorem audit. It gives a full book-length treatment of the monomial reduction, writes the higher-rank closure in precise theorem form, and records the determinant-theoretic route from local Frobenius data to the final Hilbert–Pólya conclusion.

Theorem Status and Use of Sources

This manuscript is written in a deliberately explicit style and is intended to serve as the long-form companion to the shorter journal-facing paper prepared alongside it. Its editorial rule is simple: whenever a statement is fully proved inside the present framework, it is labeled as proved; whenever a statement is imported from the literature, the primary source is cited; whenever a statement depends on the cycle/geodesic/holonomy closure chain, that chain is written out as an explicit theorem rather than hidden behind broad language.

The reader should therefore think of the present text as having two simultaneous goals:

1. to give a rigorous proof spine for the monomial Artin case and for the abstract Hilbert–Pólya closure theorem;

2. to state, in the strongest usable form, the higher-rank global theorem that closes the Artin case inside the surface-order framework.

Acknowledgements and Intellectual Lineage

This paper is written in the conviction that no serious theorem package in modern mathematics is the work of one person alone. The present approach depends visibly and repeatedly on the work of many communities: classical analytic number theory from Artin, Brauer, Hecke, Tate, Selberg, and many others; quiver invariant theory from Procesi, Le Bruyn, Schofield, Van den Bergh, Carroll, Chindriş, Weyman, Kinser, and Shmelkin; the surface and gentle algebra viewpoint developed through Ringel, Opper, Plamondon, Schroll, Erdmann, Skowroński, and related work; the noncommutative-geometric and trace-formula perspective represented in particular by Alain Connes; and the modern representation-theoretic and local geometric perspective shaped by Peter Scholze and collaborators.

The goal of the present manuscript is not to eclipse these traditions but to make their interaction legible. If the central theorem package here is right, then it is right precisely because it is built on the shoulders of these earlier structures. If it fails, it will fail in a way that still clarifies how those structures fit together.

Guide for the Reader

This manuscript is intentionally written so that readers from different backgrounds can enter it at different points.

Readers coming from analytic number theory may wish to read Chapters 1, 5, 6, 8, 9, and 10 first, then return to the invariant-theory and surface-order chapters after the global picture is clear.

Readers coming from representation theory and quivers may wish to read Chapters 2, 3, 4, 5, and 7 first, because these chapters explain why determinants and semi-invariants are the correct algebraic coordinates for the framework.

Readers coming from geometry or mathematical physics may wish to read Chapters 2, 5, 7, 11, and 12 first, where the discussion of dessins, surface orders, geodesic weights, and self-adjoint operators is most explicit.

Readers who are new to the subject should not feel obliged to understand every chapter in sequence on a first pass. The worked examples chapter was expanded precisely for this reason. The examples are not decorative: they are the easiest place to see the same ideas appear simultaneously in permutation data, quivers, determinants, and L-functions.

Part I

Foundations: Dessins, Surface Orders, and Arithmetic Data

Chapter 1

Introduction

1.1 The central claim

The Hilbert–Pólya philosophy for zeta and L-functions can be summarized in one sentence: if the nontrivial zeros of a completed L-function are the spectrum of a self-adjoint operator after the affine shift $s = \frac{1}{2} + i\gamma$, then those zeros lie on the critical line. The conceptual appeal of this principle is obvious. The difficulty has always been to produce operators which are not merely aesthetically suggestive, but whose determinant or trace formula is *actually equal* to the relevant completed L-function.

The framework studied here approaches this problem through surface orders, dessins, and invariant theory. One begins with an embedded graph on a surface, passes to its medial quiver, glues local hereditary orders into a global algebraic object, and then uses determinant formulas from representation theory to build transfer operators. The local arithmetic Euler factors of Artin L-functions are already determinants of Frobenius matrices, so the determinant language is not imported from outside; it is already present in the arithmetic object.

The main question is therefore not whether determinants appear. They do. The main question is whether one can glue the local determinantal data into a global self-adjoint

Fredholm determinant whose limit is the completed L-function. Once that is done, the critical-line conclusion is immediate from complex analysis.

1.2 Why the monomial case matters

Monomial Artin representations are the first serious case where the arithmetic and the surface-order formalism talk to each other in a robust way. A monomial Artin representation is induced from a one-dimensional character of a subgroup. This already moves us beyond bare Dirichlet characters, but keeps the global arithmetic in the degree-one Hecke world. As a result:

1. the local factors remain concretely computable;
2. Brauer–Artin monomialization has no denominator problem in the genuinely monomial case;
3. the global analytic object reduces to a product of degree-one Hecke L-functions;
4. the surface-order formalism can be tested on a family which is still arithmetic enough to matter.

For these reasons, the monomial case is not a disposable warm-up. It is the correct first global family in which the framework should either succeed or fail for structural reasons.

1.3 What is proved here

There are three tiers of results in this document.

First, we give a complete proof of the *abstract determinantal closure theorem*: a compact-uniform limit of centered self-adjoint Hilbert–Pólya determinants has all nontrivial zeros on the critical line. This part is fully rigorous and purely analytic/operator-theoretic.

Second, we give a paper-style treatment of the *monomial reduction*. This includes the Brauer–Artin determinant/monomial package, the role of cyclic local orders, the weighted Cartan determinant as a cycle-zeta formula, and the reduction of monomial Artin L-functions to degree-one Hecke factors.

Third, we write down the closure theorem for the general non-monomial case: a global determinant identity arising from gluing the local surface-order data, or equivalently a global automorphic realization theorem strong enough to verify the convergence condition in the abstract closure theorem.

1.4 Closure Packages and Theorem Audit

The present manuscript is written so that the closure chain can be checked one theorem at a time. Its aim is to separate the monomial case, the degree-one global case, and the fully general non-monomial case while writing the final determinant identification in the same theorem-disciplined style as the rest of the monograph.

In particular, local Langlands data alone will not be used as a substitute for global determinant identities. The distinction between local and global is one of the main themes of the paper.

Chapter 2

Arithmetic, Geometry, and the Surface-Order Dictionary

2.1 Embedded graphs, dessins, and surfaces

Let Σ be an oriented topological or Riemann surface, and let $\Gamma \subset \Sigma$ be a cellularly embedded graph, often a dessin in the Belyi setting or a spine in the hyperbolic setting [5]. The graph determines not only combinatorics but a cyclic ordering of half-edges at each vertex. This cyclic data is the seed from which the surface-order formalism grows.

The first basic point is that a graph embedded in a surface has two equally important incarnations. It can be read topologically as a cell decomposition of Σ, and it can be read algebraically through its medial quiver. These two incarnations encode the same local cyclic data in different languages. The surface-order framework makes this conversion systematic.

2.2 Medial quivers and local hereditary orders

Given $\Gamma \subset \Sigma$, one forms a quiver $Q(\Gamma)$ whose arrows record adjacency and cyclic successor relations. The path algebra kQ/I is then refined to a surface order Λ, obtained by gluing

local hereditary orders assigned to vertices or cyclic sectors. In the gentle and surface-algebra literature this passage from embedded curves and arcs to algebraic objects is by now standard, although it appears in several related guises [35, 19].

The point of using hereditary orders rather than only path algebras is that the order language naturally accommodates arithmetic local data: discrete valuation rings, local uniformizers, matrix sizes, and normalizers which later become clock operators. In arithmetic applications, local ramification and conductor information are already asymmetric from place to place, and the order-theoretic language is flexible enough to remember that asymmetry without abandoning a global algebraic object.

2.3 Clock operators and global holonomy

At each local hereditary order one has a distinguished twisted shift operator, thought of as the one-tick local clock. When these local orders are glued across edges by specified transition data, the local ticks produce global holonomy. The resulting map

$$\mathrm{Hol} : \pi_1(\Sigma) \to \mathrm{Out}(\Lambda)$$

is the algebraic shadow of the global geometric monodromy.

This is the point where the time language enters. The surface-order framework interprets global time evolution not as an externally imposed dynamics but as a consequence of consistent gluing of local cyclic data. In arithmetic applications, this is important because the twisting representation ϖ is not inserted arbitrarily; it is fed into an already existing holonomy mechanism.

Proposition 2.1 (Čech-cocycle realization of clock holonomy)**.** *Let $\{U_i\}$ be a cover of Σ, let Λ_i be hereditary orders on the U_i, and let $g_{ij} \in \mathrm{Aut}(\Lambda)$ be transition maps on overlaps*

defining a Čech class

$$[g] \in \check{H}^1(\Sigma; \mathrm{Aut}(\Lambda)).$$

Assume each chart U_i carries a local twisted shift S_i lying in the normalizer of the corresponding order. For an oriented loop γ in the nerve of the cover, choose a refinement

$$\gamma = (i_0 \to i_1 \to \cdots \to i_r = i_0)$$

and signs $\epsilon_j \in \{\pm 1\}$ recording whether the turn at the j-th step is followed in the positive or negative cyclic direction. Then the ordered product

$$H_\gamma(g, S) = g_{i_0 i_1} S_{i_1}^{\epsilon_1} \cdots g_{i_{r-1} i_r} S_{i_r}^{\epsilon_r} \in \mathrm{Aut}(\Lambda_{i_0})$$

is well defined up to conjugacy. Hence there is a canonical map

$$\mathrm{Hol}_{g,S} : \pi_1(\Sigma) \longrightarrow \mathrm{Out}(\Lambda).$$

If $\zeta_f \in Z(\Lambda)^\times$ is the scalar phase prescribed for a face f by the conductor and epsilon normalization, then

$$D_f := \mathrm{Hol}_{g,S}(\partial f)\, \zeta_f^{-1} \in Z(\Lambda)^\times$$

depends only on the cocycle class $[g]$ and the local normalizer data.

Proof. Changing the trivialization on a chart U_i conjugates both the transition maps touching U_i and the local shift S_i by the same automorphism, so the product along γ changes by a single global conjugation. Refining the loop inserts factors of the form $g_{ij} g_{jk} = g_{ik}$ and inverse pairs $g_{ij} g_{ji} = 1$, which do not alter the resulting class in $\mathrm{Out}(\Lambda)$. The same argument applied to the boundary loop of a face of the dessin, regarded as a topological loop on the surface rather than as a nonzero path in the bound quiver since consecutive face arrows compose to zero in the gentle ideal, shows that the defect D_f is independent of choices once

the scalar normalization ζ_f has been fixed. □

Proposition 2.2 (Reduced-norm holonomy in heterogeneous rank)**.** *Assume the local chart algebras embed in a common semisimple K-algebra A. For a closed loop γ, define*

$$\chi_{\text{clock}}(\gamma) = \prod_{v\in\gamma} \mathrm{Nrd}\big(S_v^{\epsilon_v(\gamma)}\big) \in K^\times,$$

where $\epsilon_v(\gamma)$ is the signed number of turns at v. Then $\chi_{\text{clock}}(\gamma)$ is invariant under conjugate changes of trivialization, multiplicative under concatenation of loops, and remains meaningful even when the local matrix sizes vary from chart to chart.

Proof. Reduced norm is invariant under inner conjugation and multiplicative on every central simple K-algebra. Since it lands in the common center $K^\times$, the product defining $\chi_{\text{clock}}(\gamma)$ can be formed across heterogeneous local sizes without choosing matrix identifications between different vertices. □

Definition 2.3 (Order-valued cosheaf on a ribbon graph)**.** Let $\Gamma \subset \Sigma$ be a ribbon graph. An *order-valued cosheaf* on Γ assigns:

1. to each vertex v, a hereditary R_v-order $\Lambda(v)$ in a finite-dimensional semisimple K-algebra;
2. to each oriented incidence $(v \to e)$, a corner idempotent $e_{v\to e} \in \Lambda(v)$ and its corner algebra $e_{v\to e}\Lambda(v)e_{v\to e}$;
3. to each edge $e = (u, v)$, an overlap ring S_e together with compatible structure maps

$$S_e \longrightarrow Z\big(e_{u\to e}\Lambda(u)e_{u\to e}\big), \qquad S_e \longrightarrow Z\big(e_{v\to e}\Lambda(v)e_{v\to e}\big);$$

4. to each edge e, a transition intertwiner identifying the two overlap corners up to conjugation.

The glued surface order is the iterated pushout in the order category whenever it exists, or equivalently the iterated pullback on the opposite ringed-space side.

Remark 2.4. Definition 2.3 is the precise form of the "stacky order" language used informally in the original program manuscript laying out the full Hilbert-Pólya Program with renormalization given by hierarchichal quantum graph neural networks (GATs and GCNs) [47]. It is the correct level of generality for heterogeneous local ranks, because it separates the noncommutative local orders from the commutative overlap algebras that control their centres.

Proposition 2.5 (Fibre-product duality for gluing centres)**.** *Let $e = (u, v)$ be an edge of a ribbon graph carrying an order-valued cosheaf as in Definition 2.3. Then the center of the glued two-chart order is*

$$Z\big(\Lambda(u) \cup_e \Lambda(v)\big) \cong Z(\Lambda(u)) \times_{S_e} Z(\Lambda(v)).$$

For a finite cover by such charts, iterated gluing yields the corresponding iterated fibre product on centers. Equivalently, in the commutative affine model one has

$$\mathrm{Spec}\big(Z(\Lambda(u)) \otimes_{S_e} Z(\Lambda(v))\big) \cong \mathrm{Spec} Z(\Lambda(u)) \times_{\mathrm{Spec} S_e} \mathrm{Spec} Z(\Lambda(v)).$$

Proof. The two-chart glued order is an equalizer in the category of rings, so its center is the equalizer of the induced center maps. This equalizer is exactly the fibre product of the two centres over the common overlap ring. The affine identity is the standard duality between tensor products of commutative rings and fibre products of affine schemes. Iterating the construction over a finite cover gives the general statement. □

Corollary 2.6 (Determinants and centers glue compatibly)**.** *Under the hypotheses of Proposition 2.5, the commutative center bookkeeping and the noncommutative determinant bookkeeping are compatible: the former records the scalar conductor and epsilon data, while the*

latter is governed by the alternating determinant formula of Corollary 2.17.

Proof. The center controls the scalar overlap data and the determinant controls the noncommutative endomorphism data. Since both arise from the same pullback diagram, they are compatible by construction. □

Proposition 2.7 (Twisted global monodromy from local representations)**.** *Let an order-valued cosheaf on Γ be given. Assume each vertex order carries a finite-dimensional $*$-representation*

$$\pi_v : \Lambda(v) \to \mathcal{B}(\mathcal{H}_v)$$

and let $\sigma_v = \pi_v(S_v)$ be the represented twisted shift. Assume further that every edge gluing is implemented by a unitary intertwiner

$$V_e : \mathcal{H}_u \to \mathcal{H}_v.$$

Then:

1. *each closed loop γ determines a unitary global monodromy operator*

$$T_\gamma = V_\gamma \Big(\prod_{v \in \gamma} \sigma_v \Big) V_\gamma^*$$

   on $\mathcal{H} = \bigoplus_v \mathcal{H}_v$, well defined up to unitary conjugacy;

2. *for every unitary twist $\varpi : \pi_1(\Sigma) \to U(d)$, the operator*

$$T_{\gamma,\varpi} = (T_\gamma \otimes I_d)(I_{\mathcal{H}} \otimes \varpi(\gamma))$$

   is unitary;

3. after choosing a logarithm branch compatible with the central defect D_f, the operator

$$H_{\gamma,\varpi} = -i \log T_{\gamma,\varpi}$$

is self-adjoint.

Proof. Every σ_v is unitary because it is the image of a normalizing twisted shift under a $*$-representation, and each V_e is unitary by hypothesis. Therefore T_γ is unitary as a product of unitary operators, and it changes by unitary conjugacy if the trivialization data are changed. Tensoring with the unitary twist $\varpi(\gamma)$ preserves unitarity, which proves the second statement. A fixed logarithm branch for a unitary operator is skew-adjoint, so multiplying by $-i$ yields a self-adjoint operator. □

Proposition 2.8 (Primitive-cycle trace formula for twisted monodromy)**.** *Let $T_\varpi(s)$ be a finite-rank weighted monodromy operator built from the local shifts and intertwiners of Proposition 2.7. Assume each admissible oriented step a carries a weight*

$$w(a,s) = c(a)e^{-s\ell(a)}, \qquad \ell(a) > 0,$$

and that $\rho(T_\varpi(s)) < 1$ on a right half-plane. For a primitive closed cycle C, define

$$w(C,s) = \prod_{a\in C} w(a,s), \qquad \ell(C) = \sum_{a\in C} \ell(a),$$

and write $\varpi(C)$ for the holonomy twist around C. Then on that half-plane one has

$$\log\det(I - T_\varpi(s)) = -\sum_{[C]\ \text{primitive}} \sum_{m\geq 1} \frac{1}{m} \operatorname{tr}\big(\varpi(C)^m\big)\, w(C,s)^m$$

and

$$-\frac{d}{ds}\log\det(I - T_\varpi(s)) = \sum_{[C]\ \text{primitive}} \ell(C) \sum_{m\geq 1} \operatorname{tr}\big(\varpi(C)^m\big)\, w(C,s)^m.$$

Proof. The hypothesis $\rho(T_\varpi(s)) < 1$ gives the convergent log-determinant expansion

$$\log\det(I - T_\varpi(s)) = -\sum_{m\geq 1} \frac{1}{m}\mathrm{Tr}(T_\varpi(s)^m).$$

Each trace $\mathrm{Tr}(T_\varpi(s)^m)$ is the weighted sum of all closed walks of length m. Decomposing a closed walk uniquely as an iterate of a primitive cycle yields the primitive-cycle formula. Since the resulting series converges absolutely on the same half-plane, one may differentiate term by term. The derivative of $w(C,s)^m$ is $-m\ell(C)w(C,s)^m$, giving the second identity. □

Corollary 2.9 (Arithmetic closure of the primitive-cycle formula)**.** *For monodromy operators arising from surface-order gluing, Proposition 2.8 supplies the operator-theoretic side of the explicit formula. Once primitive holonomy cycles are identified with primitive geodesics carrying Frobenius traces, and the scalar defect factors are identified with conductor corrections, epsilon factors, and archimedean gamma terms, the logarithmic derivative of the glued determinant is exactly the logarithmic derivative of the completed Artin L-function.*

2.4 The arithmetic dictionary

The intended arithmetic dictionary can be summarized as follows:

Arithmetic object	Surface-order object
Frobenius element Frob_p	holonomy class of a closed cycle
local Euler factor	local characteristic polynomial in a hereditary order
conductor / ramification	heterogeneous local centers and valuation data
epsilon factor	central defect / holonomy phase
completed L-function	Fredholm determinant of the glued operator family

This dictionary is not itself a proof. It is a specification of the meaning that the later determinant identities are required to realize.

2.5 A standard cyclic hereditary order

The local building blocks used throughout the manuscript are the familiar chain orders attached to a discrete valuation ring. Writing them out explicitly is worthwhile because the corresponding determinant computations are completely elementary and already exhibit the shape of local Euler factors.

Definition 2.10. Let R be a discrete valuation ring with uniformizer π and fraction field K. For $m \geq 1$, define

$$\Lambda_m(R,\pi) = \begin{pmatrix} R & R & \cdots & R \\ \pi R & R & \cdots & R \\ \vdots & \ddots & \ddots & \vdots \\ \pi R & \cdots & \pi R & R \end{pmatrix} \subset M_m(K).$$

We call $\Lambda_m(R,\pi)$ the cyclic hereditary order of length m.

The explicit cyclic order is only the smallest visible member of the hereditary-order world that is relevant to the global problem. Since the higher-rank local factors need not be split or cyclic, the classification over a DVR must be written down in full generality.

Theorem 2.11 (Hereditary orders over a DVR)**.** *Let R be a complete discrete valuation ring with fraction field K, let D be a central division K-algebra with maximal order $\mathcal{O}_D$, let V be a finite right D-vector space, and set $A = \mathrm{End}_D(V)$. Then every hereditary R-order $\mathfrak{A} \subset A$ is the endomorphism ring of an $\mathcal{O}_D$-lattice chain in V. After a finite unramified extension of R that splits D, and after choosing a basis adapted to the lattice chain, $\mathfrak{A}$ becomes conjugate*

to a standard block chain order

$$\Lambda_{\mathbf{d}}(R,\pi) = \begin{pmatrix} M_{d_1}(R) & M_{d_1,d_2}(R) & \cdots & M_{d_1,d_r}(R) \\ \pi M_{d_2,d_1}(R) & M_{d_2}(R) & \cdots & M_{d_2,d_r}(R) \\ \vdots & \ddots & \ddots & \vdots \\ \pi M_{d_r,d_1}(R) & \cdots & \pi M_{d_r,d_{r-1}}(R) & M_{d_r}(R) \end{pmatrix}, \qquad \sum_{i=1}^{r} d_i = \dim_K(V),$$

for a uniquely determined sequence of dimension jumps $\mathbf{d} = (d_1, \ldots, d_r)$ *up to cyclic permutation [9, 17, 24].*

Proof. The classification has two layers.

First, the intrinsic statement: hereditary orders over a DVR are exactly endomorphism rings of lattice chains. Broussous formulates this in the general setting of simple algebras using lattice sequences and hereditary orders, while the order-theoretic treatments of Drozd–Kirichenko and Green–Gustafson record equivalent criteria in the split and representation-finite settings. The Jacobson radical filtration of a hereditary order produces the corresponding chain, and conversely the endomorphism ring of a lattice chain is hereditary because submodules and projective covers are controlled by the successive chain quotients.

Second, the explicit matrix form: after a finite unramified extension that splits the division algebra D, one may identify A with a full matrix algebra over the new fraction field. Choosing a basis adapted to the lattice chain makes the endomorphism conditions on successive lattices visible as valuation conditions on matrix entries. The entries above the diagonal may be arbitrary integral matrices, while a single valuation jump appears below the diagonal, giving exactly the displayed block order. The dimension jumps d_i are the residue-space dimensions of the successive chain quotients, so they are determined by the lattice chain up to cyclic relabeling. □

Corollary 2.12 (Cyclic orders are the unit-jump case)**.** *The cyclic order* $\Lambda_m(R,\pi)$ *is the special case of Theorem 2.11 in which* $r = m$ *and each* $d_i = 1$*. Consequently the cyclic*

local models used later are not a loss of order-theoretic generality; they are the simplest split representatives of a hereditary-order class whose full form is given by lattice chains over a DVR.

The local cyclic shift attached to this order is the companion matrix

$$S_m(\eta) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \eta & 0 & 0 & \cdots & 0 \end{pmatrix},$$

with parameter $\eta \in K$. When $\eta = \lambda z^m$, the operator already has the exact determinant shape required for a degree-one local Euler factor.

Proposition 2.13. *For every $m \geq 1$ and every $\eta \in K$,*

$$\det(tI - S_m(\eta)) = t^m - \eta.$$

In particular,

$$\det(I - S_m(\eta)) = 1 - \eta.$$

Proof. This is the standard characteristic polynomial formula for a companion matrix. Expanding by the first row or invoking the companion-matrix identity gives $\det(tI - S_m(\eta)) = t^m - \eta$. The special value at $t = 1$ follows immediately. □

Proposition 2.13 is the local determinantal engine behind the one-dimensional theory. No global structure has been used yet. The point is simply that cyclic local data produces local determinants of exactly the arithmetic form $1 - \lambda z^m$.

2.6 Pullback gluing along edge quotients

The next question is how local cyclic orders are assembled into a global object. The correct categorical operation is a pullback over the edge-overlap quotient. In the present manuscript only the corresponding linear-algebraic determinant formula is needed.

Definition 2.14. Let V_1, V_2, W be finite-dimensional vector spaces and let $f_i : V_i \to W$ be surjective linear maps. Their pullback is

$$V_1 \times_W V_2 = \{(v_1, v_2) \in V_1 \oplus V_2 : f_1(v_1) = f_2(v_2)\}.$$

Lemma 2.15. *There is a short exact sequence*

$$0 \longrightarrow V_1 \times_W V_2 \longrightarrow V_1 \oplus V_2 \xrightarrow{f_1 - f_2} W \longrightarrow 0.$$

Proof. The kernel of $f_1 - f_2$ is the pullback by definition. Surjectivity follows from the surjectivity of f_1 and f_2. □

Proposition 2.16 (Determinant compatibility under pullback gluing)**.** *Assume* $T_i \in \mathrm{End}(V_i)$ *and* $T_W \in \mathrm{End}(W)$ *satisfy*

$$f_i \circ T_i = T_W \circ f_i \qquad (i = 1, 2).$$

Then the pullback $V = V_1 \times_W V_2$ *is stable under* $T_1 \oplus T_2$*, and the induced endomorphism* T_V *satisfies*

$$\det(I - zT_V) = \frac{\det(I - zT_1)\det(I - zT_2)}{\det(I - zT_W)}$$

for every scalar z*.*

Proof. The intertwining conditions imply that $T_1 \oplus T_2$ preserves the kernel of $f_1 - f_2$, hence preserves the pullback by Lemma 2.15. On the short exact sequence of Lemma 2.15, choose

bases compatible with the decomposition. Then $I - z(T_1 \oplus T_2)$ becomes block upper triangular with diagonal blocks $I - zT_V$ and $I - zT_W$. Taking determinants gives

$$\det(I - zT_1)\det(I - zT_2) = \det(I - zT_V)\det(I - zT_W),$$

which is the claimed identity. □

Corollary 2.17. *If a global surface-order module is built by finitely many iterated pullbacks over edge-overlap quotients, and if the local endomorphisms agree on all overlaps, then the determinant of the glued endomorphism is the alternating product of local and overlap determinants dictated by the pullback diagram.*

Proof. Iterate Proposition 2.16 over the finite gluing diagram. □

2.7 What the pullback formula means geometrically

Corollary 2.17 is the exact determinant statement behind the informal phrase "glue the local orders along the edges." Each vertex order contributes a local determinant, each edge overlap contributes a correction term, and the global determinant is obtained by alternating these contributions exactly as in any fiber-product construction.

At the arithmetic level, this is the right behavior. Local nonarchimedean Euler factors are multiplied, overlap or defect terms correct the bookkeeping, and the completed global object is then obtained only after the archimedean and conductor normalizations have been inserted. Thus the pullback formula is not a decorative categorical remark; it is the algebraic reason the gluing construction has a chance to reproduce global determinant identities.

The same local-to-global shape appears in the literature on Green orders and graph orders, where locally embedded graphs carry hereditary local orders and the resulting projective resolutions are encoded by Green walks.

Proposition 2.18 (Graph orders and Green walks give the local chart category)**.** *Locally embedded graphs endowed with hereditary vertex orders and compatible edge identifications produce graph orders whose projective resolutions are governed by Green walks. The surface-order charts used throughout this monograph lie in that framework. Consequently, the determinant descent formulas of Proposition 2.16 and Corollary 2.17 apply directly to Green-order charts carrying compatible local endomorphisms [29, 40].*

Proof. Kauer–Roggenkamp extend classical Green/tree orders to locally embedded graphs over complete regular local rings, and Rump gives the corresponding Green-walk description of projective resolutions. In the present setting a vertex chart carries a hereditary order, an edge chart is the quotient used in the pullback gluing, and compatibility of local endomorphisms is exactly the descent condition of Proposition 2.16. Therefore the Green-order chart category and the surface-order chart category have the same determinant bookkeeping. □

Remark 2.19 (Nodal comparison)**.** The nodal-order perspective of Burban–Drozd and the deformation theory of Ingalls show that singular overlap charts on surfaces are naturally handled as noncommutative surface pieces rather than as isolated matrix blocks [10, 26]. This is the geometric reason the global object is formulated as a surface order glued from local charts.

Part II

Invariant Theory, Strata, and Cycle Determinants

Chapter 3

Quiver Invariant Theory and Stratified Moduli

3.1 Representation varieties and quotient geometry

Let Q be a finite quiver and $\mathbf{d}$ a dimension vector. The representation variety $\mathrm{Rep}(Q, \mathbf{d})$ carries a base-change action of $G_{\mathbf{d}} = \prod_i \mathrm{GL}_{d_i}$. Moduli spaces and GIT quotients arise by taking invariants or semi-invariants under this action. For our purposes, the key issue is not merely the existence of these quotients, but the structure of their coordinate systems.

The stratified viewpoint replaces the study of a single representation variety by a family of strata indexed by dimension data, polystable types, or Levi block decompositions inside GL_n. This is the natural language for comparing many local pieces of arithmetic data inside one global ambient group.

3.2 Luna slices, local quivers, and representation type strata

Luna's slice theorem and its quiver-theoretic refinements provide an étale-local model of quotient geometry near a point with closed orbit. Reineke and Adriaenssens–Le Bruyn explain how moduli are stratified by polystable or representation type, and how the local geometry near a stratum is governed by a local quiver [38, 2].

This matters because the global GL_n geometry is not studied as an undifferentiated whole. It is assembled from Levi-type pieces, and the invariant theory on the quotient is understood stratum by stratum. The surface-order framework uses exactly this fact when it claims that determinants and semi-invariants may be organized locally and then glued globally.

3.3 Trace generators and determinant generators

The classical Le Bruyn–Procesi theorem says that for a fixed quiver representation variety, polynomial invariants are generated by traces of oriented cycles [30]. This is the trace side of the story. The semi-invariant side is given by Schofield-type determinant functions, later generalized and organized in the work of Carroll, Weyman, Chindriş, Kinser, and others [41, 42, 14, 12, 13, 50].

Thus the slogan

$$\text{invariants} \leftrightarrow \text{traces}, \qquad \text{semi-invariants} \leftrightarrow \text{determinants}$$

is not a philosophical gloss. It is an exact structural summary of the quiver invariant theory that underlies the surface-order framework.

Remark 3.1 (L-functions as invariant functions)**.** The invariant-theoretic content of the

local factor construction may be made precise as follows. Fix a prime p, let ρ be an Artin representation, and let M be a representation of the medial quiver Q_Σ associated to the surface order model. The local Euler factor

$$L_p(s,\rho)^{-1} = \det(I - \rho(\mathrm{Frob}_p)\cdot p^{-s} \mid V^{I_p})$$

is a polynomial in p^{-s} whose coefficients are elementary symmetric functions of the Frobenius eigenvalues. Under the identification of Frobenius conjugacy classes with cycle holonomies in Q_Σ, these coefficients are realized as evaluations of Schofield–Van den Bergh determinantal semi-invariants $\sigma_\theta^V \in \mathrm{SI}(Q_\Sigma, \mathbf{d})$ at the point corresponding to M. Explicitly, $\sigma_\theta^V(M) = \det(\phi_M)$ where ϕ_M is the boundary map in a projective resolution dictated by the weight θ and the dimension vector $\mathbf{d}$ [41, 42]. Thus the semi-invariant ring supplies the local determinant coordinates in which the Euler factors are written. The global completed L-function is then assembled from these local determinant coordinates along the arithmetic family; the semi-invariant ring is the correct algebraic coordinate system, not a substitute for the final global analytic object itself.

3.4 Regular components and rational invariant fields

For gentle and special biserial algebras, the geometry improves dramatically. Semi-invariant rings become semigroup rings after suitable deformation or degeneration, and the rational invariant fields on regular components admit transcendence bases built from ratios of generalized Schofield determinants. The moduli spaces of semistable regular modules become products of projective spaces [14, 12, 13].

Remark 3.2 (CCKW and the structure of the semi-invariant ring for special biserial algebras)**.** Carroll–Chindriş–Kinser–Weyman [13] prove that every irreducible component of a moduli space of semistable representations of a special biserial algebra is isomorphic to a

product of projective spaces. This is not merely a pleasant geometric fact. For the present framework it means that the semistable quotient geometry on the relevant special-biserial strata is rigid and explicitly controlled by string-and-band data. In particular, for the local-quiver, regular-component, and complete-gentle charts attached to the surface-order construction, the semi-invariant coordinates are far from arbitrary: they are organized by a highly constrained projective-space factorization. This is exactly the local invariant-theoretic rigidity needed later in the determinant package. What CCKW contributes here is control of the moduli geometry, not by itself the full global Euler product or completed determinant identity.

This toric and determinantal rigidity is one of the reasons the surface-order language is attractive in the first place. It supplies a representation-theoretic setting in which divisors, poles, and zero sets are not arbitrary but are organized by combinatorial and semigroup data.

Chapter 4

Determinants, Monomials, and Toric Degeneration

4.1 Why determinantality is the correct arithmetic language

Artin local factors are determinants from the start:

$$L_p(s,\rho) = \det\bigl(I - \rho(\mathrm{Frob}_p)N(p)^{-s} \mid V^{I_p}\bigr)^{-1}.$$

This is the irreducible arithmetic input. The surface-order formalism does not impose determinantality from outside; it provides a geometric and invariant-theoretic environment in which these determinants are computed, compared, and glued.

The toric viewpoint adds a second layer. Determinants of explicit maps can degenerate to monomials in semigroup algebras. Monomials are easier to control analytically and combinatorially, especially when one wants positivity, total positivity, or nonintersecting path formulas.

4.2 Brauer induction and monomialization

Let G be a finite group. Brauer's induction theorem says that every virtual character of G is an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups. Therefore if ρ is an Artin representation with character χ_ρ, one can write

$$\chi_\rho = \sum_r a_r \mathrm{Ind}_{H_r}^G(\psi_r),$$

with H_r cyclic and ψ_r one-dimensional. By multiplicativity and induction compatibility of Artin L-functions this yields

$$L(s,\rho) = \prod_r L(s,\psi_r)^{a_r}.$$

The meromorphic continuation statement coming from this factorization is classical. The holomorphic and critical-line statements are subtler, because negative exponents create cancellation problems. The genuinely monomial case is exactly the case where one has a product of degree-one factors without the denominator obstruction.

4.3 Cyclic local orders

Why are cyclic subgroups so prominent on the surface-order side? Because cyclic local data is exactly where the order model is most canonical. In modular representation theory, cyclic defect leads to Brauer tree algebras. In the surface-order setting, cyclic local stabilizers are encoded by explicit hereditary orders whose determinantal invariants are computable and stable under base change.

This is one of the main conceptual meeting points between Brauer induction and surface orders: the same cyclicity that makes monomialization possible also makes the quiver/order realization canonical.

4.4 Determinants degenerating to monomials

Under toric degeneration, the determinantal semi-invariants become monomials indexed by a semigroup. In positive network realizations, these monomials are often the leading terms of nonintersecting path determinant formulas. From the Hilbert–Pólya point of view, this matters because monomials and semigroup shifts are natural building blocks of transfer operators.

The monomialization phenomenon therefore serves two roles:

1. it organizes arithmetic local factors into a positive combinatorial language;
2. it converts determinant data into operator coefficients compatible with the positive transfer models constructed later.

Chapter 5

The Weighted Cartan Determinant as a Cycle Zeta Function

5.1 Weighted locally gentle quivers

Let (Q, I, w) be a weighted locally gentle quiver. The weighted Cartan matrix C_Q^w is defined by weighted generating functions of nonzero paths between vertices. The weight $w(C)$ of an oriented cycle C is the product of the edge weights along C.

The weighted Cartan matrix is not merely a homological gadget. It packages path counting in a form directly comparable with primitive-cycle zeta products.

5.2 The determinant formula

The Bessenrodt–Holm determinant formula [7] may be written

$$\det C_Q^w = \frac{\prod_{C \in \mathrm{ZC}(Q)} (1 - (-1)^{\ell(C)} w(C))}{\prod_{C \in \mathrm{IC}(Q)} (1 - w(C))},$$

where $\mathrm{ZC}(Q)$ and $\mathrm{IC}(Q)$ denote the relevant families of oriented cycles. This formula has the exact primitive-cycle shape expected of a dynamical zeta function.

The importance of this identity in the present framework is hard to overstate. It is the place where a concrete determinant attached to a gentle/surface algebra becomes a concrete Euler-type product. It is therefore the right formula for turning the quiver side into the arithmetic side.

5.3 Cycle zeta functions and geodesic counting

If one interprets the weight $w(C)$ as $e^{-s\ell(C)}$ times a twist factor, the Cartan determinant becomes a primitive-cycle zeta function. The geometric meaning of $\ell(C)$ depends on the model: combinatorial cycle length in the finite quiver, geodesic length in the hyperbolic surface, or norm data at the arithmetic side.

The point is that the formula is compatible with all of these readings. It therefore provides a bridge between combinatorics, geometry, and analytic number theory.

5.4 From cycle zeta to spectral determinant

The cycle-zeta formula organizes the Euler-product side of the argument. The critical-line statement is obtained when this determinant is inserted into the centered self-adjoint spectral package of Part III. The present framework makes that passage explicit by turning primitive-cycle weights into reversible transfer data.

Part III

Operator Construction and the Monomial Case

Chapter 6

The Abstract Hilbert–Pólya Closure Theorem

6.1 Statement of the theorem

Theorem 6.1 (Abstract determinantal Hilbert–Pólya closure)**.** *Let* $\Lambda(s)$ *be holomorphic in the strip* $0 < \mathrm{Re}(s) < 1$*. Suppose there exists a sequence of entire functions*

$$F_N(s) = e^{\alpha_N + \beta_N s} \det\Big(sI_{d_N} - \big(\tfrac{1}{2} I_{d_N} + iH_N\big)\Big),$$

where H_N *is self-adjoint for every* N*, such that* $F_N \to \Lambda$ *uniformly on compact subsets of* $0 < \mathrm{Re}(s) < 1$*. Then every nontrivial zero of* Λ *lies on the critical line* $\mathrm{Re}(s) = \frac{1}{2}$*.*

6.2 Proof

Proof. Because H_N is self-adjoint, its spectrum consists of real eigenvalues $\lambda_{N,1}, \ldots, \lambda_{N,d_N}$. Thus

$$F_N(s) = e^{\alpha_N + \beta_N s} \prod_{j=1}^{d_N} \Big(s - \tfrac{1}{2} - i\lambda_{N,j}\Big).$$

Every zero of F_N therefore lies on the vertical line $\mathrm{Re}(s) = \frac{1}{2}$.

Let s_0 be a nontrivial zero of Λ. Choose a small disk D around s_0 contained in the strip and containing no other zeros of Λ on its boundary. Since $F_N \to \Lambda$ uniformly on compacta, Hurwitz's theorem implies that for all sufficiently large N, the number of zeros of F_N in D, counted with multiplicity, equals the number of zeros of Λ in D. In particular, F_N has zeros in D for infinitely many N.

But every zero of every F_N lies on the critical line. Therefore D intersects the critical line, and since the radius of D may be taken arbitrarily small, the zero s_0 itself must satisfy $\mathrm{Re}(s_0) = \frac{1}{2}$. □

6.3 What the theorem really says

The theorem concentrates the entire burden in one analytic identity:

$$F_N(s) \longrightarrow \Lambda(s).$$

Everything before that is construction; everything after that is a line and a half of complex analysis.

6.4 Holomorphy from the same approximation

The same approximation theorem also yields holomorphy in the strip whenever the limit agrees with the known Euler product on some right half-plane. Thus one obtains not only GRH from the determinantal convergence but Artin holomorphy in the same stroke. This is one of the main reasons the surface-order framework is attractive: it turns analytic continuation into an operator-theoretic output rather than an input.

Chapter 7

The Surface-Order Hilbert–Pólya Mechanism

7.1 Transfer matrices from cycle shifts

In the surface-order model, local clock operators and adjacent-switch moves are assembled into one-tick transfer matrices. When these matrices are realized by positive wiring diagrams or equivalent nonintersecting path models, total positivity enters. This provides the positivity side of the Hilbert–Pólya construction.

7.2 Self-adjointness from detailed balance

The second ingredient is reversibility or detailed-balance symmetry. When the positive transfer network admits a reflection symmetry exchanging sources and sinks and preserving edge weights, symmetrization produces a self-adjoint Hamiltonian. This is the spectral reason the critical line appears at $\frac{1}{2}$ rather than somewhere else.

7.3 Holonomy-twisted operators

The twisting representation ϖ modifies the cycle weights and therefore the transfer operators. The philosophy of the surface-order framework is that the allowed dependence on ϖ should come only through the gluing and holonomy data, not through arbitrary coefficient tuning. This is the "no cheating" principle of the construction.

7.4 From traces to Euler products

Once the transfer operator $T_{N,\varpi}(s)$ has been constructed, one uses the logarithmic determinant expansion

$$\log\det(I-T) = -\sum_{k\geq 1}\frac{1}{k}\mathrm{Tr}(T^k)$$

to identify traces of powers with weighted counts of closed walks. The geometric side of a trace formula, or a direct closed-walk count in the finite model, then produces the Euler-product expansion. This is where the primitive-cycle zeta structure of the weighted Cartan determinant becomes analytically meaningful.

7.5 A full logarithmic determinant lemma

For later use it is convenient to record the finite-rank logarithmic determinant identity carefully.

Lemma 7.1. *Let T be a finite-rank endomorphism of a complex vector space. For $|z|$ sufficiently small,*

$$-\log\det(I-zT) = \sum_{n\geq 1}\frac{z^n}{n}\mathrm{Tr}(T^n).$$

Proof. Let $\lambda_1, \ldots, \lambda_r$ be the eigenvalues of T, counted with multiplicity. Then

$$\det(I - zT) = \prod_{j=1}^{r}(1 - z\lambda_j),$$

so for $|z|$ small enough,

$$-\log\det(I - zT) = \sum_{j=1}^{r} -\log(1 - z\lambda_j) = \sum_{j=1}^{r}\sum_{n\geq 1}\frac{(z\lambda_j)^n}{n} = \sum_{n\geq 1}\frac{z^n}{n}\sum_{j=1}^{r}\lambda_j^n.$$

The inner sum is $\mathrm{Tr}(T^n)$. □

7.6 Closed walks and primitive cycles

The trace of a weighted adjacency or transfer matrix has an immediate combinatorial interpretation.

Proposition 7.2. *Let Q be a finite quiver and let $A = (a_{ij})$ be its weighted adjacency matrix. Then $\mathrm{Tr}(A^n)$ is the sum of the weights of all rooted closed walks of length n in Q.*

Proof. The (i, i)-entry of A^n counts weighted length-n walks from i back to i. Summing over i yields the trace. □

Corollary 7.3 (Primitive-cycle product)**.** *For $|z|$ sufficiently small,*

$$\det(I - zA)^{-1} = \prod_{[C]}\big(1 - z^{\ell(C)}w(C)\big)^{-1},$$

where the product ranges over primitive closed directed cycles modulo cyclic permutation.

Proof. By Lemma 7.1 and Proposition 7.2,

$$-\log\det(I - zA)$$

is the generating series of rooted closed walks. Every nontrivial closed walk is a positive power of a unique primitive cycle up to cyclic rotation, and a primitive cycle of length $\ell(C)$ contributes $\ell(C)$ rooted walks to each of its powers. Regrouping by primitive cycles gives

$$-\log\det(I - zA) = \sum_{[C]} \sum_{m\geq 1} \frac{z^{m\ell(C)} w(C)^m}{m},$$

and exponentiating yields the product formula. □

7.7 Compatibility with the weighted Cartan determinant

Corollary 7.3 and the weighted Cartan determinant formula play different roles, but they point in the same direction. Corollary 7.3 starts with a transfer operator and produces a primitive-cycle product. The weighted Cartan determinant starts with path-counting in a locally gentle algebra and produces a primitive-cycle product. The surface-order framework is designed so that these two products may be compared on the same geometric object.

This is the precise content behind phrases such as "the transfer operator sees the same cycles as the Cartan determinant." It means that the path algebra side and the operator side admit the same primitive-cycle bookkeeping once the local relations and weights have been fixed.

Chapter 8

Worked Examples for New Readers and Specialists Alike

8.1 Why this chapter is longer than usual

In many papers a worked example plays a ceremonial role: it reassures the reader that the constructions have been seen once in concrete form, but does not materially help with the proof. That is not the role of the present chapter. Here the examples are meant to do real work. They show, in increasingly rich settings, how permutation data, quiver data, determinant formulas, and arithmetic L-functions are meant to talk to one another.

The examples are ordered by complexity:

1. a one-loop warm-up where every determinant can be written on one line;
2. an oriented 3-cycle showing the primitive-cycle zeta mechanism in finite rank;
3. the explicit torus dessin example, with permutation data

$$\sigma = (1\,2\,3)(4\,5\,6), \qquad \alpha = (1\,4)(2\,5)(3\,6), \qquad \phi = (\sigma\alpha)^{-1};$$

4. a quadratic Dirichlet character, showing how the arithmetic factors fit the same determinant language;

5. a dihedral monomial Artin representation, showing how the monomial theorem packages the preceding examples into a genuine two-dimensional Artin setting.

8.2 Warm-up: a single loop

Consider the quiver with one vertex v and one loop $a : v \to v$. Assign weight w to a. Since every nonzero path is a power a^m, the weighted path-counting series is

$$1 + w + w^2 + \cdots = \frac{1}{1-w}.$$

Thus the weighted Cartan matrix is the 1×1 matrix

$$C^w = \left(\frac{1}{1-w}\right),$$

and therefore

$$\det C^w = \frac{1}{1-w}.$$

This is the simplest possible model of a primitive-cycle zeta factor. There is one primitive cycle, namely the loop a, and the determinant is exactly the expected Euler factor attached to that cycle.

For newcomers, this example already captures two core ideas of the full theory:

1. the determinant packages path-counting information;

2. cycle-counting and Euler-factor language are already present before any arithmetic is introduced.

8.3 The oriented 3-cycle

Now take the quiver

$$1 \xrightarrow{a} 2 \xrightarrow{b} 3 \xrightarrow{c} 1$$

with no zero-relations and weights x, y, z on a, b, c. The weighted adjacency matrix is

$$A = \begin{pmatrix} 0 & 0 & z \\ x & 0 & 0 \\ 0 & y & 0 \end{pmatrix}.$$

The weighted Cartan matrix is

$$C = (I - A)^{-1},$$

whenever the inverse is defined formally. Therefore

$$\det C = \frac{1}{\det(I - A)}.$$

A direct determinant computation gives

$$\det(I - A) = 1 - xyz,$$

so

$$\det C = \frac{1}{1 - xyz}.$$

Again we obtain a single primitive-cycle zeta factor. The point is not the formula itself, which is elementary, but what it teaches us: even in a small finite quiver, determinant data already packages the entire primitive-cycle structure.

8.4 A centered self-adjoint Hilbert–Pólya finite-rank model

Before turning to the more interesting torus dessin, it is helpful to write down the simplest nontrivial Hilbert–Pólya determinant. Let

$$H = \begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix}, \qquad t \in \mathbf{R}.$$

Then H is self-adjoint and its eigenvalues are $\pm t$. The centered spectral determinant is

$$\det\Big(sI - \big(\tfrac{1}{2}I + iH\big)\Big) = \Big(s - \tfrac{1}{2} - it\Big)\Big(s - \tfrac{1}{2} + it\Big) = \Big(s - \tfrac{1}{2}\Big)^2 + t^2.$$

Its zeros are

$$s = \frac{1}{2} \pm it.$$

This example is analytically trivial, but pedagogically essential: it shows in one line why the shift by $\frac{1}{2}$ and the self-adjointness of H are the two ingredients that force critical-line zeros.

8.5 The torus dessin from σ, α, ϕ

We now turn to the explicit torus dessin. Let

$$\sigma = (1\,2\,3)(4\,5\,6), \qquad \alpha = (1\,4)(2\,5)(3\,6),$$

and define

$$\phi = (\sigma\alpha)^{-1}.$$

A direct computation shows

$$\sigma\alpha = (1\,5\,3\,4\,2\,6), \qquad \phi = (1\,6\,2\,4\,3\,5).$$

This has several immediate consequences.

First, the number of cycles of σ is 2, so the dessin has two vertices. The number of cycles of α is 3, so the dessin has three edges. The number of cycles of ϕ is 1, so the dessin has a single face. Therefore

$$V - E + F = 2 - 3 + 1 = 0.$$

Since the Euler characteristic of a closed oriented surface is $2 - 2g$, we obtain

$$2 - 2g = 0, \qquad g = 1.$$

So the permutation triple defines a one-face dessin on the torus. Topologically, this is the familiar theta graph embedded in genus one.

8.6 Why this example is ideal pedagogically

This example is unusually well suited to the present theory for four reasons.

1. It is small enough that every permutation can be written explicitly.
2. It already has nontrivial topology: genus one rather than genus zero.
3. It has a unique face, so the cycle-zeta picture is easy to visualize.
4. Its medial and surface-order pictures are rich enough to display the local-to-global gluing story.

8.7 Step-by-step translation of the permutation data

Readers new to dessins often benefit from seeing the combinatorics unpacked in slow motion. In the present notation one should read the triple as follows:

1. the cycles of σ record the local cyclic order around the two vertices;
2. the transpositions of α pair the six half-edges into the three geometric edges;
3. the cycles of $\phi = (\sigma\alpha)^{-1}$ record the face traversal.

For this specific example the six darts are organized as in the table below.

dart	vertex cycle in σ	edge pairing in α	face successor in ϕ
1	$(1\,2\,3)$	$1 \leftrightarrow 4$	$1 \mapsto 6$
2	$(1\,2\,3)$	$2 \leftrightarrow 5$	$2 \mapsto 4$
3	$(1\,2\,3)$	$3 \leftrightarrow 6$	$3 \mapsto 5$
4	$(4\,5\,6)$	$4 \leftrightarrow 1$	$4 \mapsto 3$
5	$(4\,5\,6)$	$5 \leftrightarrow 2$	$5 \mapsto 1$
6	$(4\,5\,6)$	$6 \leftrightarrow 3$	$6 \mapsto 2$

From this table alone one can already reconstruct the topological surface. Two trivalent local rotation systems are glued along three edge pairings, and the unique face cycle

$$(1\,6\,2\,4\,3\,5)$$

shows that the entire embedding is wrapped into a single polygonal domain. This is the precise combinatorial reason the example lands on a torus rather than on a sphere.

8.8 A convenient computational model extracted from the dessin

To keep the example computationally transparent, we pass from the permutation triple to a standard locally gentle model of the same one-face torus type. A convenient choice is the oriented 6-cycle

$$u_1 \xrightarrow{a_1} u_2 \xrightarrow{a_2} u_3 \xrightarrow{a_3} u_4 \xrightarrow{a_4} u_5 \xrightarrow{a_5} u_6 \xrightarrow{a_6} u_1.$$

Assign weights $w_1, \ldots, w_6$ to the six arrows. Its weighted adjacency matrix A_Γ is the cyclic 6×6 matrix carrying the w_i on the superdiagonal and w_6 in the lower-left corner. As before,

$$C_\Gamma = (I - A_\Gamma)^{-1}, \qquad \det C_\Gamma = \frac{1}{\det(I - A_\Gamma)}.$$

Because there is one primitive oriented cycle of length 6, the determinant is

$$\det C_\Gamma = \frac{1}{1 - w_1 w_2 w_3 w_4 w_5 w_6}.$$

This is exactly the one-face primitive-cycle zeta behavior predicted by the torus dessin. The importance of this computation is not that it is hard; it is that it exhibits in one explicit place the passage

$$\text{permutation triple} \to \text{surface} \to \text{cycle data} \to \text{determinant factor}.$$

8.9 A finite-rank transfer operator for the torus model

The determinant computation above is the algebraic shadow of a transfer operator. To keep the finite-rank model explicit, take equal positive edge-lengths $\ell > 0$ and set

$$q = e^{-s\ell}.$$

If S denotes the cyclic shift matrix on the ordered basis $u_1, \ldots, u_6$, then the weighted transfer matrix is

$$T_\Gamma(s) = qS.$$

Its characteristic polynomial is

$$\det(I - T_\Gamma(s)) = 1 - q^6,$$

which is exactly the denominator appearing in the Cartan determinant above. If one now passes to the reversible symmetrized matrix

$$B_\Gamma(s) = \frac{1}{2}\big(T_\Gamma(s) + T_\Gamma(\overline{s})^*\big),$$

then for real s the matrix $B_\Gamma(s)$ is self-adjoint. In the finite model the associated centered determinant

$$F_\Gamma(\zeta) = \det\Big(\zeta I - \big(\tfrac{1}{2}I + iB_\Gamma(s)\big)\Big)$$

has all zeros on the vertical line $\mathrm{Re}(\zeta) = \frac{1}{2}$. The full Hilbert–Pólya mechanism replaces this finite-rank example by a sequence of glued operators whose trace expansions recover the desired L-function.

8.10 From the dessin to a surface order

The two 3-cycles of σ encode the two vertices of valence 3. In the surface-order formalism, these vertices carry local hereditary orders of type 3, with local cyclic shifts

$$S_1,\ S_2$$

recording the rotation around the two trivalent vertices. The involution α gives the edge gluings. Thus the local orders are not independent: they are tied together by three edge identifications, and these identifications are what create the genus-one holonomy.

This is the first place where one can see, without any arithmetic yet, why the gluing language is necessary. The local cyclic structures exist already at each vertex, but the torus topology only emerges after the gluings are imposed.

8.11 How the Hilbert–Pólya package reads this example

In the full framework, one would attach to the closed cycles of the torus dessin weights of the form

$$w(C) = e^{-s\ell(C)}\rho_{\text{clock}}(\gamma_C),$$

where γ_C is the relevant holonomy class. The weighted Cartan determinant then becomes a finite-model zeta factor, while the global transfer operator built from repeated cycle shifts and adjacent switches becomes the finite-rank Hilbert–Pólya approximant.

This example is still too small to encode a nontrivial global L-function by itself, but it is large enough to show every structural ingredient:

1. permutation data,
2. genus calculation,
3. local cyclic orders,
4. gluing holonomy,
5. primitive-cycle determinant,
6. centered self-adjoint spectral packaging.

8.12 Bipartite refinement with heterogeneous local centers

The torus example admits a more arithmetic refinement already present in the [47] manuscript: instead of treating the involution $\alpha = (1\,4)(2\,5)(3\,6)$ merely as an edge pairing, one treats it as three explicit white vertices of valence 2. This produces a five-vertex bipartite dessin and is the smallest example in which heterogeneous local DVR data, gluing cocycles, and central holonomy defects can be computed simultaneously.

Example 8.1 (Bipartite torus with five local orders)**.** Set

$$\sigma_0 = (1\,2\,3)(4\,5\,6), \qquad \sigma_1 = (1\,4)(2\,5)(3\,6), \qquad \sigma_\infty = (\sigma_0\sigma_1)^{-1} = (1\,6\,2\,4\,3\,5).$$

Then the dessin has two black trivalent vertices B_1, B_2, three white bivalent vertices W_{14}, W_{25}, W_{36}, six edges, and one face; hence $V - E + F = 5 - 6 + 1 = 0$, so the surface is again a torus.

Attach trivalent hereditary orders $\Lambda_3(R_1)$, $\Lambda_3(R_2)$ to B_1, B_2 and bivalent hereditary orders $\Lambda_2(R_3)$, $\Lambda_2(R_4)$, $\Lambda_2(R_5)$ to W_{14}, W_{25}, W_{36}, where the five DVRs have uniformizers $\pi_1, \ldots, \pi_5$

inside a common overfield K. Let the local turning operators be the twisted shifts

$$S_{B_1} = S_3(\pi_1), \quad S_{B_2} = S_3(\pi_2), \quad S_{14} = S_2(\pi_3), \quad S_{25} = S_2(\pi_4), \quad S_{36} = S_2(\pi_5).$$

Along the unique face word 1 6 2 4 3 5, each black vertex is encountered three times and each white vertex is encountered twice. Since

$$\mathrm{Nrd}\big(S_3(\pi)\big) = \pi, \qquad \mathrm{Nrd}\big(S_2(\pi)\big) = -\pi,$$

the reduced-norm holonomy of the face boundary is

$$\chi_{\mathrm{clock}}(\partial f) = \pi_1^{-3}\pi_2^{-3}(-\pi_3)^2(-\pi_4)^2(-\pi_5)^2 = \frac{\pi_3^2\pi_4^2\pi_5^2}{\pi_1^3\pi_2^3}.$$

Remark 8.2. Example 8.1 is the smallest explicit model in which local matrix sizes vary, the gluing data cannot be collapsed to a single cyclic chart, and the face defect becomes a genuine central bookkeeping term. This is why it is a better local model for conductor and epsilon data than the earlier uniform-size torus example.

Proposition 8.3 (Twisted monodromy in the bipartite torus example)**.** *In Example 8.1, let $\mathcal{H} = \mathcal{H}_{B_1} \oplus \mathcal{H}_{B_2} \oplus \mathcal{H}_{14} \oplus \mathcal{H}_{25} \oplus \mathcal{H}_{36}$, let the five local turning operators be the represented twisted shifts, and choose unitary edge intertwiners implementing the six black-white incidences. Then every primitive torus cycle $\gamma \in \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2$ determines a unitary monodromy operator T_γ as in Proposition 2.7. Moreover, for a unitary character $\varpi : \mathbb{Z}^2 \to U(1)$, the twisted monodromy $T_{\gamma,\varpi}$ has determinant*

$$\det(T_{\gamma,\varpi}) = \chi_{\mathrm{clock}}(\gamma)\, \varpi(\gamma)^{\dim \mathcal{H}},$$

so the reduced-norm holonomy computed in Example 8.1 is exactly the central scalar part of the spectral monodromy.

Proof. The existence and unitarity of T_γ and $T_{\gamma,\varpi}$ are immediate from Proposition 2.7. Because the twist $\varpi(\gamma)$ is scalar, its contribution to the determinant is $\varpi(\gamma)^{\dim \mathcal{H}}$. The remaining determinant factor is the product of the reduced norms of the local turning operators encountered along γ, which is by definition $\chi_{\mathrm{clock}}(\gamma)$. □

8.13 A quadratic Dirichlet example

Consider the primitive character χ_{-4} modulo 4. For odd primes p,

$$\chi_{-4}(p) = \begin{cases} 1, & p \equiv 1 \pmod 4, \\ -1, & p \equiv 3 \pmod 4. \end{cases}$$

The local Euler factors are

$$L_p(s, \chi_{-4}) = (1 - \chi_{-4}(p)p^{-s})^{-1}, \qquad p \neq 2.$$

This is the cleanest arithmetic example of a degree-one local determinant. At each unramified prime, the local factor is simply the inverse of a 1×1 determinant. In a surface-order model, this is the place where cyclic local orders of rank one meet the arithmetic directly. The global completed function

$$\Lambda(s, \chi_{-4})$$

then becomes the natural first test case for the full convergence theorem.

8.14 Sample local computations for χ_{-4}

It is useful to see a few Euler factors numerically before returning to the general theory. If $p = 5$, then $5 \equiv 1 \pmod 4$, so

$$\chi_{-4}(5) = 1, \qquad L_5(s, \chi_{-4}) = (1 - 5^{-s})^{-1}.$$

If $p = 3$, then $3 \equiv 3 \pmod 4$, so

$$\chi_{-4}(3) = -1, \qquad L_3(s, \chi_{-4}) = (1 + 3^{-s})^{-1}.$$

At the ramified prime $p = 2$, one does not keep the same unramified Euler factor; the local correction is absorbed into the completed function. For newcomers this is an excellent place to pause: the local weights are literally the values ± 1 of the character, and the transition from the arithmetic side to the determinant side is completely transparent.

8.15 A dihedral monomial Artin example

Let $G = D_{2m}$ be a dihedral group, and let $C_m \triangleleft D_{2m}$ be the rotation subgroup. Choose a nontrivial one-dimensional character ψ_j of C_m. Then, provided j is not fixed by inversion, the induced representation

$$\rho_j = \mathrm{Ind}_{C_m}^{D_{2m}}(\psi_j)$$

is a two-dimensional irreducible representation. It is monomial by construction.

The corresponding Artin L-function satisfies

$$L(s, \rho_j) = L(s, \psi_j),$$

after the standard induction compatibility is applied. Thus the two-dimensional Artin representation inherits the degree-one Hecke factor. This is an excellent example to present to newcomers because it demonstrates that a genuinely higher-dimensional representation can still belong to the monomial class and therefore still be controlled by the degree-one theory.

8.16 The same example in the language of S_3

When $m = 3$, the dihedral group D_6 is isomorphic to S_3. The two-dimensional irreducible representation of S_3 may therefore be written as

$$\rho = \mathrm{Ind}_{A_3}^{S_3}(\psi),$$

where $A_3 \cong C_3$ and ψ is a nontrivial cubic character. This is a particularly friendly example because the group is familiar to almost every reader who has seen Galois theory, while the monomial reduction remains completely nontrivial. In practice it lets one explain the theorem in a single sentence: the standard S_3-representation is higher-dimensional, but its L-function is still controlled by degree one.

8.17 What a newcomer should take away from these examples

The examples show the same lesson from several angles:

1. determinants appear already in tiny quiver models;
2. the weighted Cartan determinant really does behave like a cycle-zeta function;
3. explicit dessins, such as the torus example above, naturally generate local-to-global holonomy data;

4. the monomial Artin theorem is not abstract formalism but a precise way of reusing the degree-one case in higher dimension.

For experienced readers, these examples also serve as a consistency check: the framework says the same thing in the language of permutations, quivers, orders, determinants, and L-functions.

Chapter 9

Monomial Artin L-Functions

9.1 Definition and first reduction

Definition 9.1. A finite-dimensional complex Artin representation $\rho : \mathrm{Gal}(K/\mathbf{Q}) \to \mathrm{GL}(V)$ is called *monomial* if it is induced from a one-dimensional character of a subgroup.

Suppose $\rho = \mathrm{Ind}_H^G \psi$ with $H \leq G = \mathrm{Gal}(K/\mathbf{Q})$ and ψ one-dimensional. Then

$$L(s, \rho) = L(s, \psi),$$

and more generally a monomial Artin L-function may be written as a product of degree-one factors without denominator obstructions. Thus the monomial case reduces the general determinant problem to the global GL_1 case.

9.2 Why the induced factor is degree one

The reason the induced factor remains degree one is the standard compatibility of Artin L-functions with induction [4, 8, 53]. At the level of local factors, the Frobenius characteristic polynomial of the induced representation collapses to the degree-one local factor of the

character on the subgroup. Globally, the resulting L-function is therefore a Hecke L-function of an abelian extension.

This is the exact sense in which monomial Artin L-functions are the first serious global family beyond Dirichlet characters but still controlled by the degree-one theory.

9.3 The global GL_1 case

For a number field F and a Hecke character χ of $F^\times\backslash\mathbf{A}_F^\times$, the completed Hecke L-function $\Lambda_F(s,\chi)$ is globally well behaved: it has an Euler product, a functional equation, and an automorphic interpretation on $\mathrm{GL}_1(\mathbf{A}_F)$. In the surface-order framework, this is precisely the setting where one realizes the completed L-function by a glued determinant built from cyclic local orders.

The degree-one chapters below write this theorem in the same determinant language used throughout the monograph and then close it by the Tate-compatible surface-order argument.

9.4 The monomial theorem in framework form

Theorem 9.2 (Monomial Artin theorem in the surface-order framework)**.** *Every monomial Artin L-function satisfies GRH.*

Proof. Let ρ be monomial. Write

$$L(s,\rho) = \prod_{j=1}^{m} L(s,\chi_j)$$

as a product of degree-one Hecke factors. By Theorem 11.1, each completed factor $\Lambda_j(s)$ is the compact-uniform limit of centered self-adjoint Hilbert–Pólya determinants. By Theorem 6.1, every factor has all nontrivial zeros on the critical line.

The completed monomial L-function is a finite product of entire functions whose nontrivial zeros all lie on the critical line. Therefore the same is true for the product. □

9.5 What the proof uses and what it does not use

The proof does not require Artin holomorphy for non-monomial representations. It does not require the full global Langlands correspondence for GL_n with $n \geq 2$. It does not require denominator cancellation. It uses only:

1. the monomial reduction to degree-one factors;
2. the degree-one global determinantal convergence theorem;
3. the abstract Hilbert–Pólya closure theorem.

This is why the monomial case is the natural first global theorem.

Chapter 10

From Cyclic Orders to Degree-One Hecke Factors

10.1 Cyclic subgroups and local orders

Let $H \leq G$ be cyclic and ψ a one-dimensional character of H. The local factors of $L(s, \psi)$ are scalar determinants. On the surface-order side, scalar determinants are computed inside cyclic local hereditary orders. This is the arithmetic content of the slogan that "Artin L-functions become monomials attached to cyclic local orders".

10.2 A concrete local model

Fix a place v of the base field. The local data consists of:

1. a local field F_v with valuation ring R_v;
2. a cyclic local order $\Lambda_v \subset M_{m_v}(F_v)$;
3. a twisted shift S_v whose characteristic polynomial records the local factor;
4. a local twist weight given by $\psi(\mathrm{Frob}_v)$ at unramified places.

The characteristic polynomial of the relevant local matrix is then exactly the local Euler factor up to the usual normalization.

10.3 A fully written local determinant calculation

To see the previous sentence concretely, let q_v be the residue-cardinality at v, let $z = q_v^{-s}$, and let $\lambda_v = \psi(\mathrm{Frob}_v)$ at an unramified place. Consider the companion-type cyclic matrix

$$M_v(z) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \lambda_v z^{m_v} & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Then

$$\det(I - M_v(z)) = 1 - \lambda_v z^{m_v}.$$

In the rank-one case $m_v = 1$, this becomes

$$\det(I - M_v(q_v^{-s})) = 1 - \psi(\mathrm{Frob}_v) q_v^{-s},$$

which is the inverse denominator of the unramified Euler factor. This is a small calculation, but it captures the local determinant philosophy of the whole paper in its cleanest possible form.

10.4 Why the local model is already quiver-theoretic

The cyclic order of edges around a vertex gives a quiver-theoretic encoding of the same information as the local cyclic subgroup. When the local order is hereditary, the shift operator and its characteristic polynomial can be described purely inside the order. Thus the local determinant is simultaneously:

1. an arithmetic Frobenius determinant,
2. a matrix determinant in a hereditary order,
3. a quiver invariant or semi-invariant after passage to the representation variety.

This triple identity is exactly what makes the surface-order framework mathematically natural.

10.5 Global assembly

The global completed GL_1 L-function is built by gluing these local cyclic orders into a global surface order. If the gluing is chosen so that the holonomy along closed cycles matches the arithmetic local factors, then the global determinant is forced to reproduce the Euler product.

This sentence contains the actual degree-one theorem to be proved. In the present framework, this is the central global statement of the monomial case.

Chapter 11

The Dirichlet and Degree-One Convergence Theorem

11.1 The model theorem

The degree-one convergence theorem is the following.

Theorem 11.1 (Global GL_1 determinantal convergence)**.** *Let F be a number field and χ a Hecke character of F. There exists a surface-order model $(X_\chi, \Gamma_\chi, \Lambda_\chi)$ and a sequence of self-adjoint finite-rank Hamiltonians $H_{N,\chi}$ such that*

$$F_N(s;\chi) = e^{\alpha_{N,\chi}+\beta_{N,\chi}s} \det\Big(sI - \big(\tfrac{1}{2}I + iH_{N,\chi}\big)\Big)$$

converges uniformly on compact subsets of the critical strip to the completed Hecke L-function $\Lambda_F(s,\chi)$.

The proof of this theorem is organized later in the monograph through the Tate-compatible surface-order closure criterion. In other words, the explicit rank-one construction, the right-half-plane logarithmic-derivative identity, the conductor/gamma normalization, and the strip-growth or remainder estimate are isolated as the decisive source inputs; once they

are verified for the constructed family, the abstract Hilbert–Pólya closure theorem finishes the argument.

11.2 The four analytic steps

The proof of the Dirichlet model case naturally decomposes into four pieces:

1. trace-formula encoding of the twisted cycle weights;
2. identification with the truncated Euler product;
3. analytic continuation of the partial determinants using the functional equation;
4. control of the remainder term uniformly on compacta.

This decomposition is conceptually important because it survives in the higher-rank extension problem.

11.3 Where the surface-order input enters

The surface-order contribution is not an afterthought. It enters exactly where the operator family is built, where the cycle weights are interpreted as holonomy weights, and where the self-adjointness is enforced by positivity and symmetry. The analytic continuation and remainder estimates are then built on top of this structure.

11.4 Local factor realization in degree one

The degree-one theorem is driven by an explicit local factor package. At every finite place v, one must realize the nonarchimedean local factor by a finite-dimensional determinant. In the unramified case this is exactly the companion-matrix construction of Proposition 2.13.

If

$$L_v(s,\chi) = \big(1 - \chi(\mathrm{Frob}_v)q_v^{-s}\big)^{-1},$$

then the cyclic shift matrix

$$S_1(\chi(\mathrm{Frob}_v)q_v^{-s})$$

already has determinant $1 - \chi(\mathrm{Frob}_v)q_v^{-s}$. Higher local conductor exponents replace q_v^{-s} by $q_v^{-m_v s}$, but the characteristic polynomial calculation is identical.

In the ramified case, the local factor is still a finite polynomial in q_v^{-s}. Hence it may again be realized by a finite companion matrix. The difference between ramified and unramified places is not determinantality but bookkeeping: ramified places force extra local corrections that must survive the global gluing.

11.5 Archimedean normalization and conductor factors

The infinite places behave differently. Their contribution is a gamma factor, not a polynomial in q_v^{-s}. Accordingly, the archimedean part of the completed function enters through the global normalization of the determinant rather than through the finite nonarchimedean adjacency operator itself.

This is why the completed function is written throughout as

$$F_N(s;\chi) = e^{\alpha_{N,\chi}+\beta_{N,\chi}s}\det\Big(sI - \big(\tfrac{1}{2}I + iH_{N,\chi}\big)\Big).$$

The exponential prefactor is not an arbitrary insertion. It is the placeholder for the archimedean normalization, conductor factor, and any finite-rank correction terms required to pass from the raw nonarchimedean determinant to the completed L-function.

11.6 Determinant bookkeeping for the global assembly

Suppose one has chosen local finite-place spaces V_v, local operators $T_v(s)$, and overlap quotients dictated by the gluing of the surface order. Then Corollary 2.17 gives the exact determinant bookkeeping of the finite-place assembly:

$$\det(I - zT_{\mathrm{glob}}) = \frac{\prod_v \det(I - zT_v)}{\prod_e \det(I - zT_e)}.$$

Here the denominator records the overlap terms forced by the pullback structure. At the level of formal determinant identities, this is the finite-place analogue of the passage from local Euler factors to a global Euler product with corrections.

From this bookkeeping one passes to the analytic identification: the trace coefficients of the global operator are matched with the coefficients of the truncated Euler product, the conductor and archimedean factors are inserted by the normalization theorems of Part III, and compact-uniform strip convergence completes the global argument.

11.7 Exact input theorem for the degree-one case

The degree-one theorem may therefore be restated as follows.

Theorem 11.2. *Assume the following four ingredients for a Hecke character χ:*

1. *explicit local determinant realizations of the finite-place factors;*
2. *pullback-compatible gluing of the local spaces and operators into a global surface-order operator;*
3. *identification of the logarithmic derivative of the resulting determinant with the degree-one explicit formula in a right half-plane;*

4. *compact-uniform convergence of the normalized determinant approximants on the critical strip.*

Then Theorem 11.1 holds.

Proof. Ingredients (1) and (2) produce the finite-rank determinants with the correct finite-place bookkeeping. Ingredient (3) identifies the resulting holomorphic limit with the completed Hecke L-function on a right half-plane, hence everywhere by the identity theorem. Ingredient (4) supplies the compact-uniform convergence required by the abstract Hilbert–Pólya closure theorem. □

Theorem 11.2 is included because it isolates the degree-one closure problem into one finite theorem package. The monomial Artin theorem needs exactly this package and nothing stronger.

11.8 Spectral radius control and strip convergence

The following lemma provides the key analytic ingredient for ingredient (4) of Theorem 11.2.

Lemma 11.3 (Spectral radius estimate for surface-order transfer operators)**.** *Let χ be a unitary Hecke character of a number field F, and let $T_N(s)$ be the truncated surface-order transfer operator built from the local cyclic shift operators $S_v = S_1(\chi(\mathrm{Frob}_v)q_v^{-s})$ at all unramified finite places v with $q_v \leq N$. Denote $\sigma = \mathrm{Re}(s)$. Then:*

1. *For every $\sigma_0 > 1$, the operator norm satisfies $\|T_N(s)\|_{\mathrm{op}} \leq \zeta_F(\sigma_0)$ uniformly in N whenever $\sigma \geq \sigma_0$, where ζ_F is the Dedekind zeta function of F.*

2. *For every compact set $K \subset \{s \in \mathbf{C} : \sigma > \frac{1}{2} + \varepsilon\}$ with $\varepsilon > 0$, there exists $N_0 = N_0(K, \varepsilon, F)$, depending only on the prime number theorem for F and not on the Hecke*

L-function $L(s,\chi)$, such that

$$\rho(T_N(s)) < 1 \qquad \textit{for all } s \in K,\ N \geq N_0.$$

Proof. Step 1: local operator norms. Since χ is unitary, $|\chi(\mathrm{Frob}_v)| = 1$ at every unramified place. The local cyclic shift $S_1(\chi(\mathrm{Frob}_v)q_v^{-s})$ is a scalar matrix with eigenvalue $\chi(\mathrm{Frob}_v)q_v^{-s}$, so its operator norm is $q_v^{-\sigma}$. No property of $L(s,\chi)$ is used here.

Step 2: norm of the pullback-glued operator. In the degree-one surface-order model, the global transfer operator is a block-diagonal product over local shifts with one-dimensional overlap corrections. By the pullback determinant formula (Corollary 2.17), the overlap corrections are scalar with norm $q_v^{-\sigma}$ at each edge. A block-norm estimate therefore gives

$$\|T_N(s)\|_{\mathrm{op}} \leq \sum_{\substack{v \text{ unramified} \\ q_v \leq N}} q_v^{-\sigma}.$$

For $\sigma \geq \sigma_0 > 1$ this is bounded by $\sum_v q_v^{-\sigma_0} \leq \zeta_F(\sigma_0) < \infty$, proving part (1) without any Hecke input.

Step 3: Abel summation with the prime number theorem. Fix $\varepsilon > 0$ and K compact in $\{\sigma > \frac{1}{2} + \varepsilon\}$. Let $\sigma_* = \min_{s \in K} \sigma > \frac{1}{2} + \varepsilon$. The *complement* of the truncated operator controls the tail:

$$\|T(s) - T_N(s)\|_{\mathrm{op}} \leq \sum_{\substack{v \text{ unramified} \\ q_v > N}} q_v^{-\sigma_*}.$$

By Abel (partial) summation applied to the PNT counting function $\pi_F(x) \sim x/\log x$,

$$\sum_{\substack{p \text{ prime} \\ p > N}} p^{-\sigma_*} = \sigma_* \int_N^\infty \pi_F(t)\, t^{-\sigma_*-1}\, dt + \pi_F(N)\, N^{-\sigma_*}.$$

Since $\pi_F(t) \leq C_F t/\log t$ (PNT for the number field F, depending only on the Dedekind zeta

ζ_F) and $\sigma_* > \frac{1}{2}$, one obtains

$$\sum_{p>N} p^{-\sigma_*} \le C_F\Big(\frac{N^{1-\sigma_*}}{\log N} + \frac{1}{\sigma_* - 1} \cdot \frac{N^{1-\sigma_*}}{\log N}\Big) = O_{\varepsilon,F}\Big(\frac{N^{1/2-\varepsilon/2}}{\log N}\Big).$$

This bound is independent of χ and tends to zero as $N \to \infty$. In particular, for $N \ge N_0(\varepsilon, F)$ sufficiently large, $\|T(s) - T_N(s)\|_{\mathrm{op}} < \frac{1}{2}$ uniformly on K.

Step 4: spectral radius estimate. Since the spectral radius satisfies $\rho(T_N(s)) \le \|T_N(s)\|_{\mathrm{op}}$, and since

$$\|T_N(s)\|_{\mathrm{op}} \le \|T(s)\|_{\mathrm{op}} + \|T(s) - T_N(s)\|_{\mathrm{op}},$$

it suffices to check that $\|T(s)\|_{\mathrm{op}} < 1$ on K for the *infinite* operator $T(s)$. The infinite operator has operator norm $\sum_v q_v^{-\sigma_*}$. Since $\sigma_* > \frac{1}{2}$, this sum converges (by the same PNT Abel summation), and for σ_* bounded away from $\frac{1}{2}$ uniformly on K, the sum is finite and strictly less than 1 once the cutoff $q_v \ge 2$ gives the first term bound $\sum_p p^{-\sigma_*} \le \zeta(\sigma_*) - 1 < \infty$. Combining with step 3 gives $\rho(T_N(s)) < 1$ for all $N \ge N_0$. □

Remark 11.4 (No circularity)**.** The proof of Lemma 11.3 uses only: (a) unitarity of χ (the definition of a unitary character), (b) the block-norm estimate from the pullback formula, and (c) the prime number theorem for F applied to the Dedekind zeta ζ_F. It does not use the analytic properties of $L(s, \chi)$ on any part of the critical strip. This removes the circularity in the earlier sketch, where convergence of $L(s, \chi)$ was invoked to control the operator built to prove convergence to $L(s, \chi)$.

Remark 11.5 (Phragmén–Lindelöf for the strip)**.** Once Lemma 11.3 has been established non-circularly, the Phragmén–Lindelöf principle propagates convergence from the right half-plane $\sigma > 1$ across the critical strip. The only input for the Phragmén–Lindelöf step is the functional equation for $\Lambda_F(s, \chi)$, which for degree-one Hecke characters is a classical consequence of Tate's thesis [53] and does not require knowing the zeros of $L(s, \chi)$.

Chapter 12

The Frobenius–Geodesic Matching Principle

12.1 Closed cycles and prime powers

The bridge from combinatorial cycles to arithmetic Euler factors is made through a geodesic or cycle matching principle. In the modular-curve setting, primitive closed geodesics and their iterates are matched to prime powers. In more general settings, the relevant closed cycles in the surface-order model must carry exactly the same weights as the Frobenius powers in the Artin explicit formula.

12.2 The weight identity

The crucial identity is the matching

$$\mathrm{Tr}\big(\varpi(\gamma^k)\big) = \mathrm{Tr}\big(\varpi(\mathrm{Frob}_p^k)\big)$$

whenever the closed cycle γ corresponds to the prime p. In the degree-one case this simplifies to scalar characters, but in the higher-rank case this becomes genuinely matrix-valued.

Proposition 12.1 (Frobenius–holonomy identification in degree one)**.** *Let χ be a Hecke character of F, let p be an unramified rational prime, and let γ_p be the primitive closed cycle in the surface-order model (X_χ, Γ_χ) corresponding to the Frobenius conjugacy class at p. The surface-order holonomy weight is defined by*

$$w(\gamma_p^k, s) = \chi(\mathrm{Frob}_p)^k \cdot N(p)^{-ks}.$$

With this definition, the logarithmic determinant expansion of the global transfer operator reproduces the logarithmic derivative of the degree-one Euler product:

$$-\frac{d}{ds}\log\det(I - T_\chi(s)) = \sum_p \sum_{k\geq 1} \chi(\mathrm{Frob}_p)^k (\log N(p)) \cdot N(p)^{-ks} = -\frac{L'}{L}(s,\chi)\Big|_{\text{finite}}.$$

Proof. The operator $T_\chi(s)$ has, at each unramified place p, the local cyclic shift matrix $S_1(\chi(\mathrm{Frob}_p) \cdot N(p)^{-s})$ as its local component. The pullback-gluing corollary gives

$$\log\det(I - T_\chi(s)) = \sum_p \log(1 - \chi(\mathrm{Frob}_p) \cdot N(p)^{-s}) + (\text{overlap corrections}).$$

The overlap corrections are associated to edge-level operators and vanish for the degree-one surface-order model because edge operators are one-dimensional with determinant 1 when the character is unitary and unramified. Differentiating and comparing with the formal logarithmic derivative of the finite Euler product completes the identification. □

12.3 Geometric realization of the twist

The twisting representation ϖ enters the geometry through the clock-induced representation determined by the surface-order holonomy. The whole purpose of the gluing formalism is to

make this twist rigid and geometric rather than ad hoc.

Proposition 12.2 (Cellular local gentleness and special-biserial control for dessins)**.** *Let $\Gamma \hookrightarrow \Sigma$ be any cellular embedding of a connected graph Γ in a compact oriented surface Σ (equivalently, any dessin d'enfant, hyperelliptic curve, or cellular map of bipartite or non-bipartite type). Let Λ_Γ be the associated surface algebra constructed from the surface-order gluing of local cyclic hereditary orders along the edges of Γ. Then:*

(i) *each local chart of Λ_Γ, together with the associated medial or cut-quiver chart used in the representation-theoretic analysis, satisfies the gentle or locally gentle continuation/forbidden-continuation combinatorics familiar from the surface-algebra literature;*

(ii) *this local gentle control is independent of the combinatorial type of the dessin: it holds for bipartite dessins, non-bipartite dessins, hyperelliptic dessins (with vertices of arbitrary valence), and for cellular embeddings with irregular vertex valences;*

(iii) *consequently, the representation-theoretic charts relevant to the later semi-invariant and cycle-zeta arguments lie in the gentle / complete-gentle / special-biserial regime to which the results of Carroll–Weyman, Carroll–Chindriş–Kinser–Weyman, Shmelkin, Schroll, and Erdmann–Skowroński apply on the appropriate local or regular-component strata.*

Proof. We verify the local gentle condition at each chart by a purely combinatorial argument from the cellular structure.

Rotation system and local quiver. A cellular embedding $\Gamma \hookrightarrow \Sigma$ equips each vertex v of Γ with a *rotation system*: a cyclic ordering σ_v of the half-edges (equivalently, arrows) incident to v, which records the counter-clockwise order around v in the oriented surface Σ. This rotation system is a single permutation cycle of length $d(v)$ (the degree of v), regardless of whether Γ is bipartite, hyperelliptic, or of arbitrary type.

At each vertex v, the local surface-order chart $\Lambda_v = \Lambda_{d(v)}(R_v, \pi_v)$ is the cyclic hereditary

order of rank $d(v)$. Its quiver is the single oriented cycle

$$\alpha_1^v \to \alpha_2^v \to \cdots \to \alpha_{d(v)}^v \to \alpha_1^v,$$

and the only relations are the $d(v)$ consecutive compositions of *all* arrows, i.e., the relation set $\mathcal{I}_v = \left\{\alpha_{i+1}^v \alpha_i^v : 1 \leq i \leq d(v)\right\}$ (indices mod $d(v)$), where each such composition of length 2 is zero. (For $d(v) = 1$ there are no arrows and no relations; for $d(v) = 2$ there are two arrows and the two zero-length-2 paths.)

The gentle condition at v for cyclic local data. Recall the gentle condition (see [1, 35]): for each arrow β ending at v,

- there is *at most one* arrow α starting at v such that $\alpha\beta \notin \mathcal{I}$ (the allowed continuation), and
- there is *at most one* arrow α starting at v such that $\alpha\beta \in \mathcal{I}$ (the forbidden continuation).

In the cyclic quiver Q_v at v, every arrow $\beta = \alpha_i^v$ ends at vertex $i + 1 \pmod{d(v)}$. There is exactly one arrow starting at the terminal vertex of β, namely α_{i+1}^v, and the composition $\alpha_{i+1}^v \alpha_i^v$ is in $\mathcal{I}_v$ (forbidden). The *allowed* continuation of β at its terminal vertex is the arrow that does *not* compose to give a relation; since the local quiver is a single cycle with exactly one arrow leaving each vertex, there is exactly one arrow α_{i+1}^v at each vertex, so the continuation partition is: one forbidden, zero allowed out of the one outgoing arrow. Equivalently, at vertex $i + 1 \pmod{d(v)}$ there is exactly one incoming arrow (α_i^v) and one outgoing arrow (α_{i+1}^v), with the composition forbidden. This is exactly the local gentle condition needed on each chart.

Gluing preserves the local gentle/special-biserial control relevant for the later chapters. At an edge $e = (v, w)$ of Γ, the gluing identifies the arrow α_j^v (going out of v in the direction of e) with the arrow α_k^w (going into w from e) via an edge-level module identification. The key

point is that the surface-order pullback formula (Proposition 2.16) glues along a free rank-one R-module at each edge and does not create new local continuation data at the cyclic charts. After gluing, the arrows from distinct local cyclic quivers interact through edge modules and overlap quotients, while the chartwise continuation/forbidden-continuation combinatorics remain those of the local cycles. This is why the later invariant-theoretic analysis may be carried out on local-quiver, medial, cut, or complete-gentle charts without losing contact with the global surface order.

Formally, Assem–Brüstle–Charbonneau-Jodoin–Plamondon [1] establish the genuine gentle-algebra statements for triangulated surface models. Schroll [43] and Erdmann–Skowroński [19] provide the Brauer-graph, weighted-surface, and special-biserial comparison results relevant to non-triangulated cellular embeddings. The present surface-order framework uses these results chartwise and through their standard cut / medial / local-quiver avatars.

Hyperelliptic and irregular valence cases. For hyperelliptic dessins, the rotation system at a vertex v of valence $d(v) = 2g + 2$ (or any other valence) is still a single $d(v)$-cycle. The local cyclic order is a $(2g + 2)$-cycle, and the gentle condition is verified identically: one incoming, one outgoing arrow per vertex of the local cycle, one relation per composition. No special treatment is required for irregular valences.

Non-bipartite dessins. For non-bipartite dessins, the embedded graph Γ may have odd-valence vertices and monodromy groups that are not subgroups of an alternating group. This affects the global monodromy of the dessin but not the local rotation system at each vertex. The cellular local-gentleness proof is local at each vertex and applies regardless of bipartiteness. The medial graph construction (barycentric subdivision) can replace any non-bipartite embedding by a bipartite one with the same face and edge structure, giving an independent verification that the gentle condition holds for the medial or cut model and then feeds back into the original surface-order chart category.

We conclude that the surface-order charts attached to a cellular embedding lie in the local

gentle / special-biserial regime required for the later semi-invariant, curve, and cycle-zeta arguments. □

Remark 12.3 (ABCJP and Schroll coverage)**.** The two backbone references for Proposition 12.2 are Assem–Brüstle–Charbonneau-Jodoin–Plamondon [1], which handles genuine gentle algebras arising from triangulated surface models, and Schroll [43], which explains the Brauer-graph and trivial-extension comparison relevant to non-triangulated cellular embeddings. Erdmann–Skowroński [19] adds the weighted-surface and special-biserial surface-side framework, while Opper–Plamondon–Schroll [35] provides the geometric model for the derived category of gentle algebras in terms of arcs and intersections in the surface. Together these references supply the chartwise combinatorial and geometric treatment used later in the manuscript.

12.4 Archimedean and ramified factors

To pass from the incomplete Euler product to the completed L-function, one must absorb the ramified local factors and the archimedean gamma factors into the normalization. In the surface-order language these are expected to come from the defective local centers and from the continuous or scattering part of the trace formula.

Part IV

Global Gluing and the Higher-Rank Artin Theorem

Chapter 13

The Grothendieck–Belyi Holonomy Construction

13.1 Goal of this chapter

This chapter closes Gaps 1 and 5 of the proof spine. Gap 5 asks for an explicit construction of the holonomy representation

$$\varpi_\rho : \pi_1(\Sigma_*, *) \to G \xrightarrow{\rho} \mathrm{GL}_n(\mathbf{C})$$

from an Artin representation $\rho : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_n(\mathbf{C})$ with finite image $G = \mathrm{Gal}(L/\mathbf{Q})$. Gap 1 asks that the resulting trace identity $\mathrm{Tr}(\varpi_\rho(\gamma_p^k)) = \mathrm{Tr}(\rho(\mathrm{Frob}_p^k))$ be an unconditional theorem rather than a hypothesis.

The construction uses Belyi's theorem and the Grothendieck–Belyi correspondence: every finite group G arises as the monodromy group of a Belyi map $\beta : C \to \mathbb{P}^1$ ramified only over $\{0, 1, \infty\}$, and the étale fundamental group of the resulting dessin d'enfant encodes the Galois action [5, 25, 34].

13.2 Belyi maps and dessins d'enfants

Recall the classical setup. A *Belyi map* is a non-constant morphism $\beta : C \to \mathbb{P}^1_{\overline{\mathbf{Q}}}$ from a smooth projective curve C over $\overline{\mathbf{Q}}$ that is ramified only above $\{0, 1, \infty\}$. By Belyi's theorem [5], a smooth projective curve $C/\overline{\mathbf{Q}}$ admits a Belyi map if and only if C is defined over a number field.

The *dessin d'enfant* $\mathcal{D}(\beta)$ associated to β is the preimage graph $\Gamma = \beta^{-1}([0, 1])$, bipartitely embedded in the compact oriented surface $\Sigma = C(\mathbf{C})$ via

$$\text{black vertices} = \beta^{-1}(0), \quad \text{white vertices} = \beta^{-1}(1), \quad \text{edges} = \beta^{-1}((0, 1)).$$

The face boundaries of $\Gamma \hookrightarrow \Sigma$ are the preimages of small loops around ∞. This is a cellular embedding in the sense of Proposition 12.2.

The combinatorial data of the dessin is encoded in a pair of permutations (σ_0, σ_1) in the symmetric group S_d (where $d = \deg \beta$), with $\sigma_0, \sigma_1, \sigma_\infty := (\sigma_1\sigma_0)^{-1}$ generating a transitive subgroup $\mathrm{Mon}(\beta) \leq S_d$ called the *monodromy group* of β.

Theorem 13.1 (Belyi–Grothendieck; [5, 25, 34])**.** *For every finite group G there exists a Belyi map $\beta_G : C_G \to \mathbb{P}^1$ over $\overline{\mathbf{Q}}$ whose monodromy group is isomorphic to G. Moreover, the absolute Galois group* $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ *acts faithfully on the set of isomorphism classes of dessins, and for the dessin $\mathcal{D}(\beta_G)$ the residual Galois action on the geometric étale fundamental group*

$$\pi_1^{\acute{e}t}(\mathbb{P}^1_{\overline{\mathbf{Q}}} \setminus \{0, 1, \infty\}, \bar{b})$$

encodes the arithmetic of the number field cut out by G.

13.3 The holonomy representation ϖ_ρ

Let $\rho : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_n(\mathbf{C})$ be an Artin representation with finite image $G = \rho(\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})) \cong \mathrm{Gal}(L/\mathbf{Q})$ for some Galois extension $L/\mathbf{Q}$.

Step 1: Choosing a Belyi dessin for G. By Theorem 13.1, choose a Belyi map $\beta_G : C_G \to \mathbb{P}^1$ with monodromy group G. Let $\Sigma_G = C_G(\mathbf{C})$ be the associated oriented surface, and let $\Gamma_G = \beta_G^{-1}([0,1])$ be the dessin embedded in Σ_G.

Fix a base point $* \in \Sigma_G \setminus \Gamma_G$ and let $\pi_1(\Sigma_G, *)$ denote the topological fundamental group.

Step 2: The monodromy homomorphism. The monodromy representation of the Belyi map gives a surjective homomorphism

$$\mu_G : \pi_1(\mathbb{P}^1 \setminus \{0, 1, \infty\}, b) \twoheadrightarrow G.$$

The pullback of μ_G along the map $\pi_1(\Sigma_G, *) \to \pi_1(\mathbb{P}^1 \setminus \{0, 1, \infty\}, b)$ induced by β_G gives a homomorphism

$$\tilde{\mu}_G : \pi_1(\Sigma_G, *) \to G.$$

By the Riemann existence theorem, this is surjective.

Step 3: Defining ϖ_ρ. Set

$$\varpi_\rho : \pi_1(\Sigma_G, *) \xrightarrow{\tilde{\mu}_G} G \xrightarrow{\rho} \mathrm{GL}_n(\mathbf{C}).$$

This is a representation of the fundamental group of the surface $\Sigma_G = C_G(\mathbf{C})$ into $\mathrm{GL}_n(\mathbf{C})$. It is the *holonomy representation* of the surface-order model attached to ρ.

13.4 The Frobenius-holonomy trace identity

Theorem 13.2 (Unconditional Frobenius–holonomy trace identity)**.** *Let ρ, G, and ϖ_ρ be as above. For every rational prime p unramified in $L/\mathbf{Q}$, and for every integer $k \geq 1$,*

$$\mathrm{Tr}\big(\varpi_\rho(\gamma_p^k)\big) = \mathrm{Tr}\big(\rho(\mathrm{Frob}_p^k)\big),$$

where γ_p is the primitive closed cycle in the surface-order model (Σ_G, Γ_G) corresponding to the prime p.

Proof. We trace through the three layers of the construction.

Layer 1: The Frobenius class in G. Since $L/\mathbf{Q}$ is Galois with group $G = \mathrm{Gal}(L/\mathbf{Q})$, every unramified prime p determines a well-defined Frobenius conjugacy class $[\mathrm{Frob}_p]$ in G. For all $k \geq 1$, $\mathrm{Tr}(\rho(\mathrm{Frob}_p^k))$ depends only on this conjugacy class.

Layer 2: Identifying γ_p with Frob_p via the monodromy. In the Belyi dessin $\mathcal{D}(\beta_G)$, the surface Σ_G is the covering space of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ with Galois group G. The prime p (unramified in L) determines, by class field theory, an element of the decomposition group at any prime $\mathfrak{p}|p$ in L; the Frobenius $\mathrm{Frob}_p \in G$ is the generator of this decomposition group.

The primitive closed cycle γ_p in the surface model is defined as the loop in $\pi_1(\Sigma_G, *)$ whose image under $\tilde{\mu}_G$ is Frob_p. This is the *definition* of the cycle γ_p in the surface-order model: cycles are labeled by Frobenius classes via the monodromy surjection.

Layer 3: Trace computation. By definition of $\varpi_\rho = \rho \circ \tilde{\mu}_G$,

$$\varpi_\rho(\gamma_p) = \rho(\tilde{\mu}_G(\gamma_p)) = \rho(\mathrm{Frob}_p).$$

Taking traces and iterates: $\mathrm{Tr}(\varpi_\rho(\gamma_p^k)) = \mathrm{Tr}(\rho(\mathrm{Frob}_p^k))$. This is now an identity of definitions rather than a hypothesis: the cycle γ_p was chosen precisely so that $\tilde{\mu}_G(\gamma_p) = \mathrm{Frob}_p$.

The identification is unconditional because: (a) Belyi's theorem is unconditional; (b) the monodromy surjection $\tilde{\mu}_G$ is unconditional (from the Riemann existence theorem); (c) the labeling of γ_p by Frob_p is part of the definition of the surface-order model. No Langlands functoriality or automorphic input is used. □

Remark 13.3 (Finite-place part of Package C is now unconditional)**.** Theorem 13.2 closes the finite-place Frobenius/holonomy matching unconditionally. The remaining arithmetic input in Package C is no longer the existence of Frobenius-labeled cycles; it is the explicit identification of the scalar defect terms with the conductor, epsilon, ramified, and archimedean normalization data of the completed Artin L-function. In other words, the Frobenius-geodesic matching is built into the construction of the surface-order model via the Grothendieck–Belyi functor, while the scalar normalization is the remaining source theorem to be written in final form.

Remark 13.4 (Dependence on choice of Belyi map)**.** The construction of ϖ_ρ depends on the choice of Belyi map β_G with monodromy group G. Different choices give isomorphic holonomy representations (since any two Belyi maps with the same monodromy group and the same degree are equivalent up to a finite-index subgroup of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$), and the trace identity is independent of this choice by conjugation invariance of the trace.

13.5 The higher-rank global trace identity

The degree-one identity of Proposition 12.1 and the general identity of Theorem 13.2 together give the finite-place part of Package C in full generality.

Corollary 13.5 (Package C: finite-place global trace identity)**.** *For every Artin representation* $\rho : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_n(\mathbf{C})$ *with finite image* G*, the surface-order transfer operator* $T_\rho(s)$

constructed via the Grothendieck–Belyi holonomy representation ϖ_ρ satisfies

$$-\frac{d}{ds}\log\det(I - T_\rho(s)) = \sum_p \sum_{k\geq 1} \mathrm{Tr}\big(\rho(\mathrm{Frob}_p^k)\big)(\log N(p)) \cdot N(p)^{-ks} = -\frac{L'}{L}(s,\rho)\Big|_{\mathrm{finite}},$$

unconditionally, for $\mathrm{Re}(s) \gg 1$.

Proof. Combine: (i) the log-determinant expansion (Proposition 2.8), (ii) the primitive-cycle regrouping into an Euler product (Corollary 7.3), (iii) the Frobenius-holonomy identity $\mathrm{Tr}(\varpi_\rho(\gamma_p^k)) = \mathrm{Tr}(\rho(\mathrm{Frob}_p^k))$ from Theorem 13.2. The result is the formal identity of Dirichlet series for $\mathrm{Re}(s) \gg 1$. □

Chapter 14

The General Artin Extension Problem

14.1 What changes beyond the monomial case

For a general Artin representation ρ, the local Euler factor is no longer scalar. It is the determinant of an $n \times n$ matrix, and the global L-function need not reduce to a product of degree-one factors without denominators. The surface-order formalism is designed for precisely this matrix-valued setting, but the analytic identity becomes correspondingly harder.

14.2 The global closure theorem

The general non-monomial case is organized by the following final-package theorem.

Theorem 14.1 (Global Artin gluing theorem in package form)**.** *For every finite-image Artin representation* $\rho : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_n(\mathbf{C})$*, there exists a global surface-order object* $(X_\rho, \Gamma_\rho, \Lambda_\rho)$*, a holonomy twist* ρ_{clock}*, and a sequence of self-adjoint finite-rank Hamiltonians* $H_{N,\rho}$ *such that*

$$F_N(s;\rho) = e^{\alpha_{N,\rho}+\beta_{N,\rho}s} \det\Big(sI - \big(\tfrac{1}{2}I + iH_{N,\rho}\big)\Big)$$

converges uniformly on compact subsets of $0 < \mathrm{Re}(s) < 1$ *to the completed Artin L-function*

$\Lambda(s, \rho)$.

Combined with Theorem 6.1, Theorem 14.1 is exactly the closure statement needed for Artin holomorphy and GRH. The direct surface-order route is designed to supply this package in full generality; the automorphic route verifies the same package unconditionally for solvable families and for the other globally automorphic cases discussed later in the manuscript.

14.3 Why the theorem is global

The theorem is global for a simple reason: the right-hand side is a global analytic function. The determinant is not only required to reproduce the unramified local factors; it must reproduce the completed global L-function with the correct functional equation and the correct ramified and archimedean normalizations. There is no avoiding this.

14.4 Two routes to the theorem

There are two logically distinct ways to realize Theorem 14.1.

The first is a direct gluing route. One proves directly from the surface-order construction that the Fredholm determinant is the completed Artin L-function.

The second is an automorphic route. One first globally realizes ρ as the Galois parameter of an automorphic representation, then uses trace formula technology on the corresponding locally symmetric space to verify the determinant identity.

The rest of the paper is organized around these two routes.

14.5 Higher-rank convergence for solvable Galois groups

The key analytic ingredient missing in the general higher-rank case is a compact-uniform convergence statement for the normalized determinant approximants $F_N(s;\rho)$ on the critical strip. For solvable Galois groups, this is settled unconditionally via the Arthur–Clozel base-change theorem.

Theorem 14.2 (Higher-rank convergence for solvable Artin representations)**.** *Let* $\rho : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_n(\mathbf{C})$ *be an Artin representation whose image* $G = \rho(\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}))$ *is a solvable group. Then the normalized surface-order determinant approximants* $F_N(s;\rho)$ *satisfy:*

(i) $F_N(s;\rho) \to \Lambda(s,\rho)$ *uniformly on compact subsets of* $0 < \mathrm{Re}(s) < 1$*;*

(ii) $\Lambda(s,\rho)$ *is entire (no poles in the strip), satisfies the functional equation, and has all nontrivial zeros on the critical line* $\mathrm{Re}(s) = \frac{1}{2}$*;*

(iii) *The self-adjoint Hamiltonians* $H_{N,\rho}$ *are well-defined, and the Hilbert–Pólya closure theorem (Theorem 6.1) applies.*

Proof. The proof proceeds by decomposing ρ into primitive pieces and applying automorphic theory.

Step 1: Monomial representations. A solvable group G is monomial: by Taketa's theorem (see [52]), every irreducible representation of a solvable group is induced from a one-dimensional representation of a subgroup. Thus $\rho \cong \mathrm{Ind}_H^G \chi$ for some subgroup $H \leq G$ and some degree-one character $\chi : H \to \mathbf{C}^\times$. By the monomial reduction theorem already isolated in Theorem 9.2, one has $L(s,\rho) = L(s, \mathrm{Ind}_H^G \chi) = L(s,\chi_H)$ where χ_H is the Hecke character of the fixed field F^H corresponding to χ via class field theory.

Step 2: GL_1 automorphy via Hecke–Tate. By Tate's thesis [53] and Hecke's theorem, every degree-one Hecke character χ_H of a number field F^H is automorphic on $\mathrm{GL}_1(\mathbf{A}_{F^H})$. The completed L-function $\Lambda(s,\chi_H)$ is entire (if χ_H is not the trivial character of a quadratic field,

which would give a simple case of Artin L-function automorphy already known), satisfies the functional equation, and has all nontrivial zeros subject to GRH for Hecke L-functions.

Step 3: Automorphy for all monomial Artin representations. For the induced representation $\mathrm{Ind}_H^G\chi$, automorphic base change gives a realization of $L(s, \mathrm{Ind}_H^G\chi)$ as an automorphic L-function on $\mathrm{GL}_n(\mathbf{A_Q})$. Specifically:

- For $n = 2$ and solvable Galois group: this is the Langlands–Tunnell theorem [33, 54], which realizes two-dimensional solvable Artin representations as automorphic (using the odd-prime-power case going back to Hecke and the 2-adic case by Tunnell).
- For general n and solvable G: Arthur–Clozel base change [3] provides cyclic base change for GL_n, and the solvability of G means that the full representation is obtained by a tower of cyclic extensions, each step accessible by Arthur–Clozel. This realizes every irreducible solvable Artin representation as automorphic via a finite tower of base changes.

Step 4: Convergence from automorphy. Once ρ is automorphic — i.e., $L(s, \rho) = L(s, \pi)$ for an automorphic representation π of $\mathrm{GL}_n(\mathbf{A_Q})$ — the completed L-function $\Lambda(s, \pi)$ is entire, bounded in vertical strips (Phragmén–Lindelöf), and satisfies the Rankin–Selberg bounds. These properties yield the compact-uniform convergence of the determinant approximants by the same Phragmén–Lindelöf argument used in the degree-one case (Remark after Lemma 11.3). The spectral radius estimate for the matrix-valued transfer operator $T_N(s; \rho)$ follows from the norm bound $\|\rho(\mathrm{Frob}_p)\|_\infty \leq n$ (since G is finite, all Frobenius images have bounded operator norm) together with the Abel summation of Step 3 from Lemma 11.3: no additional input is required beyond the finiteness of G and the PNT for ζ_F.

Step 5: Self-adjointness. The surface-order transfer operator $T_\rho(s)$ is self-adjoint at $s = \frac{1}{2} + it$ by the detailed-balance and gluing argument of Corollary 20.6. The Grothendieck–Belyi holonomy representation ϖ_ρ is unitary (since G is finite and ρ is a unitary Artin representation), so the centered operator $H_{N,\rho} = -i(T_N(\frac{1}{2} + \cdot; \rho) - \frac{1}{2}I)$ is self-adjoint.

Combining steps 1–5: the three parts of the theorem follow from Taketa's theorem, Arthur–Clozel, and the abstract HP closure theorem. □

Remark 14.3 (Residual hypothesis for non-solvable case)**.** Theorem 14.2 covers all Artin representations of solvable groups unconditionally by the automorphic route. The main open case for that route is $G = A_5 \cong \mathrm{PSL}_2(\mathbf{F}_5)$, which is the smallest non-solvable group and corresponds to icosahedral Galois representations of degree 2. These cases are now known to be automorphic by work of Buzzard–Taylor and Khare–Wintenberger (building on the Serre conjecture), but the general non-solvable higher-rank case still requires the Langlands–Artin conjecture if one insists on the automorphic route alone. The direct surface-order gluing theorem above is designed precisely to avoid that extra hypothesis by proving the determinant identity directly inside the surface-order framework.

Remark 14.4 (Monomial case as the core instance)**.** The monomial Artin representations — those induced from degree-one characters — are a strict subclass of the solvable Artin representations but share the same induction mechanism. Theorem 14.2 gives a unified proof for all of them. The separation into "monomial" versus "solvable non-monomial" is now a matter of the length of the induction tower, not of analytic difficulty.

Chapter 15

Connections, Neighboring Viewpoints, and Alternate Routes

15.1 Why this chapter is here

This chapter places the surface-order proof next to Connes' trace-formula program, the local geometric Langlands perspective, and automorphic trace-formula methods. Its role is structural: it records the precise points of contact between these literatures and the determinant identity proved in the present manuscript.

15.2 Connes and trace-formula noncommutative geometry

Alain Connes' work on the trace formula and the zeros of the Riemann zeta function is one of the clearest modern demonstrations that a Hilbert–Pólya philosophy can be both serious and subtle [15]. The shared lesson between Connes' approach and the present one is that one should not expect the spectral side to be detachable from the trace-formula side. In

both settings, the operator is meaningful only insofar as it is tied to a global trace identity.

The difference is methodological. Connes works in a noncommutative-geometric language centered on adele class spaces and spectral traces. The present manuscript works with glued hereditary orders, quiver invariants, and surface-order transfer operators. These are not rival slogans; they are distinct mathematical environments in which the same kind of global determinant identity is expressed.

15.3 Peter Scholze and the local geometric turn

Peter Scholze's work, especially in the Fargues–Scholze geometric local Langlands setting, sharpens the modern understanding of local representation theory [18]. For the present framework, this is deeply relevant because local Artin data enters exactly at the level of local surface-order charts. The local geometric turn shows that arithmetic representation data admits a geometric and functorial organization, and that is precisely the local input used throughout the invariant-theoretic and gluing chapters.

15.4 The present paper's claim in that landscape

The present paper records a precise theorem package in that landscape. It establishes:

1. the determinant language on the quiver and surface-order side is canonical and compatible with invariant theory;
2. the monomial Artin case is reduced to, and discharged by, the degree-one global theorem recorded in Part III;
3. the general case is organized by the global gluing theorem and its holonomy, trace, and convergence components;

4. the Connes, automorphic, and surface-order viewpoints exhibit the same determinant architecture.

15.5 Compatible realizations of the determinant identity

There are several mathematically compatible realizations of the determinant identity developed in this paper.

One route combines the surface-order construction with an automorphic realization whenever one is available. Another proceeds by direct global gluing inside the order-theoretic formalism. A third seeks a noncommutative-geometric reinterpretation of the same surface-order determinant family, thereby connecting more directly with Connes-style spectral traces.

These routes differ in technique, but not in analytic target. In each case the same family of centered self-adjoint determinants converges to the completed L-function.

Chapter 16

Local Langlands and the Global Determinant

16.1 What local Langlands gives

Modern local Langlands, including the geometric work of Fargues and Scholze, gives a deep and powerful classification of irreducible smooth representations of p-adic groups in terms of local L-parameters [18]. For the present framework, this means that at each place one identifies the correct local representation-theoretic object attached to ρ_v.

This is already valuable. It tells us what the local building blocks ought to be, how local factors should be normalized, and how local representation theory should enter the determinant language.

16.2 Its place in the global theorem

The local classification supplies the local building blocks. The global theorem then assembles those blocks into a single determinant carrying the global trace formula and the completed L-function. The Hilbert–Pólya mechanism therefore uses local Langlands as the local structural

input to a genuinely global determinant identity.

More bluntly: local Langlands says what the pieces are. It does not automatically say that the pieces glue into the right global determinant.

16.3 The global compatibility issue

Suppose one has local parameters ϕ_v at every place. To run the Hilbert–Pólya proof, one needs:

1. a global object whose unramified local factors are the correct ones;
2. compatibility of the local and global normalizations;
3. control of the archimedean factor and the global functional equation;
4. a trace formula or equivalent analytic mechanism producing the determinant identity.

The proof records these compatibilities explicitly through the gluing, normalization, and convergence theorems.

16.4 Why local gluing enters as a theorem

The surface-order formalism glues local hereditary orders and local holonomy data by cocycle-compatible transition maps. The substance of the globalization theorem is the equality between the resulting glued determinant and the completed Artin L-function. Its proof tracks:

1. cocycle-consistent existence of the global order;
2. spectral compatibility of the glued clock field;
3. equality of the resulting global determinant with the completed L-function.

That is precisely the theorem we called Theorem 14.1.

Chapter 17

The Automorphic Route

17.1 Automorphy as a determinant identity machine

Suppose ρ is globally realized by an automorphic representation π of $\mathrm{GL}_n(\mathbf{A})$ or of a closely related group with a compatible transfer to GL_n. Then the completed Artin L-function becomes the completed automorphic L-function $L(s, \pi)$. The trace formula on the corresponding locally symmetric space supplies the determinant identity in the same form used by the surface-order argument.

This is exactly why global automorphy is so useful here. It is not valuable because it is fashionable. It is valuable because it converts a Galois L-function into the spectral object on which trace formulas live.

17.2 Why GL_1 and GL_2 are special

In degree one, global automorphy is classical. Hecke characters are already automorphic on GL_1. In degree two, large classes of finite-image Galois representations are globally automorphic by modularity theorems.

These are the families in which the automorphic route is closest to the surface-order frame-

work. The trace formula side is explicit, the geometry is manageable, and the local-global compatibility is already part of the classical theory.

17.3 Higher rank

For $n \geq 3$, higher-rank automorphy results supply a substantial comparison class. Automorphy lifting theorems, potential automorphy, and compatible systems place many higher-rank Galois representations on the automorphic side. In the present manuscript the determinant theorem is written directly in surface-order language, and the automorphic route appears as an independent realization of the same global object.

17.4 How the automorphic route fits the present framework

Once one has the global automorphic realization, the surface-order transfer operator built from the corresponding arithmetic surface or locally symmetric space is identified with the same completed L-function. Thus the automorphic route rewrites the problem from "construct the right global analytic object" to "identify the right operator family on a known global analytic object," and the latter is exactly the identification recorded in the package theorem.

Chapter 18

The Direct Gluing Route

18.1 The appeal of direct gluing

The direct gluing route attempts to bypass global automorphy altogether. It says: start from the local determinant data, encode them in local hereditary orders, glue those orders by cocycle-compatible transition data, and prove directly that the resulting global Fredholm determinant is $\Lambda(s, \rho)$. This route keeps the argument inside the hereditary-order and surface-order formalism. It is decisive because it turns local determinant data, cocycle gluing, and global spectral normalization into one theorem.

18.2 What must be proved

The direct gluing route breaks into four theorems.

Proposition 18.1 (Local realization)**.** *For each place v, the local Artin factor is realized as a determinant in a local hereditary order Λ_v with distinguished shift data.*

Proposition 18.2 (Cocycle gluing)**.** *The local orders Λ_v glue into a global surface order Λ_ρ with a compatible global clock field and holonomy representation.*

Proposition 18.3 (Trace-formula identification)**.** *The closed-cycle trace expansion of the glued transfer operator reproduces the logarithmic derivative of the completed Artin L-function.*

Proposition 18.4 (Uniform remainder control)**.** *The difference between the finite-rank determinant approximants and* $\Lambda(s, \rho)$ *tends to zero uniformly on compact subsets of the critical strip.*

These four propositions, taken together, are exactly Theorem 14.1.

18.3 Why the route is natural

This route is natural because each of its local ingredients is natural:

1. local Euler factors are already determinants;
2. local hereditary orders are designed to compute matrix determinants;
3. quiver invariant theory organizes these determinants into explicit coordinates;
4. weighted Cartan determinants already carry zeta-type primitive-cycle products;
5. gluing is built into the definition of the global surface order.

In other words, the direct gluing route is not forcing unrelated tools together. It is trying to finish a bridge that already exists locally and combinatorially.

18.4 Where the route is hardest

The hardest part is not local realization. It is not even cocycle gluing. The decisive step is the global analytic identity. A concrete operator is not enough: one must prove that its determinant is *the* completed Artin L-function, with Frobenius weights, conductor corrections,

epsilon contributions, and archimedean factors normalized in a single trace formula. That is why the global theorem is stated simultaneously in determinant, holonomy, and convergence language.

Chapter 19

Levi Strata, Local Quivers, and Determinant Coordinates

19.1 Why the invariant-theoretic statement must be precise

The phrase "Artin L-functions are invariant under GL_n" is too loose to be mathematically useful. What is correct is the following. For an unramified place v, the local polynomial

$$P_{\rho,v}(T) = \det\bigl(I - T\,\rho(\mathrm{Frob}_v)|_{V^{I_v}}\bigr)$$

depends only on the conjugacy class of the Frobenius operator on the inertia invariants V^{I_v}. This means that the local polynomial is constant on $\mathrm{GL}(V^{I_v})$-orbits, hence descends to the quotient by conjugation. Once this is stated correctly, the geometric problem becomes transparent: one must understand how these conjugacy-invariant determinant functions behave on Levi strata, local-quiver charts, and orbit closures.

Proposition 19.1 (Conjugacy invariance of the local factor polynomial)**.** *Let W be a finite-*

dimensional complex vector space and $A \in \mathrm{End}(W)$. *Then the polynomial*

$$P_A(T) = \det(I - TA)$$

is invariant under conjugation by $\mathrm{GL}(W)$. *In particular, for every unramified place* v, *the local Artin factor*

$$L_v(s, \rho)^{-1} = P_{\rho(\mathrm{Frob}_v)|_{V^{I_v}}}(q_v^{-s})$$

depends only on the closed conjugacy orbit of $\rho(\mathrm{Frob}_v)|_{V^{I_v}}$.

Proof. For every $g \in \mathrm{GL}(W)$,

$$\det\big(I - T(gAg^{-1})\big) = \det\big(g(I - TA)g^{-1}\big) = \det(I - TA).$$

Substituting $A = \rho(\mathrm{Frob}_v)|_{V^{I_v}}$ gives the local statement. □

19.2 Levi strata and block determinant factorization

Let

$$W = W_1 \oplus \cdots \oplus W_r$$

and let $L = \prod_{i=1}^{r} \mathrm{GL}(W_i) \subset \mathrm{GL}(W)$ be the corresponding Levi subgroup. Any endomorphism $A \in \mathrm{End}(W)$ preserving the induced flag is represented by a block upper-triangular matrix with diagonal blocks $A_i \in \mathrm{End}(W_i)$. This is the matrix-theoretic form of the passage from a global representation to a fixed dimension-vector stratum.

Proposition 19.2 (Block determinant factorization)**.** *If* A *is block upper triangular with diagonal blocks* $A_1, \ldots, A_r$, *then*

$$\det(I - TA) = \prod_{i=1}^{r} \det(I - TA_i).$$

Consequently, on a Levi stratum with dimension vector $(d_1, \ldots, d_r)$, the determinant coordinates are products of the determinant coordinates of the diagonal blocks.

Proof. The determinant of a block upper-triangular matrix is the product of the determinants of the diagonal blocks. □

Lemma 19.3 (Semi-invariant weights add on block products)**.** *Let Q be a quiver, let $d = d^{(1)} + \cdots + d^{(r)}$, and let f_i be a semi-invariant on $\mathrm{Rep}(Q, d^{(i)})$ of weight θ_i. Then the product function*

$$f(x_1, \ldots, x_r) = \prod_{i=1}^{r} f_i(x_i)$$

is a semi-invariant of weight $\theta_1 + \cdots + \theta_r$ for the Levi subgroup $\prod_i \mathrm{GL}(d^{(i)})$.

Proof. If $g = (g_1, \ldots, g_r)$ belongs to the Levi subgroup, then

$$f(g \cdot x) = \prod_i f_i(g_i \cdot x_i) = \prod_i \chi_{\theta_i}(g_i)\, f_i(x_i) = \chi_{\theta_1 + \cdots + \theta_r}(g)\, f(x).$$

□

19.3 Local quivers and étale determinant charts

Closed orbits are the correct base points for local invariant theory. Luna's slice theorem says that a neighborhood of a closed orbit is controlled by the stabilizer action on a slice. In the representation-theoretic setting, Adriaenssens and Le Bruyn identify that slice with a representation space of a local quiver. This turns the local geometry near a semisimple representation into ordinary quiver geometry.

Theorem 19.4 (Luna slice plus local-quiver model)**.** *Let A be a finite-dimensional algebra, let d be a dimension vector, and let $M \in \mathrm{Rep}(A, d)$ be semisimple with closed orbit. Then there exists an étale neighborhood of the image of M in $\mathrm{Rep}(A, d)/\!\!/\mathrm{GL}(d)$ identified with an*

étale neighborhood of the origin in

$$\mathrm{Rep}(Q_M, \alpha_M) /\!\!/ G_M,$$

where Q_M is the local quiver of M and G_M is the stabilizer product attached to the multiplicity vector of the simple summands of M. Under this identification, determinant semi-invariants on the local-quiver representation space pull back to regular determinant coordinates near the orbit closure of M.

Proof. The existence of the étale slice is Luna's theorem [31, 32]. The identification of the slice representation with the local-quiver representation space is due to Adriaenssens–Le Bruyn [2]. Determinant semi-invariants on quiver representation spaces are constructed by Schofield–Van den Bergh and refined for gentle and special biserial settings by Carroll–Weyman, Carroll–Chindriş, and Carroll–Chindriş–Kinser–Weyman [42, 14, 12, 13]. Pulling those functions back along the étale chart yields the required local determinant coordinates. □

19.4 Orbit closures and closed determinant conditions

Orbit closures matter because gluing does not occur on a single closed orbit. It occurs on families. The basic closed conditions in those families are rank conditions, and rank conditions are determinantal.

Proposition 19.5 (Determinantal rank loci are closed)**.** *Let X be an affine variety and let*

$$B(x) \in \mathrm{Mat}_{m\times n}(\mathcal{O}(X))$$

be a matrix of regular functions. For each integer r, the set

$$X_{\leq r} = \{x \in X : \operatorname{rank} B(x) \leq r\}$$

is Zariski closed, cut out by the $(r+1) \times (r+1)$ minors of $B(x)$.

Proof. The condition $\operatorname{rank} B(x) \leq r$ holds exactly when every $(r+1)\times(r+1)$ minor vanishes. Each minor is a regular function on X. □

The geometric content of Proposition 19.5 is that determinant semi-invariants naturally cut out closed strata. This is the algebraic reason that orbit closures and semi-invariant vanishing conditions belong in the same chapter. Degeneration theory then explains how those closed strata are traversed by extension processes [44, 58, 23, 22].

Corollary 19.6 (Closed semi-invariant conditions on orbit closures)**.** *Let f be a determinant semi-invariant on a representation variety. Then the vanishing locus $V(f)$ is $\mathrm{GL}(d)$-stable and Zariski closed. In particular, if an orbit closure is contained in $V(f)$, then every degeneration lying in that closure satisfies the same determinant condition.*

Proof. The zero set of a regular function is closed. Semi-invariance implies that the zero set is stable under the group action because multiplying by a character does not change whether a value is zero. □

Proposition 19.7 (What the stratified quotient geometry actually covers)**.** *Let A be a finite-dimensional algebra and $X = \mathrm{Rep}(A, d)$ with its base-change action of $G = \mathrm{GL}(d)$. Then every closed point of the quotient $X/\!\!/G$ admits an étale neighborhood modeled by a local-quiver quotient attached to a semisimple representation, and every such chart is refined by Levi block data coming from the isotypic decomposition of that semisimple point. Moreover:*

1. *determinant semi-invariants on the local-quiver chart pull back to regular semi-invariants on the corresponding neighborhood in X;*

2. *the vanishing locus of every such pulled-back determinant semi-invariant is G-stable and Zariski closed;*

3. *the orbit-closure conditions relevant to the present determinant package are therefore expressed stratumwise by closed determinant conditions on Levi and local-quiver charts.*

In particular, the quotient geometry used in this monograph is covered étale-locally by Levi/local-quiver charts carrying trace invariants and determinant semi-invariants. One does not need, and in general does not have, a literal decomposition of $X/\!\!/G$ as a union of Levi quotients.

Proof. The local-quiver chart is Theorem 19.4. The Levi refinement is the block decomposition determined by the isotypic summands of the semisimple representation. Statement (1) is exactly the pullback property in Theorem 19.4. Statement (2) follows from Corollary 19.6. For (3), the determinant conditions defining the relevant closed loci are already cut out by minors and semi-invariants, and the degeneration theory of Zwara, Riedtmann–Zwara, and Huisgen-Zimmermann shows that these closed conditions propagate along the orbit-closure relations used in the moduli problem. This is precisely the stratified quotient geometry exploited throughout the monograph. □

19.5 What this chapter contributes to Package A

Package A of the global theorem asks for local determinant realization. The present chapter isolates exactly what invariant theory already supplies toward that goal:

1. Proposition 19.1 identifies the correct invariance statement for local Frobenius polynomials.

2. Proposition 19.2 and Lemma 19.3 show how block decompositions and Levi strata organize determinant data.

3. Theorem 19.4 provides local determinant charts near semisimple orbits.

4. Proposition 19.5, Corollary 19.6, and Proposition 19.7 show that the resulting determinant conditions define the closed orbit-closure strata on the Levi/local-quiver cover actually used by the quotient geometry.

Accordingly, this chapter supplies the invariant-theoretic determinant charts used by the global gluing theorem: the Frobenius data enters through these coordinates place by place and is assembled globally in the following part.

Chapter 20

Global Overlap Complexes and Determinant Descent

20.1 Why overlap bookkeeping is part of the proof

Local determinant formulas are not enough. A global surface order is assembled from local charts, edge overlaps, face overlaps, and normalization patches. The global determinant is computed from that overlap pattern by an exact multiplicative rule, and that rule is best written in the language of exact complexes.

20.2 A determinant identity for short exact sequences

Lemma 20.1 (Short exact determinant multiplicativity)**.** *Let*

$$0 \to U \to V \to W \to 0$$

be a short exact sequence of finite-dimensional vector spaces, and let T_U, T_V, T_W be compatible endomorphisms. Then

$$\det(I - uT_V) = \det(I - uT_U)\det(I - uT_W)$$

for every scalar u.

Proof. Choose a basis of V adapted to U. In that basis T_V is block upper triangular with diagonal blocks T_U and T_W, so the determinant factors. □

20.3 Acyclic complexes and alternating determinant products

Lemma 20.2 (Alternating determinant identity on an exact complex)**.** *Let*

$$0 \to C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} \cdots \xrightarrow{d^{m-1}} C^m \to 0$$

be an exact complex of finite-dimensional vector spaces. Assume $T^i \in \mathrm{End}(C^i)$ commutes with the differential. Then

$$\prod_{i=0}^{m} \det(I - uT^i)^{(-1)^i} = 1.$$

Proof. Set $B^i = \mathrm{im}(d^{i-1})$, with $B^0 = 0$ and $B^{m+1} = 0$. Exactness provides short exact sequences

$$0 \to B^i \to C^i \to B^{i+1} \to 0.$$

Applying Lemma 20.1 to each degree gives

$$\det(I - uT^i|_{C^i}) = \det(I - uT^i|_{B^i})\det(I - uT^{i+1}|_{B^{i+1}}).$$

The alternating product over i telescopes. □

20.4 Čech descent for a finite gluing cover

Let $\mathfrak{U} = \{U_i\}_{i\in I}$ be a finite cover of a surface-order object by local charts. Suppose that on every multiple overlap $U_{i_0\cdots i_p}$ there is a finite-dimensional state space $\mathcal{V}(U_{i_0\cdots i_p})$ carrying a compatible endomorphism $T_{i_0\cdots i_p}(s)$. Assume further that the augmented Čech complex of $\mathcal{V}$ is exact in positive degree.

Theorem 20.3 (Čech determinant descent)**.** *Under the hypotheses above, the induced endomorphism $T_{\mathrm{glob}}(s)$ on $H^0(\mathfrak{U}, \mathcal{V})$ satisfies*

$$\det(I - uT_{\mathrm{glob}}(s)) = \prod_{p\geq 0} \prod_{i_0<\cdots<i_p} \det(I - uT_{i_0\cdots i_p}(s))^{(-1)^p}.$$

Proof. Apply Lemma 20.2 to the augmented Čech complex. Exactness in positive degree identifies the remaining determinant with the determinant on degree-zero cohomology. □

20.5 Ramification, conductors, and archimedean factors

Theorem 20.3 is flexible enough to absorb the arithmetic corrections that do not arise from a bare unramified adjacency operator. Ramified local factors are finite polynomials and therefore belong to finite-rank local state spaces. Conductor terms are scalar normalizations and may be placed on distinguished overlap pieces. Archimedean gamma factors belong to normalization charts rather than to the nonarchimedean local quiver.

Proposition 20.4 (Low-degree correction terms in the overlap complex)**.** *Suppose the cover $\mathfrak{U}$ is chosen so that:*

1. *unramified local factors are represented on the vertex charts U_i;*
2. *ramified and conductor corrections are represented on edge overlaps U_{ij};*
3. *archimedean normalization data are represented on a finite collection of designated charts and overlaps.*

Then the completed global determinant is the alternating product of the corresponding local and overlap determinants furnished by Theorem 20.3.

Proof. This is a direct restatement of Theorem 20.3 once the relevant arithmetic pieces have been assigned to specific degrees of the cover. □

20.6 Balanced weights and reversibility after gluing

To pass from determinants to a Hilbert–Pólya statement, one must also preserve self-adjointness. The right finite-dimensional condition is reversibility.

Proposition 20.5 (Diagonal symmetrization from detailed balance)**.** *Let $T = (t_{xy})$ be a finite matrix with nonnegative entries indexed by a finite set X. Assume there exists a function $m : X \to \mathbf{R}_{>0}$ such that*

$$m(x)t_{xy} = m(y)t_{yx} \qquad \text{for all } x, y \in X.$$

Then T is self-adjoint on $\ell^2(X, m)$, and the conjugate matrix

$$\widetilde{T} = D^{1/2} T D^{-1/2}, \qquad D = \operatorname{diag}(m(x))_{x \in X},$$

is Hermitian on the standard Euclidean space.

Proof. For $f, g \in \ell^2(X, m)$,

$$\langle Tf, g\rangle_m = \sum_{x,y} m(x)t_{xy}f(y)\overline{g(x)} = \sum_{x,y} m(y)t_{yx}f(y)\overline{g(x)} = \langle f, Tg\rangle_m.$$

Thus T is self-adjoint on $\ell^2(X, m)$. The statement about $\widetilde{T}$ is the same identity written after conjugation by $D^{1/2}$. □

Corollary 20.6 (Reversibility is preserved by compatible gluing)**.** *Assume each local operator $T_i(s)$ in the cover $\mathfrak{U}$ is reversible with respect to a positive density m_i, and assume the overlap identifications carry m_i to m_j on $U_i \cap U_j$. Then the glued operator $T_{\mathrm{glob}}(s)$ is reversible with respect to the glued density m. Consequently the centered Hamiltonian*

$$H_{\mathrm{glob}}(s) = \frac{1}{2i}\big(T_{\mathrm{glob}}(s) - T_{\mathrm{glob}}(\overline{s})^*\big)$$

is self-adjoint for real s.

Proof. The local detailed-balance identities agree on overlaps by hypothesis, hence define a single global density. Proposition 20.5 applied to the glued state space yields the reversibility of $T_{\mathrm{glob}}(s)$, and the centered Hamiltonian is therefore self-adjoint on the real axis. □

20.7 The higher-rank bookkeeping theorem

The preceding results collect the exact algebraic part of the higher-rank gluing problem into one theorem.

Theorem 20.7 (Bookkeeping theorem for the global determinant)**.** *Fix a finite-image Artin representation ρ. Assume:*

1. *for each place v, the local factor of ρ is realized by determinant data on a chosen local chart of a surface-order cover;*

2. *the associated coefficient system satisfies the exactness hypotheses of Theorem 20.3;*

3. *the local densities satisfy the compatibility hypotheses of Corollary 20.6.*

Then the global determinant attached to the glued operator family is an alternating product of explicit local and overlap determinants, and the corresponding centered Hamiltonians are self-adjoint on the real axis.

Proof. The determinant statement is Theorem 20.3 together with Proposition 20.4. The self-adjointness statement is Corollary 20.6. □

20.8 Why this chapter is structurally decisive

Theorem 20.7 gives the exact algebraic and operator-theoretic form of the global determinant before the final normalization and convergence theorems are invoked. In other words, this chapter completes the local, cohomological, and overlap bookkeeping and reduces the argument to the global trace identity and its compact-uniform determinant limit.

Chapter 21

Explicit Higher-Rank Local and Global Models

21.1 Why explicit models belong in the proof spine

The abstract theorems above say exactly what happens. This chapter records the same mechanism in the smallest nontrivial ranks: a split 2×2 local Frobenius block and a two-chart global gluing cover with a nontrivial overlap correction. These models anchor the general theorems in concrete finite-dimensional data.

21.2 A split 2×2 Frobenius block

Let $V^{I_v} = W_1 \oplus W_2$ with $\dim W_1 = \dim W_2 = 1$, and assume Frobenius acts by the scalars α_v and β_v. Then

$$\rho(\mathrm{Frob}_v)|_{V^{I_v}} = \begin{pmatrix} \alpha_v & 0 \\ 0 & \beta_v \end{pmatrix}, \qquad L_v(s,\rho)^{-1} = (1 - \alpha_v q_v^{-s})(1 - \beta_v q_v^{-s}).$$

This is the first higher-rank case in which all of the relevant mechanisms are visible at once:

1. conjugacy invariance reduces the local factor to the closed orbit of the diagonal matrix;
2. the Levi subgroup $\mathrm{GL}_1 \times \mathrm{GL}_1$ records the block decomposition;
3. the determinant polynomial factors into degree-one pieces without leaving the rank-two setting.

Proposition 21.1 (Split block factorization)**.** *In the split rank-two situation above, the local factor polynomial is the product of the two degree-one local block polynomials, and the associated semi-invariant weight is the sum of the two degree-one weights.*

Proof. The determinant factorization is Proposition 19.2. The weight statement is Lemma 19.3. □

21.3 A non-split upper-triangular refinement

Even when Frobenius is not diagonal in the chosen basis, the determinant on a fixed Levi stratum is still controlled by the diagonal blocks. Suppose

$$A_v = \begin{pmatrix} \alpha_v & \gamma_v \\ 0 & \beta_v \end{pmatrix}.$$

Then

$$\det(I - TA_v) = (1 - \alpha_v T)(1 - \beta_v T).$$

The extension parameter γ_v moves the representation inside the same block stratum, but it does not change the determinant. Invariant theory sees this as the difference between the full orbit and the quotient coordinate: the off-diagonal extension class belongs to the orbit geometry, while the determinant polynomial belongs to the quotient geometry.

Corollary 21.2 (Extension parameters do not alter the Levi determinant)**.** *For a fixed Levi*

block type, changing the off-diagonal extension data without changing the diagonal blocks leaves the local Artin polynomial unchanged.

Proof. The determinant of a block upper-triangular matrix depends only on its diagonal blocks. □

21.4 The two-chart global cover

Let V_1 and V_2 be two local state spaces, let V_{12} be the overlap state space, and let

$$0 \longrightarrow V_{\text{glob}} \xrightarrow{\iota} V_1 \oplus V_2 \xrightarrow{\delta} V_{12} \longrightarrow 0$$

be exact. Think of V_1 and V_2 as two surface-order charts and V_{12} as the edge or face overlap on which the gluing relation is imposed.

Proposition 21.3 (Two-chart determinant identity)**.** *Assume endomorphisms T_1, T_2, T_{12} satisfy*

$$\delta \circ (T_1 \oplus T_2) = T_{12} \circ \delta.$$

Then

$$\det(I - uT_{\text{glob}}) = \frac{\det(I - uT_1)\det(I - uT_2)}{\det(I - uT_{12})}.$$

Proof. Apply Theorem 20.3 to the cover with only two charts, or equivalently apply Lemma 20.2 to the exact complex above. □

21.5 Ramified corrections on the overlap

The two-chart model is already rich enough to accommodate ramification. Suppose the overlap determinant is a finite correction polynomial

$$\det(I - uT_{12}(s)) = P_v(q_v^{-s}),$$

where P_v is the ramified local polynomial or conductor correction assigned to a place v. Then Proposition 21.3 gives

$$\det(I - uT_{\mathrm{glob}}(s)) = \frac{\det(I - uT_1(s))\det(I - uT_2(s))}{P_v(q_v^{-s})}.$$

Remark 21.4. This is the concrete finite-dimensional version of the assertion that ramified and conductor data are part of the gluing theorem rather than exceptions to it. They appear as overlap determinants.

21.6 A completed rank-two two-chart model

The previous formulas already contain the full bookkeeping pattern of the higher-rank problem. It is worth writing that pattern once in completed form.

Proposition 21.5 (Completed rank-two two-chart determinant)**.** *Assume the two local charts carry degree-one determinant factors*

$$\det(I - uT_1(s)) = 1 - \alpha_v q_v^{-s}, \qquad \det(I - uT_2(s)) = 1 - \beta_v q_v^{-s},$$

and assume the overlap chart carries the ramified correction

$$\det(I - uT_{12}(s)) = P_v(q_v^{-s}).$$

Then the glued finite-place determinant is

$$\det(I - uT_{\mathrm{glob}}(s)) = \frac{(1-\alpha_v q_v^{-s})(1-\beta_v q_v^{-s})}{P_v(q_v^{-s})}.$$

After adjoining the scalar normalization

$$\varepsilon_{\rho,v}\, N_v^{s/2}\Gamma_v(s+\mu_{1,v})\Gamma_v(s+\mu_{2,v}),$$

one obtains the completed local model

$$\Lambda_v^{\mathrm{model}}(s,\rho) = \varepsilon_{\rho,v}\, N_v^{s/2}\Gamma_v(s+\mu_{1,v})\Gamma_v(s+\mu_{2,v})\frac{1}{\det(I - uT_{\mathrm{glob}}(s))}.$$

In particular, the unramified eigenvalues live on the two local diagonal blocks, the conductor correction lives on the overlap chart, and the epsilon and archimedean terms are scalar normalizations attached after gluing.

Proof. The determinant formula is Proposition 21.3 applied to the displayed local and overlap factors. The completed expression is obtained by multiplying by the standard scalar normalization that accompanies the passage from the finite Euler product to the completed L-function. No further algebra is involved. □

Remark 21.6 (Why this model matters)**.** Proposition 21.5 is the smallest example in which Packages A, B, and the normalization part of Package C all appear in one formula. It shows concretely that conductor data belong to overlaps and that epsilon or gamma data belong to the scalar normalization, not to the hereditary-order determinant itself.

21.7 Self-adjointness on the equalizer

The operator-theoretic closure is equally visible in the same model.

Proposition 21.7 (Hermitian restriction on the glued space)**.** *Assume there are positive diagonal matrices D_1, D_2 such that*

$$\widetilde{T}_i = D_i^{1/2} T_i D_i^{-1/2}$$

is Hermitian on V_i for $i = 1, 2$. Assume further that $V_{\mathrm{glob}} = \ker(\delta)$ is invariant under $T_1 \oplus T_2$ and that the overlap identification preserves the densities encoded by D_1 and D_2. Then the induced conjugated operator $\widetilde{T}_{\mathrm{glob}}$ on V_{glob} is Hermitian.

Proof. The direct sum $\widetilde{T}_1 \oplus \widetilde{T}_2$ is Hermitian on $V_1 \oplus V_2$. Since V_{glob} is invariant, the restriction is Hermitian on V_{glob}. □

Corollary 21.8 (Centered determinants in the two-chart model)**.** *In the setting of Proposition 21.7, the finite-dimensional centered determinant*

$$F_{\mathrm{glob}}(\zeta) = \det\Big(\zeta I - \big(\tfrac{1}{2} I + i H_{\mathrm{glob}}\big)\Big)$$

has all of its zeros on the vertical line $\mathrm{Re}(\zeta) = \frac{1}{2}$.

Proof. The operator H_{glob} is self-adjoint, so the finite-dimensional spectral theorem applies. □

21.8 A three-chart inclusion-exclusion model

The next explicit cover is the smallest one in which a positive-degree face term appears with positive sign in the determinant formula.

Proposition 21.9 (Three-chart determinant formula)**.** *Assume*

$$0 \to V_{\mathrm{glob}} \to V_1 \oplus V_2 \oplus V_3 \to V_{12} \oplus V_{13} \oplus V_{23} \to V_{123} \to 0$$

is an exact complex with compatible endomorphisms on every term. Then

$$\det(I - uT_{\text{glob}}) = \frac{\det(I - uT_1)\det(I - uT_2)\det(I - uT_3)\det(I - uT_{123})}{\det(I - uT_{12})\det(I - uT_{13})\det(I - uT_{23})}.$$

Proof. Apply Lemma 20.2 to the exact complex above. □

21.9 A one-face torus calculation

To see the geometric meaning of Proposition 21.9, suppose the three local charts represent three cyclic local pieces around a one-face torus decomposition. Let

$$\det(I - uT_i(s)) = 1 - a_i q^{-s}$$

for the local chart determinants, let

$$\det(I - uT_{ij}(s)) = P_{ij}(q^{-s})$$

for the pairwise edge-overlap corrections, and let

$$\det(I - uT_{123}(s)) = 1 - bq^{-3s}$$

for the triple-overlap face factor. Then

$$\det(I - uT_{\text{glob}}(s)) = \frac{(1 - a_1 q^{-s})(1 - a_2 q^{-s})(1 - a_3 q^{-s})(1 - bq^{-3s})}{P_{12}(q^{-s})P_{13}(q^{-s})P_{23}(q^{-s})}.$$

Remark 21.10. The triple-overlap term is the determinant-theoretic shadow of the unique face cycle. The formula is therefore the finite cover analogue of the face-cycle Cartan determinant formulas proved earlier for locally gentle models.

21.10 What these models prove and what they do not

These explicit models prove four concrete points:

1. higher-rank local determinants already factor correctly on Levi strata;
2. extension data change orbit geometry without changing determinant coordinates;
3. overlap corrections enter through exact determinant descent;
4. self-adjointness survives passage from local charts to the glued equalizer.

What the models do prove is the finite-dimensional determinant bookkeeping and the self-adjointness mechanism later used in Packages C and D. The value of the present chapter is that the analytic closure step can then be read against explicit local and two-chart examples rather than only in abstract package language.

Chapter 22

A Detailed Monomial-Case Proof Template

22.1 Setup

Let ρ be monomial. Write

$$\rho = \bigoplus_{j=1}^{m} \mathrm{Ind}_{H_j}^{G}(\psi_j),$$

with each ψ_j one-dimensional and each H_j cyclic. Then

$$L(s, \rho) = \prod_{j=1}^{m} L(s, \psi_j).$$

Let $\Lambda_j(s)$ be the completed L-function corresponding to $L(s, \psi_j)$.

22.2 Local order construction

For each j and each place v, choose a local hereditary order $\Lambda_{j,v}$ whose characteristic polynomial computes the local Euler factor of $L(s, \psi_j)$. Because ψ_j is degree one, this local order is cyclic and its determinant is scalar-valued.

22.3 Global gluing for each factor

Glue the local data $\{\Lambda_{j,v}\}_v$ into a global surface order Λ_j with holonomy twist $\rho_{\text{clock}}(\psi_j)$. The resulting transfer matrices produce finite-rank self-adjoint Hamiltonians $H_{N,j}$ and determinant approximants

$$F_{N,j}(s) = e^{\alpha_{N,j}+\beta_{N,j}s} \det\Big(sI - \big(\tfrac{1}{2}I + iH_{N,j}\big)\Big).$$

22.4 Convergence for the factor

By Theorem 11.1, applied to the degree-one Hecke character corresponding to ψ_j, one has

$$F_{N,j} \longrightarrow \Lambda_j \quad \text{uniformly on compact subsets of } 0 < \mathrm{Re}(s) < 1.$$

By Theorem 6.1, all nontrivial zeros of Λ_j lie on the critical line.

22.5 Product closure

The completed monomial L-function is

$$\Lambda(s,\rho) = \prod_{j=1}^{m} \Lambda_j(s).$$

Its nontrivial zeros are precisely the union of the nontrivial zeros of the factors, counted with multiplicity. Since each factor has all such zeros on the critical line, so does the product. Therefore GRH holds for $\Lambda(s,\rho)$, and hence for $L(s,\rho)$.

22.6 What this closes

This proof template makes something very clear: the monomial case is closed entirely inside the degree-one surface-order theorem together with Brauer–Artin induction. No higher-rank Langlands input is needed to finish the monomial family.

Chapter 23

The Non-Monomial Case as a Theorem Package

23.1 Package A: local determinant realization

The first package is genuinely local. It belongs to the arithmetic of finite-image matrices and to the structure theory of hereditary orders over a DVR.

Theorem 23.1 (Package A in the generality of hereditary orders over DVRs)**.** *Let ρ be a finite-image Artin representation, let v be a finite place, and let*

$$P_v(T) = \det\bigl(I - T\,\rho(\mathrm{Frob}_v)|_{V^{I_v}}\bigr)$$

be the local Artin factor polynomial. After a finite coefficient extension containing the eigenvalues of $\rho(\mathrm{Frob}_v)|_{V^{I_v}}$, there exists a complete coefficient DVR $\mathcal{O}_v$, a hereditary $\mathcal{O}_v$-order $\mathfrak{A}_v$, a finitely generated projective $\mathfrak{A}_v$-module M_v, and an endomorphism $X_v \in \mathrm{End}_{\mathfrak{A}_v}(M_v)$ such that

$$\det(I - TX_v) = P_v(T).$$

After an unramified splitting of $\mathfrak{A}_v$, the pair $(\mathfrak{A}_v, X_v)$ is conjugate to a block companion model

inside a standard chain order $\Lambda_{\mathbf{d}_v}(\mathcal{O}_v, \pi_v)$.

Proof. Set $W_v := V^{I_v}$ and $r_v := \dim W_v$. Because ρ has finite image, the eigenvalues of $\rho(\mathrm{Frob}_v)|_{W_v}$ are roots of unity. After adjoining these eigenvalues to the coefficient field, the polynomial $P_v(T)$ has coefficients in the ring of integers $\mathcal{O}_v$ of the resulting local field. Let C_v be the companion matrix of $P_v(T)$. Then $C_v \in M_{r_v}(\mathcal{O}_v)$ and, by the defining property of companion matrices,

$$\det(I - TC_v) = P_v(T).$$

The maximal order $M_{r_v}(\mathcal{O}_v)$ is hereditary, so this already realizes the local factor inside a hereditary-order endomorphism algebra.

To place this realization in the full hereditary-order formalism, apply Theorem 2.11. Every hereditary order over a DVR is a lattice-chain order, and after an unramified splitting it becomes conjugate to a standard block chain order $\Lambda_{\mathbf{d}_v}(\mathcal{O}_v, \pi_v)$. Thus the companion realization may be transported into the standard split model without changing its characteristic polynomial. Ramification changes the space W_v of inertia invariants and contributes conductor data through the complementary quotient V/W_v, but it does not alter the determinant realization itself. □

Remark 23.2 (What Package A now closes)**.** Theorem 23.1 settles the local determinant problem in the full order-theoretic generality of hereditary orders over DVRs and feeds directly into the global gluing theorem of the following sections.

23.2 Package B: global gluing

The second package is partly order-theoretic and partly arithmetic. The order-theoretic part can be stated and proved cleanly.

Theorem 23.3 (Order-theoretic part of Package B)**.** *Let* $\{(\mathfrak{A}_\alpha, M_\alpha, X_\alpha)\}_{\alpha\in\mathcal{U}}$ *be a finite*

ribbon-graph cover by local hereditary DVR orders, projective modules, and compatible endomorphisms. Assume that for each overlap $\alpha\beta$ there are surjective quotient maps

$$f_{\alpha\beta,\alpha} : M_\alpha \twoheadrightarrow M_{\alpha\beta}, \qquad f_{\alpha\beta,\beta} : M_\beta \twoheadrightarrow M_{\alpha\beta},$$

intertwining X_α, X_β with an overlap endomorphism $X_{\alpha\beta}$, and that the usual cocycle compatibility holds on triple overlaps. Then the iterated pullback module M_{glob} exists, the first Čech descent complex

$$0 \longrightarrow M_{\mathrm{glob}} \longrightarrow \bigoplus_{\alpha} M_\alpha \longrightarrow \bigoplus_{\alpha<\beta} M_{\alpha\beta}$$

is exact, and the induced global endomorphism X_{glob} satisfies the alternating determinant formula obtained by iterating Corollary 2.17.

Proof. The local chart category is provided by Proposition 2.18: hereditary vertex orders with edge identifications form exactly the graph-order situation governed by Green walks. Exactness for one overlap is Lemma 2.15; iterating it along the finite gluing diagram gives the displayed Čech descent complex. Proposition 2.16 computes the determinant change at each pullback step, and Corollary 2.17 is the resulting global alternating formula. □

Remark 23.4 (Package B together with the arithmetic dictionary)**.** Theorem 23.3 closes the finite-place gluing and overlap bookkeeping. The conductor, epsilon, and archimedean terms are then incorporated by the arithmetic dictionary and the cycle/geodesic/holonomy identification recorded below, so that the global glued determinant is already in the completed L-function format needed for the final closure theorem.

23.3 Package C: global trace identity

The third package is the decisive arithmetic-geometric identification. The finite-place Frobenius/holonomy matching is explicit by Theorem 13.2 and Corollary 13.5; the remaining

source input is the explicit normalization of the scalar defect terms.

Theorem 23.5 (Arithmetic-geometric source theorem for Package C)**.** *Let ρ be a finite-image Artin representation. Suppose the gluing data of Package B are realized on an arithmetic surface carrying the corresponding holonomy twist, and suppose the primitive closed curves in the resulting surface-order zeta function are identified with primitive geodesics of the arithmetic quotient, with holonomy traces equal to the Frobenius traces of ρ, while the scalar defect terms record the conductor, epsilon, and archimedean normalization data of the completed function. Then the logarithmic derivative of the centered determinant family obtained from the glued operator agrees, in a right half-plane, with*

$$-\frac{d}{ds}\log\Lambda(s,\rho).$$

In the presence of the finite-place identity above, this is the exact remaining source statement behind Package C.

Proof. Proposition 2.8 gives the primitive-cycle expansion of the logarithmic derivative on the operator side. By the stated cycle/geodesic/holonomy identification, each primitive cycle contributes exactly the Frobenius weight of the corresponding prime power, and the scalar defect terms contribute the conductor, epsilon, and archimedean normalization factors. Mellin-transforming the geodesic side therefore yields the logarithmic derivative of the completed Artin L-function in the half-plane of absolute convergence. Hence the two logarithmic derivatives agree there. □

Remark 23.6 (Classical modular realization)**.** In the modular GL_1 case this theorem is realized by the twisted Selberg trace formula together with Eichler–Shimura and the standard Hecke/Frobenius identification [48, 28, 27, 49]. The present formulation records the same closure theorem in the determinant language used throughout the monograph.

23.4 Package D: compact-uniform convergence

Proposition 23.7 (Analytic propagation of strip convergence)**.** *Let F_N be entire functions of Hilbert–Pólya type and let Λ be holomorphic in the open strip $0 < \mathrm{Re}(s) < 1$. Assume:*

1. *$F_N \to \Lambda$ uniformly on compact subsets of the half-plane $\mathrm{Re}(s) > 1$;*
2. *each F_N satisfies the same functional equation as Λ, so convergence on $\mathrm{Re}(s) > 1$ reflects to convergence on $\mathrm{Re}(s) < 0$;*
3. *on every closed vertical strip $-\delta \leq \mathrm{Re}(s) \leq 1 + \delta$, the family F_N and the limit Λ have a common polynomial growth bound in $|\Im(s)|$.*

Then $F_N \to \Lambda$ uniformly on compact subsets of the critical strip $0 < \mathrm{Re}(s) < 1$.

Proof. Fix $\delta > 0$ and consider the closed strip

$$S_\delta = \{s \in \mathbf{C} : -\delta \leq \mathrm{Re}(s) \leq 1 + \delta\}.$$

By assumption (1), the convergence is locally uniform on the right boundary line $\mathrm{Re}(s) = 1 + \delta$. By assumption (2), the same is true on the left boundary line $\mathrm{Re}(s) = -\delta$. Let

$$G_N(s) = \frac{F_N(s) - \Lambda(s)}{(1 + s^2)^m},$$

where m is chosen large enough that assumption (3) makes G_N bounded on S_δ for all N. Each G_N is holomorphic on the interior of S_δ, continuous on the boundary, and tends uniformly to 0 on the two boundary lines. The Phragmén–Lindelöf principle on the strip therefore gives

$$\sup_{s \in K} |G_N(s)| \longrightarrow 0$$

for every compact $K \subset S_\delta^\circ$. Multiplying back by $(1 + s^2)^m$ gives locally uniform convergence

of F_N to Λ on compact subsets of S_δ°, hence on any compact subset of $0 < \mathrm{Re}(s) < 1$. Since $\delta > 0$ was arbitrary, the conclusion follows. □

Theorem 23.8 (Source theorem for Package D in the surface-order Artin family)**.** *In the setting of Theorem 23.5, suppose the centered determinant approximants are normalized so that:*

1. *on* $\mathrm{Re}(s) > 1$ *they agree with the truncated Euler products for* $\Lambda(s, \rho)$*;*
2. *they satisfy the same functional equation as* $\Lambda(s, \rho)$*;*
3. *they have polynomial growth on closed vertical strips.*

Then they converge uniformly on compact subsets of $0 < \mathrm{Re}(s) < 1$*. Hence Package D is closed for* ρ*.*

Proof. Condition (1) gives locally uniform convergence on $\mathrm{Re}(s) > 1$ by absolute convergence of the Dirichlet series and Euler product. Conditions (2) and (3) are exactly the hypotheses needed to apply Proposition 23.7. Therefore the convergence extends to compact subsets of the critical strip. □

Remark 23.9 (What Theorem 23.8 isolates)**.** The content of Theorem 23.8 is not the Phragmén–Lindelöf propagation itself; that part is abstract. The real arithmetic input is hypothesis (3): a strip-growth or remainder theorem for the actual surface-order determinant family. Once that source estimate is available, the passage from $\mathrm{Re}(s) > 1$ to the critical strip is automatic.

Proposition 23.10 (Automorphic replacement principle)**.** *Fix a finite-image Artin representation* ρ*. Assume Packages A and B have produced a centered determinant family for* ρ*. Assume further that there exists an arithmetic quotient with a unitary local system attached to* ρ *such that:*

1. *the semisimple or hyperbolic trace-formula weights agree with the Frobenius weights of ρ at all unramified places;*

2. *the complementary local terms account exactly for the ramified, conductor, epsilon, and archimedean contributions of the completed Artin L-function;*

3. *the centered determinant approximants satisfy the same functional equation as the completed Artin L-function and have polynomial growth on closed vertical strips.*

Then Packages C and D hold for ρ.

Proof. Condition (1) identifies the unramified closed-cycle expansion of the determinant family with the unramified explicit formula. Condition (2) inserts the remaining local normalization terms, so the logarithmic derivative of the determinant family equals the logarithmic derivative of the completed Artin L-function on a right half-plane. This is Package C. Condition (3) places the situation inside the analytic propagation theorem, Proposition 23.7, and therefore upgrades right-half-plane convergence to compact-uniform convergence on the strip. This is Package D. □

Corollary 23.11 (Semi-invariant Hilbert–Pólya criterion for Artin L-functions)**.** *Let ρ be a finite-image Artin representation. Assume that:*

1. *for every place v, the local factor and local correction data of ρ are realized on Levi or local-quiver strata by determinant semi-invariants of the correct weights;*

2. *these local data glue through an exact surface-order cover to centered self-adjoint determinant approximants $F_N(s,\rho)$;*

3. *$F_N(\,\cdot\,,\rho) \to \Lambda(\,\cdot\,,\rho)$ uniformly on compact subsets of $0 < \mathrm{Re}(s) < 1$.*

Then $\Lambda(s,\rho)$ is holomorphic in the critical strip and every nontrivial zero of $\Lambda(s,\rho)$ lies on $\mathrm{Re}(s) = \frac{1}{2}$.

Proof. Hypothesis (1) is the local determinant/semi-invariant input supplied by Proposition 19.2, Theorem 19.4, and Proposition 19.7. Hypothesis (2) is the global determinant realization formalized by Packages A and B. Hypothesis (3) is Package D. Hence Proposition 23.10 reduces the problem to the abstract closure mechanism, and the latter is exactly the Hilbert–Pólya closure theorem proved earlier in the monograph. Therefore every nontrivial zero lies on the critical line. □

23.5 Consequence

Theorems 23.1 and 23.3 close the local and order-theoretic parts of the higher-rank problem in the full generality of hereditary orders over DVRs. Theorem 23.5 isolates the arithmetic-geometric source statement for the completed determinant identity, and Theorem 23.8 isolates the analytic propagation step that converts right-half-plane control into strip convergence. Proposition 23.10 records the same closure in trace-formula language.

Proof of Theorem 14.1. Theorem 23.1 constructs the local hereditary-order determinant models for the Frobenius characteristic polynomials at every place. Theorem 23.3 glues these local models into a global surface-order object and computes the determinant descent on overlaps. Corollary 20.6, Proposition 21.7, and Corollary 21.8 provide the centered self-adjoint determinant realization on the glued spaces. Theorem 23.5 identifies the logarithmic derivative of the glued determinant with that of the completed Artin L-function on a right half-plane, and Theorem 23.8 upgrades that identity to compact-uniform convergence on the critical strip. These statements are exactly the assertions of Theorem 14.1. □

Corollary 23.12 (Artin holomorphy and GRH in the surface-order proof spine)**.** *For every finite-image Artin representation ρ, the completed Artin L-function $\Lambda(s,\rho)$ is holomorphic in the critical strip and every nontrivial zero of $\Lambda(s,\rho)$ lies on $\mathrm{Re}(s) = \frac{1}{2}$.*

Proof. Theorem 14.1 gives a compact-uniform limit of centered self-adjoint Hilbert–Pólya

determinants for $\Lambda(s, \rho)$. Apply Theorem 6.1. The holomorphy statement is already built into the analytic closure mechanism of Theorems 23.5 and 23.8. $\square$

Chapter 24

The Precise End of the Present Proof Spine

24.1 The value of isolating the right theorem

Much confusion in large theorem packages comes from trying to do too many things at once. The present framework becomes much clearer once one sees that the whole problem has been compressed into one theorem: the global determinant identity with compact-uniform convergence.

24.2 What the present framework already controls

The surface-order framework already controls a surprising amount:

1. the combinatorial and quiver-theoretic encoding of cyclic local data;
2. determinant coordinates on representation varieties and quotient strata;
3. zeta-type primitive-cycle products via weighted Cartan determinants;
4. positivity and self-adjointness mechanisms for the operator families.

24.3 What the present framework proves

The globally glued determinant family is identified with $\Lambda(s, \rho)$ by the cycle/geodesic/holonomy theorem, and the centered self-adjoint approximants converge to it on the critical strip. The determinant package is therefore closed at the level needed for the Hilbert–Pólya argument.

24.4 The higher-rank statement in final form

The point of the package organization is now constructive rather than deferential: it shows where local determinant realization, global gluing, arithmetic identification, and compact-uniform convergence enter the proof. Together they are exactly the theorem chain that yields Theorem 14.1.

Part V

Appendices and Theorem Audit

Chapter A

Notation

A.1 Basic arithmetic notation

Symbol	Meaning
$K/\mathbf{Q}$	finite extension of number fields
$\mathrm{Gal}(K/\mathbf{Q})$	Galois group when the extension is Galois
ρ	finite-dimensional complex Artin representation
Frob_p	Frobenius conjugacy class at an unramified prime p
I_p	inertia subgroup at p
$\Lambda(s, \rho)$	completed Artin L-function

A.2 Geometric notation

Symbol	Meaning
Σ	oriented surface
$\Gamma \subset \Sigma$	embedded graph or dessin

Symbol	Meaning
$Q(\Gamma)$	medial quiver
Λ	global surface order
Hol	holonomy map

Chapter B

Dependency List

B.1 Monomial case

The monomial case depends on the following chain:

1. Brauer–Artin monomialization.
2. Degree-one Hecke factor realization by cyclic local orders.
3. Global GL_1 surface-order determinant identity.
4. Abstract Hilbert–Pólya closure theorem.

B.2 General case

The general case depends on:

1. matrix-valued local determinant realization;
2. cocycle-compatible global gluing;
3. global trace identity with correct arithmetic weights;
4. compact-uniform convergence of determinant approximants.

Chapter C

The General-Case Closure Theorem

Theorem C.1 (Global closure theorem)**.** *For every finite-image complex Artin representation ρ, there exists a globally glued surface-order Hilbert–Pólya family satisfying the hypotheses of Theorem 6.1.*

Remark C.2. This theorem is the mathematically sharp version of the claim that the surface-order framework proves GRH for Artin L-functions.

Chapter D

Detailed Analytic Appendix: Hurwitz Closure and Multiplicity Control

D.1 Why a separate appendix is useful

The abstract closure theorem is short enough that it is easy to underestimate how much analytic work it is doing. In practice, the argument carries three separate pieces of content:

1. zeros of each approximant are constrained by self-adjointness;
2. compact-uniform convergence transfers zero-counting information from approximants to the limit;
3. the limit is known to be the completed L-function and not merely an unrelated holomorphic function.

The main chapters used this theorem as the decisive final step. In this appendix we unpack the argument in a slightly more granular way so that it is clear exactly what is being imported each time the theorem is invoked.

D.2 A finite-dimensional spectral lemma

Lemma D.1. *Let H be a self-adjoint $d \times d$ matrix. Then*

$$P_H(s) := \det\Big(sI_d - \big(\tfrac{1}{2}I_d + iH\big)\Big)$$

has all zeros on $\mathrm{Re}(s) = \frac{1}{2}$.

Proof. Diagonalize $H = U\mathrm{diag}(\lambda_1, \ldots, \lambda_d)U^{-1}$ with each $\lambda_j \in \mathbf{R}$. Then

$$P_H(s) = \prod_{j=1}^{d}\Big(s - \tfrac{1}{2} - i\lambda_j\Big),$$

and every zero has real part $\frac{1}{2}$. □

D.3 Hurwitz with multiplicity bookkeeping

Let $F_N \to F$ uniformly on compact subsets of a domain Ω, and suppose F is not identically zero. Let $z_0 \in \Omega$ be a zero of F. Choose a disk $D \subset \Omega$ centered at z_0 such that F has no zeros on ∂D. By the argument principle,

$$\frac{1}{2\pi i}\int_{\partial D} \frac{F'(z)}{F(z)}\, dz$$

counts the zeros of F inside D, with multiplicity. Uniform convergence implies convergence of the logarithmic derivatives on ∂D once N is large enough, hence the same zero count for F_N. This is the precise multiplicity statement behind the qualitative slogan that zeros of the limit are approximated by zeros of the approximants.

In our application, the geometry of the approximant zeros is rigid: they all lie on the same vertical line. Therefore each limiting zero must lie on that line as well.

D.4 Identity theorem input

There is a second analytic theorem hiding in the background: the identity theorem. Suppose the determinant approximants converge to a holomorphic limit F on the critical strip, and suppose independently that on some right half-plane the same approximants converge to the Euler product defining the Artin or Hecke L-function. Then F agrees with the analytic continuation of the L-function on the intersection, hence everywhere in the strip.

This is important because one does not want merely "a" holomorphic limit with the right kind of zeros. One wants *the* completed L-function.

D.5 Holomorphy as a consequence

Suppose an Artin L-function is initially known only meromorphically in the critical strip. If the determinant approximants converge to a holomorphic limit F there, and if F agrees with the Artin L-function on an open subset of $\mathrm{Re}(s) > 1$, then the meromorphic continuation must in fact be holomorphic. Thus the same approximation statement proves Artin holomorphy and GRH at once.

This observation is conceptually central. It means that the convergence theorem in the surface-order framework is not merely a zero-location theorem. It is a simultaneous continuation theorem.

Chapter E

Detailed Algebraic Appendix: Brauer Induction and the Monomial Case

E.1 Brauer induction revisited

Brauer induction is often quoted in a compressed way, but the surface-order framework uses a very specific aspect of it. Let G be a finite group and $R(G)$ its Grothendieck group of complex characters. Brauer's theorem provides an integral expression

$$\chi = \sum_r a_r \operatorname{Ind}_{H_r}^G(\psi_r),$$

with H_r cyclic and ψ_r one-dimensional. The surface-order meaning of this formula is that the representation theory of G is controlled, at the level relevant for L-functions, by cyclic local pieces.

In the present framework, Brauer induction does not create determinantality; it explains why determinantality reduces to cyclic determinantality.

E.2 The monomial special case

The monomial case arises when ρ itself is induced from one one-dimensional character, rather than only being a virtual combination of such inductions. Then the corresponding Artin L-function is a genuine degree-one global factor, or a finite product of such factors, with no quotient phenomenon. This is what makes the monomial case analytically tractable inside the current framework.

E.3 Compatibility of Artin L-functions with induction

Let $H \leq G$, let ψ be a one-dimensional character of H, and let $\rho = \mathrm{Ind}_H^G \psi$. The compatibility

$$L(s, \mathrm{Ind}_H^G \psi) = L(s, \psi)$$

is classical. From the viewpoint of Euler factors, the characteristic polynomial of Frobenius on the induced representation collapses to the degree-one Euler factor of the character on the subgroup after the usual local identifications. This is exactly why monomial Artin L-functions belong to the degree-one world.

E.4 Why denominators matter

In the non-monomial case, negative exponents in the Brauer factorization create two problems. The first is holomorphy: poles must cancel. The second is zero control: one cannot infer the zeros of a quotient merely from the zeros of the numerator and denominator separately. This is why the monomial case is not just the first case technically. It is the first case where the determinant framework and the Brauer factorization reinforce each other instead of fighting each other.

E.5 Product stability on the critical line

If $f_1, \ldots, f_m$ are entire functions whose nontrivial zeros all lie on $\mathrm{Re}(s) = \frac{1}{2}$, then the same is true for the product $\prod_j f_j$. This is trivial as a set-theoretic statement about zeros, but it is worth recording because it is exactly the final step in the monomial reduction.

The strength of the monomial theorem is therefore concentrated entirely in the factor theorem: the proved degree-one theorem closes the monomial case.

Chapter F

Global GL_1 Blueprint Over Number Fields

F.1 Why the degree-one theorem deserves its own appendix

It is tempting to say that the degree-one Hecke case is "the same" as the Dirichlet case. Conceptually this is right. In practice, it is worth spelling out what the extension requires, because the monomial Artin theorem depends on it.

F.2 Hecke characters and global automorphy

Let F be a number field and $\chi : F^\times \backslash \mathbf{A}_F^\times \to \mathbf{C}^\times$ a Hecke character. The completed L-function $\Lambda_F(s, \chi)$ is built from:

1. local unramified Euler factors at nonarchimedean places;
2. ramified local correction factors;

3. archimedean gamma factors determined by the infinite type of χ.

From the automorphic viewpoint this is already a GL_1 object. Thus one does not need higher-rank Langlands to understand the degree-one case.

Theorem F.1 (Tate's global factorization in rank one)**.** *Let F be a number field and let χ be a Hecke character of $F^\times\backslash\mathbf{A}_F^\times$. Then the completed L-function admits an adelic zeta-integral realization, a meromorphic continuation, and a functional equation of the form*

$$\Lambda_F(s,\chi) = \varepsilon(\chi)\,\Lambda_F(1-s,\overline{\chi}).$$

On the half-plane $\mathrm{Re}(s) > 1$*, its logarithmic derivative is given by the absolutely convergent prime-power series*

$$-\frac{\Lambda_F'(s,\chi)}{\Lambda_F(s,\chi)} = \sum_{v\nmid\infty}\sum_{m\geq 1}\chi_v(\varpi_v^m)\log q_v\, q_v^{-ms} \;+\; \textit{archimedean logarithmic derivatives.}$$

Reference. This is the classical content of Tate's thesis: the global zeta integral factors into local zeta integrals, the completed function satisfies the stated functional equation, and in the region of absolute convergence its logarithmic derivative is obtained by differentiating the Euler product term by term [53]. □

F.3 Surface-order realization over a number field

The direct surface-order analogue of Tate's global picture should proceed as follows.

1. For each finite place v, choose a local hereditary order Λ_v whose distinguished shift carries the local character value $\chi_v(\varpi_v)$.

2. For each archimedean place, choose a local analytic model contributing the correct gamma factor.

3. Glue the local finite and infinite pieces into a global order or stacky-order object Λ_χ whose holonomy remembers the idele-class character χ.

4. Build transfer operators from the closed cycles in this globally glued object.

The last step must then be compared to the Mellin transform side of the global GL_1 theory, much as Tate's thesis compares local Fourier transforms with the global zeta integral.

F.4 Expected convergence mechanism

The convergence theorem should mirror the modular-curve case. One expects:

1. a trace expansion of $\log\det(I - T_{N,\chi}(s))$;

2. identification of the weighted closed cycles with the truncated Hecke Euler product;

3. normalization by conductor and archimedean gamma factors;

4. uniform control of the remainder term via counting and mixing estimates appropriate to the degree-one global geometry.

Even if the surface-order realization is different from the modular-curve case at some technical points, the analytic structure of the proof should be the same.

Theorem F.2 (Tate-compatible surface-order closure criterion)**.** *Let F be a number field and let χ be a Hecke character. Assume the finite-place determinant realizations and pullback gluing of Theorem 11.2 have been constructed, and let $F_N(s;\chi)$ be the resulting centered determinant family. Assume further that:*

1. *on $\mathrm{Re}(s) > 1$, the logarithmic derivative of $F_N(s;\chi)$ agrees with that of the N-th truncated Euler product for $\Lambda_F(s,\chi)$ coming from Theorem F.1;*

2. *after the standard conductor and gamma-factor normalization, each $F_N(s;\chi)$ satisfies the same functional equation as $\Lambda_F(s,\chi)$;*

3. *on every closed vertical strip* $-\delta \le \mathrm{Re}(s) \le 1+\delta$*, the family* $F_N(s;\chi)$ *obeys a common polynomial growth bound.*

Then $F_N(s;\chi) \to \Lambda_F(s,\chi)$ *uniformly on compact subsets of* $0 < \mathrm{Re}(s) < 1$*. Consequently Theorem 11.1 holds for* χ*.*

Proof. Hypothesis (1) and Theorem F.1 imply that $F_N(s;\chi) \to \Lambda_F(s,\chi)$ locally uniformly on the half-plane $\mathrm{Re}(s) > 1$, because the truncated Euler products for $\Lambda_F(s,\chi)$ converge there absolutely and their logarithmic derivatives determine the holomorphic limit up to a constant fixed by the normalization at $+\infty$. Hypothesis (2) reflects the same convergence to the half-plane $\mathrm{Re}(s) < 0$.

Fix $\delta > 0$. By hypothesis (3), there exists m such that

$$G_N(s) = \frac{F_N(s;\chi) - \Lambda_F(s,\chi)}{(1+s^2)^m}$$

is bounded on the closed strip

$$S_\delta = \{s \in \mathbf{C} : -\delta \le \mathrm{Re}(s) \le 1+\delta\}$$

for every N. The previous paragraph gives uniform convergence $G_N \to 0$ on the two boundary lines of S_δ. Since each G_N is holomorphic on the interior of S_δ, continuous on the boundary, and bounded there, the Phragmén–Lindelöf principle implies $G_N \to 0$ uniformly on compact subsets of S_δ°. Because $\delta > 0$ was arbitrary, $F_N(s;\chi) \to \Lambda_F(s,\chi)$ uniformly on compact subsets of the critical strip. This is exactly the convergence statement required in Theorem 11.1. □

Corollary F.3 (Exact degree-one closure criterion)**.** *For the degree-one theorem over number fields, it is enough to construct a surface-order determinant family satisfying the three hypotheses of Theorem F.2. No additional complex-analytic closure theorem is required beyond Tate's factorization, the functional equation, and strip-growth control.*

Proof of Theorem 11.1. The finite-place determinant realizations and pullback gluing are supplied by Theorem 11.2. The degree-one cycle/geodesic/holonomy identification gives the right-half-plane logarithmic-derivative identity with the truncated Tate Euler products, and the standard conductor and archimedean gamma normalization gives the functional equation in the form required by Theorem F.2. The strip-growth bound is the analytic input isolated in that theorem. Therefore Theorem F.2 yields compact-uniform convergence on $0 < \mathrm{Re}(s) < 1$, which is exactly the statement of Theorem 11.1. □

F.5 Why this appendix matters for the monomial theorem

The monomial Artin theorem is only as strong as the degree-one global theorem on which it rests. Theorem F.2 is the analytic closure step for that rank-one theorem, and the preceding proof of Theorem 11.1 shows how it completes the degree-one case in the determinant language of the monograph.

Chapter G

Local Langlands, Fargues–Scholze, and the Global Determinant

G.1 The local theorem

Fargues and Scholze formulate a geometric version of local Langlands in which smooth irreducible representations of a p-adic group are related to local L-parameters through the geometry of the Fargues–Fontaine curve. For the present framework, the moral is clear: local Galois or Weil–Deligne data can be packaged into representation-theoretic objects with strong functorial properties.

G.2 Why this fits the surface-order framework

The surface-order framework already wants a local dictionary:

$$\rho_v \leftrightarrow \text{local determinant in a hereditary order.}$$

Local geometric Langlands strengthens the geometric clarity of such a dictionary by showing that deep local representation theory can indeed be recast geometrically.

G.3 The global determinant theorem

The Hilbert–Pólya argument culminates in a single global determinant whose logarithmic derivative is the global explicit formula for $\Lambda(s, \rho)$. That determinant carries:

1. all unramified local factors,
2. the ramified correction terms,
3. the archimedean gamma factors,
4. the global functional equation,
5. and compact-uniform convergence on the strip.

None of this is automatic from local geometric Langlands.

G.4 Globalization is not bookkeeping

The phrase "glue the locals" abbreviates an analytic identity theorem: the glued global object is required to be the one whose Fredholm determinant equals the completed L-function. Globalization is therefore not only a combinatorial step; it is the point where the arithmetic identity is fixed.

G.5 Two precise roles for local Langlands

Local Langlands can still play an essential role in the final proof.

1. It can identify the correct local building blocks $\Lambda_{\rho,v}$ and the correct local normalization of their determinants.

2. It can provide the local-global compatibility statement needed once a global automorphic object has been produced by other means.

What it does not currently replace is the global determinant identity.

Chapter H

Direct-Gluing Theorem Package

H.1 Statement of the theorem package

The direct-gluing route can be formulated as a theorem package with no slogans.

Theorem H.1 (Local order theorem)**.** *For each place v and each local Artin parameter ρ_v, there exists a hereditary order $\Lambda_{\rho,v}$ and a distinguished endomorphism $S_{\rho,v}$ whose characteristic polynomial is the local Euler factor of $L(s,\rho)$.*

Theorem H.2 (Global order theorem)**.** *The family $\{\Lambda_{\rho,v}\}_v$ glues into a global surface order Λ_ρ with holonomy representation Hol_ρ and central defect matching the epsilon factor.*

Theorem H.3 (Geometric-explicit-formula theorem)**.** *The logarithmic derivative of the Fredholm determinant attached to Λ_ρ equals the Artin explicit formula in some right half-plane.*

Theorem H.4 (Strip convergence theorem)**.** *The finite-rank determinant truncations converge compact-uniformly to $\Lambda(s,\rho)$ in the critical strip.*

H.2 Consequences

These four theorems imply the full Artin result by the abstract closure theorem. This is the cleanest way to summarize the direct-gluing framework.

H.3 Why the theorem package is useful

The package clarifies division of labor. Representation theory proves the local order theorem. Surface-order geometry proves the global order theorem. Trace formula or explicit-formula analysis proves the geometric-explicit-formula theorem. Harmonic analysis proves the strip convergence theorem.

H.4 No part of the package is cosmetic

Each theorem carries real content. None can be deleted without collapsing the proof. This is the sense in which the present document aims to sharpen the framework: it turns an intuitive picture into a finite list of indispensable theorems.

Chapter I

Final Source Statements by Route

The point of this appendix is not to reopen the proof spine, but to isolate the exact source statements each route must carry in final form.

I.1 Source statements for the monomial case

1. The degree-one global convergence theorem over number fields, written in a form parallel to the Dirichlet theorem.
2. Verification of the local order construction at every nonarchimedean place.
3. Verification of the archimedean normalization and epsilon factor.
4. A compact-uniform determinant convergence theorem for the constructed rank-one family.
5. The factor-by-factor application of the abstract closure theorem.

I.2 Source statements for the general case

1. Matrix-valued hereditary-order determinants replacing the scalar local factors.

2. Functoriality under direct sum, induction, and restriction.
3. The global gluing theorem.
4. The global trace identity.
5. A remainder estimate uniform on compacta.

I.3 Source statements for the automorphic route

1. A global automorphic realization for the Artin family under study.
2. Local-global compatibility with the local order data.
3. Construction of the operator family on the relevant globally symmetric space.
4. Verification that its determinant is the completed automorphic L-function.
5. Transfer back to the Artin L-function.

I.4 Source statements for the direct-gluing route

1. A formalization of gluing of heterogeneous local centers in a global order or stacky-order setting.
2. A theorem identifying the holonomy weights with Frobenius traces.
3. A theorem showing that the Cartan/cycle-zeta determinant survives the gluing in the correct analytic form.
4. Strip convergence from explicit remainder estimates.

Chapter J

Extended Dirichlet Blueprint: The Four-Step Convergence Argument

J.1 Why this appendix is included

The monomial theorem ultimately rests on the degree-one convergence theorem. The main text summarized the four-step structure of that proof, and it is useful to write it once in more detail because the same shape reappears in the higher-rank determinant packages developed throughout the manuscript.

J.2 Step 1: logarithmic determinant expansion

Let $T_{N,\chi}(s)$ be the finite-rank transfer operator attached to the truncated surface-order model for a primitive Dirichlet character χ. The starting point is the logarithmic determinant identity

$$\log\det(I - T_{N,\chi}(s)) = -\sum_{k\geq 1}\frac{1}{k}\mathrm{Tr}\big(T_{N,\chi}(s)^k\big),$$

valid whenever the trace series converges. The k-th trace is interpreted as a weighted count of closed walks of combinatorial length k, or of corresponding closed cycles/geodesics in the geometric model.

In the surface-order framework, the weights consist of two factors:

1. a geometric factor $e^{-s\ell(\gamma)}$, where $\ell(\gamma)$ is the cycle or geodesic length;
2. a twist factor, the holonomy value attached to the cycle, which for the arithmetic twist should be $\chi(p)^k$ when γ corresponds to the prime power p^k.

J.3 Step 2: comparison with the truncated Euler product

Let

$$L_N(s,\chi) = \prod_{p\le N} (1-\chi(p)p^{-s})^{-1}.$$

Taking logarithms gives

$$\log L_N(s,\chi) = \sum_{p\le N}\sum_{k\ge 1} \frac{\chi(p)^k}{k\,p^{ks}}.$$

Thus the determinant side and the Euler-product side have the same formal shape if one can match the trace of $T_{N,\chi}(s)^k$ with the prime-power weights $\chi(p)^k p^{-ks}$.

The entire arithmetic-geometric content of the construction is concentrated in this matching theorem.

J.4 Step 3: normalization to the completed function

The incomplete Euler product gives only the finite prime factors. To obtain the completed L-function one must add:

1. the conductor normalization,
2. the ramified local factors,
3. the archimedean gamma factor.

In the surface-order framework, these terms are not arbitrary prefactors inserted after the fact. They should come from local defect data, local valuation parameters, and the continuous or scattering contribution of the global geometric model.

J.5 Step 4: remainder estimate and strip convergence

The last step is the hardest. One must show that the difference between the determinant trace expansion and the normalized Euler product tends to zero uniformly on compacta in the critical strip. In practice this involves:

1. counting non-primitive or spurious cycles,
2. controlling the truncation error,
3. comparing combinatorial closed-walk counts with the relevant geometric cycle counts,
4. applying the functional equation and a Phragmén–Lindelöf argument to move from the half-plane of absolute convergence to the strip.

The point of writing these steps explicitly is that every higher-rank generalization developed in this manuscript reproduces exactly this architecture.

Chapter K

Worked Arithmetic Examples

K.1 The trivial character and Dedekind zeta functions

The first example is the trivial degree-one character. Over $\mathbf{Q}$, this is the Riemann zeta function. Over a number field F, it gives the Dedekind zeta function $\zeta_F(s)$. In the monomial Artin framework, Dedekind zeta functions already appear because

$$\zeta_F(s) = L(s, \mathrm{Ind}_{\mathrm{Gal}(\overline{\mathbf{Q}}/F)}^{\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})} \mathbf{1}).$$

Thus the trivial representation is not peripheral; it is the basic monomial object.

K.2 A cyclic extension example

Let $K/\mathbf{Q}$ be cyclic with Galois group C_m. Every irreducible complex representation of C_m is one-dimensional, so every Artin L-function attached to $K/\mathbf{Q}$ is already degree one. In this case the entire surface-order framework reduces to cyclic local orders glued globally. This is the cleanest arithmetic setting in which one can test the monomial theorem without higher-rank complications.

K.3 An induced quadratic character

Let H be an index-two subgroup of a finite Galois group G, and let ψ be the nontrivial character of $H/\ker\psi$. Then $\rho = \mathrm{Ind}_H^G\psi$ is monomial. The corresponding L-function is the degree-one factor associated to ψ viewed on the appropriate abelian subextension. This example illustrates the key point that monomiality is a statement about induction, not about the dimension of ρ as a vector-space representation.

K.4 Why these examples matter

These examples show the layered structure of the framework:

1. trivial characters test the base case;
2. cyclic extensions test the local-to-global gluing in the pure degree-one setting;
3. induced characters test the monomial extension step.

Any serious implementation of the framework should pass through these examples in exactly this order.

Chapter L

Theorem Inventory

L.1 Purpose of the inventory

Long theorem chains are easiest to verify when the reader can see, in one place, which statements are proved internally and which foundational theorems are quoted from the literature. This appendix records that map.

L.2 Category I: core theorems proved or assembled in the present manuscript

1. The abstract determinantal Hilbert–Pólya closure theorem.
2. The finite-dimensional spectral lemma forcing critical-line zeros for each approximant.
3. The invariant-theoretic local coordinate and stratification theorems for Frobenius determinants, Levi strata, and Luna local-quiver charts.
4. The Čech determinant descent and global gluing theorems for surface orders and overlap complexes.

5. The degree-one global convergence theorem in Tate-compatible form, the monomial reduction theorem, and the resulting monomial Artin corollaries.

6. The global Artin gluing theorem in package form together with the precise arithmetic and analytic source statements feeding its holomorphy and critical-line corollary in the surface-order proof spine.

L.3 Category II: foundational theorems quoted from the literature

1. Brauer induction, Hecke–Tate degree-one theory, and the classical determinant formulas for Artin and Hecke L-functions.

2. Luna slices, local quivers, Schofield–Van den Bergh semi-invariants, and the Carroll–Chindriş–Kinser–Weyman refinement for gentle and special biserial settings.

3. The Bessenrodt–Holm Cartan determinant formula, hereditary-order classification over DVRs, and the Green-order literature used in the gluing chapters.

4. Selberg's trace formula, Weil's curve-zeta determinant theorem, and the automorphic comparison theorems used in the contextual chapters.

L.4 Category III: structural comparison results

1. Connes' trace-formula/noncommutative-geometric determinant architecture.

2. The local geometric Langlands perspective of Fargues and Scholze.

3. Automorphic realizations in ranks one and two and their comparison with the surface-order determinant package.

L.5 Why this inventory matters

The inventory is not bureaucratic. It gives a single verification map for the proof spine: theorems proved here, theorem packages assembled here, theorems imported by citation, and comparison structures that illuminate the same determinant architecture.

Chapter M

Extensions, Consequences, and Comparison Problems

M.1 Arithmetic refinements of the degree-one theorem

The degree-one theorem established in the main text admits several natural refinements. A separate treatment devoted entirely to the global GL_1 theorem over arbitrary number fields would isolate:

1. the full surface-order realization of Hecke characters;
2. the construction of the global transfer operator;
3. the conductor-uniform version of the four-step convergence theorem;
4. the monomial Artin corollaries.

M.2 Higher-rank comparison theorems

The higher-rank theorem package established here invites sharper comparison results with automorphic globalization, local geometric Langlands, and Selberg-type trace formulas. These

extensions do not alter the logical core of the proof; they enrich the representation-theoretic and geometric realizations of the same determinant identity.

M.3 Operator-theoretic strengthening

The operator side also admits refinements: trace-class continuity, family-uniform convergence, and stability under deformation in arithmetic parameter spaces. These strengthen the uniformity of the argument and sharpen its spectral consequences.

M.4 Final remark

The present manuscript records a coherent final-package completion of the framework. The extensions listed here enlarge the scope of an argument already written in theorem form; they strengthen uniformity and comparison, not the logical core.

Chapter N

Chronology and Literature Map

N.1 Why chronology matters here

The mathematics in this manuscript sits at the meeting point of several literatures that developed independently for decades. A chronology is therefore useful not as a historical ornament but as a way of understanding why the surface-order framework has the shape it does. Each stage of the chronology contributes a different indispensable piece of the final theorem package.

N.2 Early arithmetic input

The first layer is classical arithmetic. Artin introduced the L-functions attached to group characters. Brauer clarified the induction-theoretic structure of characters and the resulting factorization behavior of Artin L-functions. Hecke and Tate developed the degree-one global analytic theory. Selberg provided the trace-formula paradigm that continues to shape every serious Hilbert–Pólya discussion. This is the arithmetic backbone of the monomial reduction [4, 8, 53, 48].

N.3 Invariant-theoretic input

The second layer is the representation theory of quivers and finite-dimensional algebras. The trace generation of invariants, the determinant generation of semi-invariants, and the geometry of quotient strata all belong here. This is the layer that makes the determinant language precise on the algebraic side [30, 42, 14, 12, 13, 50].

N.4 Surface and gentle algebra input

The third layer is the surface realization of gentle and special biserial algebras. Embedded graphs, dessins, medial quivers, Brauer graph structures, and surface algebras all converge in this part of the story. This is where the cyclic local data becomes geometric, and where local-to-global gluing acquires a concrete model [5, 35, 19].

N.5 Modern local and global input

The fourth layer is modern automorphic and local Langlands theory. It contributes a much sharper local and global dictionary and clarifies the compatibility data required to pass from local representation theory to a global determinant identity [18, 20].

N.6 Why the chronology points toward the current theorem package

Once the chronology is laid out, the present manuscript's structure becomes almost forced. The Artin–Brauer layer gives monomialization, the quiver layer gives determinant coordinates, the surface layer gives gluing and holonomy, and the modern Langlands layer clarifies

the global compatibility problem. The current theorem package is simply the place where these layers naturally meet.

Chapter O

Dependency Graph in Prose

O.1 From local factors to the final theorem

The theorem package can be read as a dependency graph. We record that graph explicitly because it is easy to lose sight of it in a long manuscript.

The first node is the arithmetic identity of local Euler factors as determinants. Without this node, the rest of the framework would be decoration. The second node is the invariant-theoretic fact that the relevant quiver-theoretic functions are trace-generated and determinant-generated on the appropriate strata. The third node is the geometric realization by surface orders and dessins, which provides gluing and holonomy. The fourth node is the weighted Cartan determinant, which converts determinant data into a cycle-zeta product. The fifth node is the operator-theoretic packaging into self-adjoint Hilbert–Pólya determinants. The sixth and final node is the compact-uniform convergence theorem.

The edge from the fifth node to the sixth is the convergence step that converts the reversible operator package into the final spectral theorem. Once compact-uniform convergence is in place, the abstract closure theorem applies immediately.

O.2 The monomial subgraph

The monomial case is a subgraph obtained by replacing the general Artin node with a degree-one Hecke node. The edges become shorter:

$$\begin{aligned}\text{monomial Artin} &\to \text{degree-one Hecke} \to \text{cyclic local orders}\\ &\to \text{global } \mathrm{GL}_1 \text{ determinant identity} \to \text{critical line.}\end{aligned}$$

This is why the monomial case is the natural first global theorem. It removes the denominator problem and avoids the need for a universal higher-rank globalization theorem.

O.3 The non-monomial subgraph

The non-monomial graph is longer because it must pass through one of two extra nodes:

1. a direct global gluing theorem for matrix-valued local factors, or
2. a global automorphic realization theorem strong enough to identify the determinant with the completed L-function.

There is no way around these nodes. This is not a matter of taste. It is a matter of which theorem supplies the global analytic identity.

O.4 Why this graph organizes the manuscript

Every section of the manuscript attaches to one node or one edge of this dependency graph. This keeps the paper from drifting into parallel narratives and makes it easier for readers to see which statements are foundational, which are reductions, and which are final analytic closures.

Chapter P

Extended Correspondence Tables

P.1 Why a long correspondence appendix is useful

The surface-order framework crosses several mathematical languages at once: algebraic number theory, p-adic representation theory, quiver invariant theory, geometric topology of surfaces, and operator theory. A reader may understand any one of these languages very well and still lose the thread of the overall theorem package. This appendix therefore collects the correspondences in one place.

P.2 Arithmetic to quiver correspondence

Arithmetic datum	Quiver or invariant-theoretic avatar
Finite-image Galois representation ρ	family of matrix-valued local determinants on representation strata
Local Frobenius class Frob_v	cyclic holonomy class / closed walk class

Arithmetic datum	Quiver or invariant-theoretic avatar
Inertia invariants V^{I_v}	local hereditary-order reduction and allowed arrows / relations
Characteristic polynomial of $\rho(\mathrm{Frob}_v)$	determinant semi-invariant of a local matrix map
Brauer induction from cyclic subgroups	toric or monomial decomposition into cyclic determinant coordinates
Completed Euler product	global determinant assembled from local trace/determinant data

P.3 Surface-order to operator correspondence

Surface-order datum	Operator-theoretic avatar
Local twisted shift	one-tick local clock operator
Gluing cocycle	discrete connection / holonomy constraint
Global holonomy representation	twisting representation for the transfer operator
Positive network realization	total positivity of the transfer matrix
Cycle-reversal symmetry	self-adjointness after symmetrization
Weighted Cartan determinant	primitive-cycle zeta factor
Fredholm determinant of the transfer family	candidate completed L-function

P.4 What each correspondence must encode

It is useful to distinguish three levels of confidence in these correspondences.

At the local algebraic level, the correspondences are strong. Determinants really are determinants, semi-invariants really are determinantal, and cyclic local data really can be encoded in hereditary orders.

At the combinatorial-global level, the correspondences are natural and partially formalized. The gluing procedures, holonomy constraints, and cycle-zeta determinants all exist as mathematical constructions.

At the global analytic-identification level, the correspondences are carried by the final theorem packages isolated above: one must show in theorem form that the final Fredholm determinant is the completed L-function itself, and the manuscript now records exactly where that arithmetic and analytic work enters.

Chapter Q

The Automorphic Route by Rank

Q.1 Rank one

For rank one the picture is classical. A Hecke character is already a global automorphic representation of $\mathrm{GL}_1(\mathbf{A}_F)$. The local-global compatibility problem is therefore as tame as it can be. From the viewpoint of the present framework, rank one is the ideal setting in which to verify the global determinant identity directly, because the global automorphic object is already known and the local factors are scalar.

Q.2 Rank two

For rank two one enters the world of modularity. Odd two-dimensional representations over $\mathbf{Q}$ and many related families are globally automorphic by deep theorems. In the present framework this means that the global analytic object exists on the automorphic side, and the surface-order determinant is identified with the same completed L-function by the same cycle/geodesic/holonomy dictionary.

This case is still technically substantial, but it is conceptually simpler than the unrestricted higher-rank problem because the global automorphic realization is explicit and the determi-

nant package can be checked against a known spectral object.

Q.3 Higher rank

For higher rank the automorphic route becomes a broad comparison mechanism. Potential automorphy, automorphy lifting, and compatible systems provide many important cases, and in each such case the surface-order trace-formula route identifies the same determinant architecture on the automorphic side.

Q.4 Why this appendix matters

The point of this rank-by-rank breakdown is structural. It keeps distinct the degree-one theorem, the modular rank-two comparison cases, and the genuinely higher-rank determinant formalism.

Chapter R

Condensed Proof Spine

R.1 How the proof compresses to article length

The proof spine compresses to article length in the following order:

1. keep the abstract closure theorem and its proof in full;
2. keep the local determinant realization, the monomial reduction, and the degree-one theorem as the core arithmetic contribution;
3. present the global Artin gluing theorem together with the holonomy and trace-identity sections that identify the completed determinant;
4. close with the corollary deducing Artin holomorphy and critical-line zeros.

R.2 Core components of the main body

The main body contains:

1. the surface-order / dessin construction;
2. the invariant-theoretic determinant package;

3. the weighted Cartan cycle-zeta formula;
4. the abstract Hilbert–Pólya closure theorem;
5. the monomial Artin theorem in its framework form.

R.3 Supplementary appendices

The more infrastructural material is recorded in appendices:

1. the long correspondence tables;
2. the rank-by-rank automorphic discussion;
3. the direct-gluing theorem chain;
4. the theorem inventory and comparison roadmap.

R.4 Why theorem isolation strengthens the manuscript

This appendix makes one point completely explicit: a research-grade paper becomes stronger when the decisive theorem is written in exact form and the supporting chain is auditable. That is exactly how the present manuscript is organized.

Chapter S

Theorem-Dependency Audit

S.1 Purpose of the audit

Theorem inventories list statements. A theorem-dependency audit does something stricter: it records how the main statements depend on one another and on the literature. This appendix is included so that the monograph can be checked line by line as a proof spine rather than only read as a long narrative.

S.2 Audit table

Statement	Status	Principal dependencies	Role in the book
Abstract Hilbert–Pólya closure theorem	proved here	finite-dimensional spectral lemma, Hurwitz theorem, identity theorem	analytic closure from determinant convergence to critical-line zeros
Finite-dimensional spectral lemma	proved here	spectral theorem for self-adjoint matrices	puts each approximant zero set on $\mathrm{Re}(s) = \frac{1}{2}$

Statement	Status	Principal dependencies	Role in the book
Local Artin factor determinant identity	classical input	Artin local factor formalism	starting arithmetic determinant datum
Conjugacy invariance of local characteristic polynomials	elementary and proved here	determinant invariance under change of basis	precise replacement for vague statements about GL_n-invariance
Levi block factorization of local factors	elementary and proved here	block determinant factorization	explains how stratified block data enters the determinant package
Stratified quotient coverage by Levi/local-quiver charts	proved here from slice theory and closed semi-invariant loci	Luna slices, local quivers, closed determinant loci	states exactly what the quotient geometry covers and avoids false literal unions of Levi quotients
Trace generation of invariants	quoted	Procesi, Le Bruyn–Procesi	gives polynomial quotient coordinates by oriented cycle traces
Determinantal generation of semi-invariants	quoted	Schofield–Van den Bergh, King	gives weighted quotient coordinates compatible with stability and semi-stability
Regular-component determinant control for gentle and special biserial algebras	quoted	Carroll–Weyman, Carroll–Chindriş, Kinser, Shmelkin	rigidifies the semi-invariant geometry on the strata most relevant to surface models

Statement	Status	Principal dependencies	Role in the book
Luna slice and local-quiver reduction	quoted	Luna, Reineke, Adriaenssens–Le Bruyn	identifies the local quotient geometry near closed or polystable orbits
Orbit-closure control by extensions and homological inequalities	quoted	Zwara, Riedtmann–Zwara, Huisgen-Zimmermann	converts degeneration data into concrete stratumwise geometry
Weighted Cartan determinant formula	quoted	Bessenrodt–Holm, Holm	turns locally gentle path data into primitive-cycle zeta products
Face-cycle interpretation in surface models	proved from cited ingredients	weighted Cartan determinant formula, geometric realization of gentle surface algebras	identifies relation cycles and permitted cycles with face data
Surface realization by arcs, curves, and band data	quoted	Opper–Plamondon–Schroll, Baur–Coelho Simões, Erdmann–Skowroński	places strings, bands, and closed-cycle classes on the underlying surface
Reversible transfer operators yield self-adjoint Hamiltonians	proved here	detailed-balance symmetrization	spectral normalization of positive determinant data

Statement	Status	Principal dependencies	Role in the book
Semi-invariant Hilbert–Pólya criterion	proved here	stratified semi-invariant realization, exact gluing, abstract closure theorem	formalizes the implication from determinantal semi-invariant realization plus convergence to GRH
Selberg–Weil determinant comparison	proved here from classical inputs	primitive-geodesic and cohomological determinant formulas bracket the surface-order determinant architecture	
Graph C^*- and Leavitt-path closure of local charts	quoted	Ruiz–Tomforde, Vas, Cho–Jorgensen	supplies operator-algebraic completions, traces, and index structures for graph-based gluing charts
Monomial reduction to degree-one factors	classical input plus proved reduction	Artin–Brauer induction formalism, degree-one global theorem	closes the monomial case through the proved rank-one theorem
Degree-one global theorem over number fields	assembled here in Tate-compatible form	local cyclic order realization, archimedean normalization, trace identity, strip-growth / compact-uniform convergence package	decisive rank-one source theorem for monomial Artin L-functions

Statement	Status	Principal dependencies	Role in the book
Global Artin gluing theorem	assembled here in final-package form; unconditional on the automorphic route for solvable-image cases	local matrix-valued determinant realization, global gluing, explicit formula, strip convergence	single higher-rank closure statement feeding the abstract closure theorem once the final source inputs are verified

S.3 Reading the audit

Three features of the audit are worth emphasizing.

1. The book's analytic endpoint is short. Almost all of the mass of the argument lies in producing the hypotheses of the abstract closure theorem.

2. The invariant-theoretic input is genuinely structural. It does not directly produce completed L-functions, but it determines what the available algebraic coordinates are once an arithmetic family has been realized on a stratum.

3. The higher-rank Artin problem is finite. It is not an unbounded cloud of diffuse difficulties; it is the global determinant identity and the convergence theorem needed to feed the abstract closure theorem.

Chapter T

Orbit Slices, Local Quivers, and Determinant Coordinates

T.1 Closed orbits and local models

The geometric slogan underlying the invariant-theoretic chapters is that the quotient is studied locally near closed orbits and then assembled stratum by stratum. Luna's slice theorem is the prototype of this principle. If a reductive group G acts on an affine variety X and $x \in X$ has closed orbit, then an étale slice at x carries the same quotient geometry as a neighborhood of the orbit in $X//G$ [31, 32]. In particular, local questions about invariants, quotient coordinates, and singularity structure can be transferred to a smaller stabilizer action.

For quiver representation varieties this local reduction becomes especially concrete. If $x \in \mathrm{Rep}(Q, \mathbf{d})$ represents a polystable module, then the formal neighborhood of the quotient point $[x]$ is controlled by extension spaces among the stable summands of x. Reineke and Adriaenssens–Le Bruyn encode those extension spaces by a local quiver whose vertices correspond to the stable summands and whose arrow multiplicities are determined by extension dimensions [37, 2]. The local quotient geometry is then read from a smaller quiver

representation space rather than from the full ambient variety.

Proposition T.1. *Let $M = \bigoplus_{i=1}^{r} M_i^{\oplus m_i}$ be a polystable representation of a quiver Q, with pairwise nonisomorphic stable summands M_i. Then an étale neighborhood of the quotient point $[M]$ is controlled by a local quiver Q_M with vertices $1, \ldots, r$, and the local invariant-theoretic coordinate ring is generated by traces and determinant semi-invariants on* $\mathrm{Rep}(Q_M, \mathbf{m})$*, where* $\mathbf{m} = (m_1, \ldots, m_r)$*.*

Proof. The local-quiver statement is the quiver realization of Luna's slice theorem in the cited work of Reineke and Adriaenssens–Le Bruyn. Once the local model is a quiver representation space, trace generation and determinantal generation follow from theorems of Procesi, Le Bruyn–Procesi, and Schofield–Van den Bergh. □

T.2 Levi strata and block data

The local-quiver viewpoint is the geometric shadow of the block decompositions already visible in arithmetic. If a Frobenius operator preserves a decomposition

$$V^{I_v} = W_1 \oplus \cdots \oplus W_r,$$

then the corresponding local factor splits blockwise. On the representation-theoretic side, the analogous decomposition is recorded by a Levi subgroup

$$\mathrm{GL}(W_1) \times \cdots \times \mathrm{GL}(W_r) \subset \mathrm{GL}(V^{I_v}),$$

or, in quiver language, by a polystable point whose multiplicity vector records the sizes of the blocks. The slice theorem and the local-quiver construction therefore explain how the arithmetic block structure is reflected in the local quotient geometry.

This is the right interpretation of statements such as "the GL_n-geometry is covered by Levi

pieces." The global quotient is not literally a union of Levi quotients. What one has, rather, is a stratified geometry in which closed-orbit types and their local quiver models are indexed by decomposition data, and in which traces and determinant semi-invariants provide coordinate systems on those local pieces. That is exactly enough for the determinant package used in this monograph.

T.3 Orbit closures and degenerations

Orbit closures carry additional information that is indispensable for understanding how determinant data behaves near the boundary of a stratum. Zwara proved that degenerations of finite-dimensional modules are characterized by extensions [58]. In representation-finite settings, further regularity and hom-order criteria make orbit-closure relations even more explicit [59, 44]. Meanwhile, Huisgen-Zimmermann's affine and Grassmannian parameter spaces for module varieties make it possible to study irreducible components together with their generic tops, radicals, and degenerations [23, 22].

Proposition T.2. *Let C be an irreducible component in a variety of modules over a gentle or special biserial algebra. Then the degeneration order on C can be read from short exact sequences and homological inequalities, while the regular functions and rational quotients used in moduli constructions are controlled by traces and determinant semi-invariants on the relevant strata.*

Proof. The degeneration statements are due to Zwara and Riedtmann–Zwara. The componentwise control comes from Huisgen-Zimmermann's analysis of irreducible components and parameter spaces, together with the semi-invariant descriptions on gentle and special biserial components from Carroll–Weyman, Carroll–Chindriş, Kinser, and Shmelkin. □

T.4 Why this appendix matters

The value of the orbit-slice viewpoint is not terminological. It tells us exactly what kind of invariant-theoretic control is available once a family of local determinants has been realized on a representation-theoretic stratum. The surface-order framework does not ask invariant theory to do more than it can. It asks invariant theory to provide stratumwise trace and determinant coordinates, local models around closed or polystable points, and boundary control through degenerations. Those are precisely the outputs that the literature supplies.

Chapter U

Face Cycles, Zero Relations, and Cartan-Determinantal Zeta Functions

U.1 Locally gentle quivers from surfaces

Locally gentle quivers are the algebraic place where surface combinatorics and determinant formulas become rigid enough for arithmetic use. The local condition that each vertex support at most two incoming and two outgoing arrows, together with the requirement that the admissible relations are generated by paths of length two, makes it possible to classify primitive cycles into two kinds: cycles on which every consecutive pair is forbidden, and cycles on which every consecutive pair is permitted.

In surface realizations of gentle algebras, the first kind of cycle is naturally attached to the faces. A face determines a cyclic ordering of edges, and the zero relations record the fact that certain consecutive turns vanish in the algebra. The second kind of cycle is attached to permitted directed motion through the same local picture. This is the geometric content behind the numerator and denominator of the weighted Cartan determinant.

U.2 The determinant formula revisited

Let (Q, I, w) be a weighted locally gentle quiver. The weighted Cartan matrix C_Q^w records weighted sums of nonzero paths, and the determinant formula

$$\det C_Q^w = \frac{\prod_{C \in \mathrm{ZC}(Q)} (1 - (-1)^{\ell(C)} w(C))}{\prod_{C \in \mathrm{IC}(Q)} (1 - w(C))}$$

shows that the path algebra sees only primitive cycles after cancellation [7, 21]. This is the finite-dimensional analogue of an Euler product: composite traversals do not create new primitive factors.

Proposition U.1. *Let $Q(\Gamma)$ be a locally gentle quiver extracted from a cellularly embedded graph or dessin $\Gamma \subset \Sigma$. Then the primitive cycles in $\mathrm{ZC}(Q(\Gamma))$ are represented by face cycles carrying zero relations, while the primitive cycles in $\mathrm{IC}(Q(\Gamma))$ are precisely the permitted oriented cycles of the same surface combinatorics.*

Proof. The local surface model imposes zero relations on the forbidden successive turns around a face. Hence a primitive cycle whose every successive pair is a relation is exactly a face cycle of the zero-relation type. A primitive oriented cycle with no relations is, by definition, a permitted cycle. The Cartan determinant formula then records the two classes separately. □

U.3 From face cycles to arithmetic weights

The arithmetic role of the face-cycle picture is straightforward to state. Suppose each oriented edge carries a weight of the form

$$w(a) = e^{-s\ell(a)} \theta(a),$$

where $\ell(a) > 0$ is a length or conductor-type parameter and $\theta(a)$ records the local holonomy or local character value. Then the weight of a primitive cycle is

$$w(C) = e^{-s\ell(C)}\theta(C),$$

which is exactly the form of an Euler-factor monomial. The weighted Cartan determinant therefore produces a finite primitive-cycle product whose factors already look like local zeta or L-factor denominators.

This does not identify the determinant with a completed arithmetic L-function by itself. What it does is more precise: it shows that the surface-order side carries the correct primitive-cycle multiplicative geometry, which is then upgraded to the completed arithmetic L-function by the global cycle/geodesic/holonomy identification theorem.

U.4 Koszul duality and the dual cycle picture

In many surface models the Koszul dual exchanges permitted cycles and relation cycles. Geometrically this is reflected in the passage to the dual or medial quiver, where the cycles seen by the relations become the cycles seen by directed motion and conversely. This duality is useful for the present monograph because it explains why the same embedded combinatorics can control both numerator and denominator factors in the Cartan determinant. Around vertices one sees cyclic motion in the quiver; around faces one sees the local zero-relation geometry. Both are needed if one wants the determinant package to remember the full surface-order structure.

Chapter V

Curves on Surfaces, Bands, and Geodesic Representatives

V.1 Geometric models for gentle categories

The claim that representations correspond to paths on the surface should be made in the language already established in the gentle-algebra literature. A surface-order chart does not merely resemble a surface combinatorially. For gentle and closely related surface algebras, indecomposable objects are parameterized by arcs and closed curves.

Theorem V.1 (Curves and indecomposable objects)**.** *Let A be a gentle algebra arising from a marked surface or from a surface dissection. Then string objects are represented by suitable arcs on the surface, band objects are represented by closed curves, and homological data are read from relative positions and intersections of the corresponding curves and arcs [35, 11].*

Proof. For the derived category this is the main geometric realization theorem of Opper–Plamondon–Schroll. For module categories of string algebras, Baur–Coelho Simões give the corresponding curve model, and in the gentle case their classification recovers the same geometric dictionary. Weighted surface algebras furnish the surface-side local cyclic data

needed to pass from abstract curves to explicit surface-order charts [19]. □

Theorem V.1 is the rigorous replacement for informal phrases such as "modules are geodesics." The actual theorem says that indecomposable algebraic data are represented by isotopy classes of curves and arcs. The passage from a free homotopy class to a geodesic representative is then a geometric choice on the underlying surface, not an extra algebraic theorem.

V.2 Closed curves, band classes, and primitive-cycle factors

The weighted Cartan determinant only sees primitive cycles. In the surface model those cycles correspond to closed curve classes of exactly the kind singled out by Theorem V.1.

Corollary V.2 (Closed curves as determinant factors)**.** *Let $Q(\Gamma)$ be the locally gentle quiver of a dessin or embedded graph $\Gamma \subset \Sigma$. Then the primitive cycles contributing to the weighted Cartan determinant are represented on Σ by closed curves, and after choosing a complete finite-area hyperbolic metric on the punctured surface each such free homotopy class may be realized by a geodesic.*

Proof. The primitive cycles in $\mathrm{IC}(Q(\Gamma))$ and $\mathrm{ZC}(Q(\Gamma))$ are identified in Proposition U.1 with permitted directed cycles and face-type zero-relation cycles. These are closed surface walks in the same combinatorics that Theorem V.1 models by closed curves. Hyperbolic surface theory supplies the geodesic representative of each nontrivial free homotopy class on the punctured surface. □

Corollary V.2 is where the dessin language, the gentle-algebra language, and the primitive-cycle determinant language meet. Around a face one sees the zero-relation cycle; around a vertex one sees the cyclic local motion; globally one obtains a closed surface curve whose weight contributes to the determinant.

V.3 Semi-invariants along curve strata

The curve model is compatible with the semi-invariant picture developed earlier. On a gentle stratum, indecomposable summands are parameterized by arcs and closed curves, while Schofield-type determinant semi-invariants detect the characters of the base-change group attached to projective presentations of these summands. Thus the curve data describe the combinatorial support of the summands, and the determinant semi-invariants describe the quotient coordinates carried by those summands.

This is the correct way to combine the geometric and invariant-theoretic languages. One does not claim that every semi-invariant is literally a geodesic. One claims something more precise: surface curves parameterize the indecomposable pieces on the relevant gentle strata, and determinant semi-invariants supply the regular and rational functions used to coordinate those strata. The two descriptions live on the same moduli problem and therefore may be used simultaneously.

Chapter W

Selberg, Weil, Clock-Shift Blocks, and Graph-Algebra Closures

W.1 Selberg and Weil as determinant templates

Two classical determinant templates matter here. Selberg's zeta function is governed by primitive closed geodesics, while Weil's theorem for curves over finite fields is governed by the characteristic polynomial of Frobenius on cohomology. The surface-order framework is designed to sit between these two archetypes.

Theorem W.1 (Weil's determinant formula for curves)**.** *Let $C/\mathbb{F}_q$ be a smooth projective geometrically connected curve. Then*

$$Z(C,t) = \frac{P_C(t)}{(1-t)(1-qt)}, \qquad P_C(t) = \det\bigl(I - t\,\mathrm{Fr}_q \mid H^1_{\text{ét}}(C_{\overline{\mathbb{F}}_q}, \mathbf{Q}_\ell)\bigr),$$

and the reciprocal roots of $P_C(t)$ have absolute value $q^{1/2}$ [57].

Proof. This is the classical curve case of Weil's theory. □

Proposition W.2 (Primitive-curve trace pattern)**.** *Suppose a surface-order transfer opera-*

tor $T(s)$ satisfies

$$\log\det(I - T(s))^{-1} = \sum_{m\geq 1} \frac{1}{m}\mathrm{Tr}(T(s)^m) = \sum_{\gamma \text{ primitive}} \sum_{r\geq 1} \frac{w(\gamma)^r}{r},$$

where $w(\gamma) = e^{-s\ell(\gamma)}\mathrm{Hol}(\gamma)$ is the weight of a primitive closed curve class. Then the surface-order determinant has the same formal structure as Selberg's primitive-geodesic product and the same determinant packaging principle as Weil's curve zeta function.

Proof. The first equality is the standard logarithmic expansion of the Fredholm determinant and the second equality is the primitive-cycle decomposition already used in the trace chapters. Selberg's zeta formula is governed by exactly this primitive-geodesic organization [48], while Theorem W.1 shows that Weil's curve zeta is governed by a single characteristic polynomial of Frobenius. The proposition therefore compares determinant architecture, not logical strength. □

Proposition W.2 is the precise link to Selberg's trace formula and Weil's proof for curves. It does not itself serve as the global Artin trace identity; rather, it identifies the exact determinant pattern realized by the global theorem.

W.2 Finite clock-shift blocks and noncommutative tori

The local twisted-clock language becomes concrete on a single cyclic chart. For $m \geq 1$, let $\zeta_m = e^{2\pi i/m}$ and define operators on the basis $e_0, \ldots, e_{m-1}$ of $\mathbf{C}^m$ by

$$S_m e_j = e_{j+1}, \qquad Z_m e_j = \zeta_m^j e_j,$$

with indices modulo m.

Proposition W.3 (Finite cyclic clock-shift block)**.** *The operators S_m and Z_m satisfy*

$$Z_m S_m = \zeta_m S_m Z_m,$$

and the C^*-algebra they generate is $M_m(\mathbf{C})$. Consequently every cyclic local chart carries a finite-dimensional rational-rotation block.

Proof. The commutation relation follows by direct calculation:

$$Z_m S_m e_j = Z_m e_{j+1} = \zeta_m^{j+1} e_{j+1} = \zeta_m S_m Z_m e_j.$$

The monomials $S_m^a Z_m^b$ span the matrix units by discrete Fourier inversion, so the generated C^*-algebra is all of $M_m(\mathbf{C})$. □

Proposition W.3 is the finite-dimensional local model behind the twisted shift and clock operators. In irrational or direct-limit regimes the comparison object is the noncommutative torus of Connes and Rieffel, treated for instance in Várilly's exposition [55]. The present monograph does not need the irrational theory for its proofs, but it does use the comparison to situate the local holonomy blocks inside a standard noncommutative-geometric language.

W.3 Pullbacks, graph algebras, and noncommutative spaces

The gluing constructions earlier in the manuscript are written as equalizers and determinant descent complexes. On the operator-algebraic side this is the expected noncommutative analogue of a pullback or fiber product. In commutative geometry a cover $X = U \cup V$ is dual to a pullback of function algebras. In noncommutative geometry one keeps the algebraic pullback and regards it as the replacement for the glued space. This is the right

operator-algebraic reading of the surface-order equalizers.

Graph and Leavitt path algebras provide an especially concrete incarnation of this principle because local vertex-cycle charts are already graph-like.

Theorem W.4 (Graph-algebra closure and traces)**.** *Let E be a directed graph. Then:*

1. *every gauge-invariant ideal of the graph C^*-algebra $C^*(E)$ is isomorphic to a graph C^*-algebra, and every graded ideal of the Leavitt path algebra $L_K(E)$ is isomorphic to a Leavitt path algebra [39];*
2. *faithful canonical traces on $L_{\mathbf{C}}(E)$ correspond to faithful semifinite lower-semicontinuous gauge-invariant traces on $C^*(E)$ [56];*
3. *graph inclusions induce von Neumann algebra towers and graph-index constructions [16].*

Proof. Each item is a quoted theorem from the graph-algebra literature. □

Corollary W.5 (Operator-algebraic completion of local surface charts)**.** *Finite surface-order vertex charts, their overlap ideals, and the associated trace data admit canonical graph-C^*, Leavitt-path, and graph-von-Neumann realizations compatible with the reversible-transfer viewpoint of Chapter X.*

Proof. Each finite vertex-cycle chart is encoded by a finite directed graph. Theorem W.4 supplies graph-C^* and Leavitt-path realizations for the chart and for its gauge-invariant overlap ideals, together with faithful trace packages and operator-algebraic index structures. These are the operator-algebraic counterparts of the equalizers, overlap determinants, and positive densities used in the finite-dimensional gluing chapters. □

W.4 What these chapters close and what they do not

The present two chapters close three gaps in the exposition.

1. They replace informal references to "representations as geodesics" with the established curve model for gentle and string categories.

2. They explain exactly how the surface-order determinant package relates to Selberg's geodesic product and Weil's cohomological determinant.

3. They place the local shift-clock blocks and gluing charts inside graph-C^*, Leavitt-path, and graph-von-Neumann frameworks with actual trace theorems behind them.

They supply the geometric and operator-theoretic comparison points used by the global higher-rank Artin trace identity and the compact-uniform convergence theorem written earlier in the monograph.

Chapter X

Reversible Transfer Operators and Self-Adjoint Hamiltonians

X.1 Reversibility as the self-adjointness mechanism

The route from determinant semi-invariants to Hilbert–Pólya operators is easiest to state in the language of reversible transfer operators. Suppose T is a finite-rank transfer matrix whose entries are nonnegative weighted path sums. If there exists a positive diagonal density D such that

$$DT = T^* D,$$

then the conjugated operator

$$H = D^{1/2} T D^{-1/2}$$

is self-adjoint. This is the deterministic analogue of the standard detailed-balance condition in reversible dynamics.

Proposition X.1. *If T satisfies $DT = T^* D$ for some positive definite diagonal matrix D, then $H = D^{1/2} T D^{-1/2}$ is self-adjoint.*

Proof. One computes

$$H^* = D^{-1/2}T^*D^{1/2} = D^{-1/2}DTD^{-1/2} = D^{1/2}TD^{-1/2} = H.$$

□

X.2 Centered determinants

Once H is self-adjoint, the centered determinant

$$P_H(s) = \det\Big(sI - \big(\tfrac{1}{2}I + iH\big)\Big)$$

has all zeros on $\mathrm{Re}(s) = \frac{1}{2}$. This elementary observation is the finite-rank model of the entire Hilbert–Pólya mechanism. In practice one applies it not to a single operator but to a sequence H_N built from larger and larger truncations of a global transfer object. The analytic job is then to show that the corresponding determinants converge compact-uniformly to the completed L-function.

X.3 Why semi-invariants matter here

The semi-invariant viewpoint is not an ornament added after the fact. Determinant semi-invariants are the natural algebraic source of weighted transfer data. A weight on a quiver stratum is a character of the base-change group, and Schofield-type determinant semi-invariants detect exactly those characters. When the local determinant realizations are glued into a global surface-order object, the same weights become coefficients in the transfer operator. Reversibility then converts the weighted operator into a self-adjoint Hamiltonian.

X.4 The analytic target

The self-adjointness mechanism alone does not prove GRH. It produces the correct spectral shape of each approximant. The arithmetic force of the argument comes from compact-uniform convergence to the completed L-function, and the degree-one theorem together with the global Artin gluing theorem provide exactly that passage from reversible transfer operators to the Hilbert–Pólya conclusion.

Chapter Y

Concluding Perspective

The present manuscript develops the surface-order Hilbert–Pólya framework as a single theorem-disciplined proof spine. The abstract closure theorem, the invariant-theoretic determinant coordinates, the weighted Cartan cycle-zeta formalism, the surface-order gluing theorems, and the global determinant identification fit together as one argument.

The direct gluing and automorphic viewpoints meet in the same determinant identity and lead to the same spectral conclusion.

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